

AVOIDING TRANSPARENCY THROUGH OFFSHORE REAL ESTATE: EVIDENCE FROM THE UNITED KINGDOM*

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Abstract

In the middle of the 2010s, countries implemented the most comprehensive policy ever enacted to tackle financial tax evasion: the Common Reporting Standard (CRS). We study the effect of this policy on the investment patterns of offshore account holders. The CRS is a multilateral automatic exchange of information standard that introduces cross-border reporting on financial assets but not on real estate. Using administrative data on real estate purchases made by foreign companies in the UK, our paper provides evidence that offshore users shifted financial assets to real estate assets to avoid the new reporting requirements. We show that real estate investments from tax havens that are more exposed to the CRS significantly increase after the introduction of the policy. We estimate that around \$45 billion has been invested in the UK real estate market between 2013 and 2016 in response to the CRS. Our results suggest that real estate assets account for a growing share of offshore portfolios, partly because of the increased transparency in ownership of cross-border financial assets. JEL codes: H26, R30, D31.

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1 Introduction

How to enforce taxes on cross-border investments is a major challenge of globalization, as financial assets held in tax havens represent about 8% of households' financial wealth (Zucman, 2013). The vast majority of offshore wealth remains unreported (Johannessen and Zucman, 2014; Alstadsæter et al., 2019), leading to substantial tax losses (Pellegrini et al., 2016). In an attempt to gather more information about this offshore wealth and tackle tax evasion, the OECD endorsed the Common Reporting Standard (CRS) in 2014.¹ The CRS is a standard of multilateral automatic exchange of information (AEOI) designed to limit the possibilities for taxpayers to hold undeclared assets, by introducing third-party reporting of cross-border financial assets between participating jurisdictions. More than 100 jurisdictions have agreed to exchange information today, making it the most comprehensive policy ever enacted to enhance tax transparency. The CRS has led to a reduction of cross-border bank deposits held in tax havens of at least 11% (Menkhoff and Miethe, 2019; Casi et al., 2020; O' Reilly et al., 2021; Beer et al., 2019). One major loophole of the policy however, is that it only covers financial assets. This creates new incentives for evaders to invest their offshore holdings in non-financial assets, in order to avoid the reporting requirements introduced by the CRS.

This paper provides evidence that investors rebalance their offshore portfolios towards real estate assets in response to the CRS. We provide quasi-experimental evidence of the effects of the policy on the offshore real estate market—i.e. foreign-owned real estate assets—in the United Kingdom. The UK is an ideal setting to investigate this issue for at least three reasons. First, London provides amenities that are very attractive for the ultra-wealthy population including high-quality housing stock, a well-connected city and good international schools (Knight Frank, 2016). Second, the real estate market in the UK – and in particular in London – is often considered as a “safe haven” for foreign capital, meaning an asset for which demand is not curtailed in periods of political or economic uncertainty. Third, UK properties can be purchased through vehicles registered in secrecy jurisdictions, thus attracting investors who want to remain anonymous (Tax Transparency UK, 2015). Overall, this means that if the CRS prompted a “flight-to-safety” of offshore evaders towards real estate assets, the UK is likely the destination of a substantial share of this new flow of capital.

To identify the effect of the CRS on real estate investments, we draw on unique administrative data on foreign corporate ownership of UK properties. As most of the purchases are made through shell companies incorporated in tax havens, the ultimate owner's identity often remains undisclosed. We break new ground in determining the ultimate origin of corporate real estate investments in the UK by combining this dataset with beneficial ownership data from the Offshore Leaks (which includes the Panama Papers, the Pandora Papers, and other leaks) and from the Luxembourg company register. To investigate the effects of the CRS on real estate investments, we conduct a difference-in-differences analysis around the announcement of the policy. We find that the policy caused an increase

¹ See O' Reilly et al. (2021) for a detailed timeline of the expansion of tax transparency and the introduction of the CRS.

in real estate purchases by companies based in tax havens commonly used by individuals from countries adopting the CRS.

Our paper makes two main contributions. The first is to provide new evidence on the size and the nature of the offshore real estate market in the UK. We find that the offshore property market in the UK is large. We estimate that almost \$240 billion worth of properties were held from abroad in 2018, representing about 4% of the total stock of English and Welsh real estate. About 60% of this real estate wealth is held through corporations, while 40% is held directly by individuals, and foreign ownership is higher for corporate ownership (9%) than direct ownership (2%). Approximately two-thirds of foreign-owned properties are located in London, totaling an amount of \$150 billion.²

The UK offshore real estate market is also highly secretive. More than 75% of these properties were owned through a tax haven, primarily through Jersey and the British Virgin Islands. This is particularly the case for corporate ownership, as more than 94% of the companies owning UK real estate in 2018 are incorporated in a tax haven. To trace back ultimate ownership of UK properties, we match our administrative dataset of transactions to the Offshore leaks, the Luxembourg company register, and the newly implemented UK Register of Overseas Entities which contains information on the beneficial owners of foreign companies holding properties in the UK. We identify the ultimate owners of more than 20% of the properties in our sample. We find that most investments channeled through shell companies come from residents of the United Kingdom, the Middle East (Arabian Peninsula, Israel, Jordan) and the United States. The finding that a significant fraction of offshore real estate is ultimately held by UK residents is suggestive of a “home bias” in investment decisions of offshore users (Coeurdacier and Rey, 2013).

Our second contribution is to provide quasi-experimental evidence of the effects of the CRS on offshore real estate investments in the UK. We start by documenting a significant heterogeneity in cross-country use of tax havens. For example, Western European residents are more likely to incorporate a company in Luxembourg, while residents from the Asia-Pacific region are more likely to use the British Virgin Islands. Leveraging these country-specific preferences, we compute the level of exposure of tax havens to the CRS based on whether their clients are from CRS-adopting countries. This allows us to build credible estimates of the origin of individuals investing through tax havens in the UK. Using a difference-in-differences design with a continuous treatment variable corresponding to CRS exposure, we show that a higher exposure is associated with significantly higher real estate investments after a large group of countries commit to implementing the CRS in 2013-2014. To investigate the origins of this increase, we examine the transaction trends above/below the £1 million cut-off threshold. We find that the surge in real estate investments is primarily driven by the purchase of high-value properties, which corroborates recent evidence that most offshore users come from the top of the income and wealth distribution (e.g. Alstadsæter et al., 2019; Guyton et al., 2021). Then, we confirm that our results are not driven by tax reforms in the UK property market during the post-CRS period –i.e.

²To put this number in perspective, Alstadsæter et al. (2022) estimate that foreign-owned real estate in Dubai also amounted to around \$150 billion in 2020.

between 2014 and 2017— by comparing investments in residential real estate, which were affected by numerous tax changes over this period, to purchases of commercial properties, mostly unaffected by the reforms. We find that the purchasing dynamics around the introduction of the CRS are similar across both residential and commercial property classes.

The shift of offshore investments towards real estate assets appears to be substantial. In a back-of-the-envelope calculation, we estimate that around \$45 billion have been invested in real estate in England and Wales in the post-CRS period because of the threat the CRS poses for people hiding financial assets offshore. To gain insight into the overall magnitude of asset shifting in response to this transparency policy, we compare this figure to the estimated effect of the Common Reporting Standard (CRS) on financial assets. Our findings suggest that around 9% of the offshore financial wealth that flowed out of tax havens due to increased transparency was ultimately reinvested in real estate in the UK.³ One consequence of this result is the growing importance of real estate assets in offshore portfolios. In conjunction with the observation that offshore wealth is heavily concentrated among the top of the income and wealth distribution (Alstadsæter et al., 2019; Guyton et al., 2021; Londoño Vélez and Ávila-Mahecha, 2021; Lejour et al., 2023), our findings have important implications for the composition of wealth at the top, suggesting that the share of real estate assets may have been significantly underestimated.

Our results also have direct implications for the design of automatic exchange of information policies. These policies should include all types of assets in their scope to avoid tax evaders taking advantage of coverage gaps through portfolio reshuffling. However, governments first need to gather information about the owners of domestic assets, in particular real estate, to be able to share it through automatic exchange of information agreements. Our finding that the Common Reporting Standard increased investments in the UK real estate market could also have important consequences for UK residents. This influx of investments could affect the neighborhoods that are favored by offshore buyers, through property prices (Sá, 2016), rental prices, share of vacant properties or relocation of local populations.

Our paper adds to the literature studying the amount of wealth held offshore (Zucman, 2013; Pellegrini et al., 2016; Vellutini et al., 2019; Henry, 2012), its distribution across countries (Alstadsæter et al., 2018) and within countries (Alstadsæter et al., 2019; Guyton et al., 2021; Londoño Vélez and Ávila-Mahecha, 2021; Lejour et al., 2023; Johannesen et al., 2020, 2023). Our main contribution to this literature is to focus on real estate assets and to highlight their importance in offshore portfolios. Previous papers have studied some aspects of foreign ownership of real estate in Norway, (Alstadsæter and Økland, 2022), Dubai (Alstadsæter et al., 2022), France (Morel and Uri, 2021), or the UK (Bomare, 2019; Johannesen et al., 2022; Bourne et al., 2022). Our paper provides a full assessment of offshore ownership in the

³Where did the rest of the money go? Part of it was repatriated through voluntary disclosure schemes implemented in various countries. Additionally, while the UK serves as a significant cross-border real estate market, it is likely that other property markets such as the US, Dubai, and others have also absorbed shifted wealth. Evidence suggests that tax evaders have employed alternative evasion strategies, beyond real estate investments, following the implementation of the CRS. For instance, some individuals have relocated unreported assets to the US (Casi et al., 2020) or have taken advantage of citizenship-by-investment programs (Langenmayr and Zyska, 2023). The use of trusts also presents a means to evade reporting under the CRS (Knobel and Meinzer, 2014).

United Kingdom, encompassing both residential and commercial properties, and both direct and corporate ownership.

Second, our paper relates closely to the expanding literature on the impacts of policies designed to enhance tax transparency (see [Slemrod \(2019\)](#) for an overview). Specifically, it contributes to the body of research evaluating the effectiveness of the CRS in enforcing tax compliance ([Menkhoff and Miethe, 2019](#); [Casi et al., 2020](#); [O' Reilly et al., 2021](#); [Alstadsæter et al., 2023](#); [Bomare and Collin, 2023](#)). Our study confirms that individuals engaging in non-compliant behavior adapt their strategies in response to changes in the global tax landscape, seeking alternative means to conceal their assets and to continue under-reporting ([Johannesen, 2014](#); [Johannesen and Zucman, 2014](#); [Casi et al., 2020](#); [Martínez-Toledano and Roussille, 2023](#)). While [De Simone et al. \(2020\)](#) provides indirect evidence that FATCA, the US policy of Automatic Exchange of Information, prompted a shift from foreign financial assets into the real estate market, we study transaction-level administrative data that provide the first direct evidence of this shifting effect.

Third, we also contribute to a literature studying the drivers and the effects of foreign investment in real estate markets (e.g. [Badarinza and Ramadorai, 2018](#); [Sá, 2016](#); [Cvijanovic and Spaenjers, 2020](#); [Hilber and Vermeulen, 2016](#)). A set of studies focuses in particular on the effects of transparency measures on real estate markets. While [Collin et al. \(2022\)](#) find no effect of rules increasing transparency in the U.S. on foreign purchases of properties, other studies show that offshore real estate investments in the UK ([Johannesen et al., 2022](#); [Collin et al., 2023](#)) and in Singapore ([Argawal et al., 2020](#)) reacted to national policies increasing transparency. Our paper shows that policies aimed at curbing offshore tax evasion can have the unintended effect of increasing real estate investments, highlighting the importance of studying the international tax and secrecy environment when investigating foreign investments in properties. It also makes an important methodological contribution, by proposing a new way to assign capital flows to their ultimate source. Using this methodology, we document that a substantial part of foreign investments in the UK real estate market can actually be traced back to domestic individuals. This approach can be generalized to other contexts where the identity of investors is ultimately obscured.

The rest of the paper is organized as follows. Section 2 describes our data. Section 3 provides descriptive evidence on the nature of the UK offshore real estate market. In section 4, we show how corporate real estate investments in the UK responded to the CRS, and section 5 compiles robustness checks of our main results. Section 6 proposes to estimate the shifting effect from financial to UK real estate assets. Section 7 concludes.

2 Data

The UK is a popular destination for global real estate investments ([Sá, 2016](#); [Badarinza and Ramadorai, 2018](#)). London is particularly attractive, because of its opportunities for high-end investments and the amenities it offers to the wealthy population ([Knight Franck, 2016](#)). This section outlines the data sources we use to study offshore real estate in the UK, namely real estate held from abroad. We draw primarily on administrative data on indirect

ownership of real estate, i.e. properties held through foreign companies. We complement these datasets with corporate registries and leaks, which provide information on the country of the owners of the purchasing companies. To present a comprehensive picture of the offshore real estate market in the UK, we also rely on an administrative dataset of direct foreign ownership of UK properties.

2.1 Foreign ownership of properties in the UK

Indirect ownership of property. Our primary data source is the Overseas Companies Ownership Dataset (OCOD), managed by the Land Registry. The OCOD provides comprehensive snapshots of real estate ownership by foreign companies in England and Wales. The dataset includes information on the date and location of purchases made by foreign companies, the recorded price (available in 30% of cases), tenure classification (Freehold or Leasehold), and detailed information about the corporate buyer, including its name, country of incorporation, and registered address. The first available snapshot dates back to November 2015, with quarterly updates until November 2017, transitioning to monthly updates thereafter. We focus on transactions made from 2000 until the end of 2020.

The OCOD has been used to study the effects of foreign investments on the UK real estate market ([Sá, 2016](#)), to predict the country of origin of investors using corporate wrappers to buy properties ([Bomare, 2019](#)), to study the motives of owning real estate through a company ([Johannessen et al., 2022](#)) and to analyze the effects of a new measure aimed at eliminating anonymous ownership of properties ([Collin et al., 2023](#)).

Combining machine learning techniques with the use of external transaction data sources, we transform the raw data to make it at the transaction level, parse the property addresses and classify them between residential and commercial properties. We provide details of this extensive processing work in Appendix section [B.1](#).

One caveat of the OCOD is that approximately 70% of the transactions in the dataset do not include information on the price paid. This can happen for several reasons. First, HM Land Registry only started collecting price data for transactions involving non-private buyers from October 2013.⁴ Second, companies often purchase portfolios of properties, and if the price paid for each property is not specified in the transaction document,⁵ the Land Registry excludes the price information from the register. To recover missing price information, we make use of another dataset managed by the Land Registry, the Price Paid Data (PPD). The PPD provides details on the majority of residential and some non-residential property purchases in the UK since 1995, including price, address, and transaction date. After standardizing address information in both datasets, we match the OCOD to the PPD using the property’s address and postcode. The matching process allows us to recover the price of an additional 24% of the transactions. The matching rate is not 100% because the PPD excludes a substantial share of properties present in the OCOD, including non-residential properties. Moreover, addresses have different structures in both datasets and we adopt a conservative

⁴For some transactions happening before that date the price information can be non-missing if the applications were lodged for registration to the Land Registry after 2013.

⁵The transaction document is Form TR5: Transfer of a portfolio of titles (whole or part).

matching technique to avoid false positives. We predict the price of the remaining 46% of the purchases using the sample of transactions where the price is available. More precisely, we use the characteristics of the purchase (type of property, location, date of the purchase, etc) to infer the price paid, estimating an OLS model with 5-fold cross-validation.⁶

Descriptive statistics. Our final sample consists of 103,848 transactions conducted over the period from 2000 to 2020. Table I displays the characteristics of the properties bought by foreign companies. Overall, the average value of UK properties owned through foreign companies is high. The average price in our sample is approximately £1.1 million, and reaches £1.7 million when restricting to prices directly observed in the OCOD and the PPD.⁷ In contrast, the average price of residential properties in the UK between 2000 and 2020 was £230,000. Appendix figure A.1 plots the price distribution for all UK residential properties and those owned exclusively through foreign companies. The offshore real estate price distribution exhibits a notably thicker right tail, indicating that highly expensive properties are much more common in this segment than in the overall residential property market. Taken together, these results suggest that properties bought through foreign companies are likely to be located at the high end of the UK real estate market.

Our analysis sample of OCOD properties is composed of about 34,000 commercial properties (23%) and 80,000 residential properties (77%). More than 48% of these properties are located in the Greater London area. Figure I displays more precisely the concentration of these transactions, as a share of all residential transactions over the 2000-2019 period.⁸ Purchases through foreign companies are concentrated in Greater London, in the Liverpool-Manchester area, and the Birmingham area. Foreign companies also buy a relatively high share of properties near the border with Scotland. Zooming in, Figure I shows the concentration of offshore purchases across wards. Foreign companies buy relatively more in the Centre and the North-West of the Greater London area, which corresponds to the most expensive parts of the city.

Direct ownership of property. To provide a comprehensive picture of the offshore UK real estate market, we also use a dataset on direct foreign property ownership. This dataset was obtained by the Centre for Public Data (CFPD) through a Freedom of Information request and was recently published.⁹ It provides information on the number of property titles held in England and Wales by individuals with an overseas correspondence address. The data consists of

⁶We detail our matching algorithm between the OCOD and the PPD and our inference method in Appendix section B.1.

⁷This is partly due to our conservative price prediction method which is likely to provide a lower bound for the true price (the transaction prices we use for the prediction are winsorized at the 97% level). In the main analysis, we show that our results are robust to the price inference, as they are qualitatively similar when restricting the estimation sample to transactions with non-missing price information.

⁸We compute this figure excluding transactions from 2020 because some transactions that will appear in the OCOD are registered with a long delay. Note that the OCOD includes both commercial and residential transactions, while we show the percentage of transactions of the OCOD over the number of residential transactions only. This is because the Price Paid Dataset provides extensive information on residential transactions, and there is no equivalent for commercial transactions to our knowledge.

⁹The data is accessible at <https://www.centreforpublicdata.org/property-data-overseas-individuals>.

snapshots of the number of foreign directly owned properties by local geographic areas. The dataset is available every two years between January 2010 and January 2020, with a final snapshot in August 2021.

The last snapshot reveals that more than 180,000 properties were registered to individuals with an overseas correspondence address, which represents 0.7% of all registered titles in England and Wales.¹⁰ This figure has more than doubled since 2010 and now surpasses the number of properties owned by foreign companies. It thus constitutes a very important element of the UK offshore real estate market.

However, there is no information on the value of the properties held. To estimate the price of each property, we use the average price of residential properties in the district at the time of the snapshot, using data from the Office for National Statistics. Given that foreign owners are likely to buy more expensive properties than the average, even after controlling for location, our estimates of the value of direct ownership of properties from abroad are likely to be conservative.

2.2 Beneficial ownership data

Foreign ownership datasets. To get more information on the foreign owners of UK real estate, we use several files leaked from offshore service providers between 2013 and 2021. These files include the Offshore Leaks, the Bahamas Leaks, the Paradise Papers, the Panama Papers, and the Pandora Papers. The documents, which originate from law firms and offshore financial institutions, provide data on the beneficial owners of thousands of shell companies that they created or managed for their clients. The International Consortium of Investigative Journalists analyzed the files and published the names, addresses, and countries of residence of the entities' owners. We also make use of data that was leaked in 2019 from the Cayman National Bank and Trust in the Isle of Man (CNBIOM), a bank in the Isle of Man that hosted the accounts of about 1,400 companies.¹¹ Finally, we exploit the recently released OpenLux database, which is the result of an investigation led by the French newspaper *Le Monde*. OpenLux offers a snapshot of the Luxembourg company registry as of December 2019 and provides information on 260,000 Luxembourg companies and over 70,000 company owners. Throughout the paper, we refer to these leaked and Openlux data as the beneficial ownership data, or as the Leaks data.

Table A.1 provides a summary of the characteristics of the beneficial ownership data sources. The table includes information on the time coverage of the data, and the number of companies and unique beneficial owners observed. In total, we have an insight into the structure of more than one million companies and the holdings of more than 500,000 identified beneficial owners.¹² The three most important data sources are the Panama Papers, the Paradise Papers and

¹⁰“New data on property in England and Wales owned by overseas individuals”, Centre for Public Data analysis, November 2021, last consulted on July 15, 2023.

¹¹The complete CNBIOM data have been thoroughly analyzed in Collin (2021a), to offer a comprehensive picture of offshore users in the Isle of Man. Bomare and Collin (2023) use this dataset to analyze the effects of the CRS on offshore clients.

¹²Note that we do not have information about the beneficial owners of all the companies listed in the leaks and the OpenLux data; sometimes we only have access to the identity of the administrators, the directors, or no information at all about its ownership/management.

OpenLux.

The foreign ownership datasets display patterns in the use of tax havens across regions. Figure II shows where individuals from different countries incorporate companies.¹³ We display the proportion of ultimate beneficial owners (UBO) in the leaks from each world’s region on top of the figures. The individuals creating the most shell companies are from the Asian-Pacific region (28%), Western Europe (23%), North America and the English-speaking world (10%). The United Kingdom represents 8% of the owners alone. The graph highlights the heterogeneity of tax haven use, by world region. While owners from the Arabian Peninsula, the Asia-Pacific area, South Asia and South and Central America incorporate their shell companies mostly in the British Virgin Islands, European nationals and United Kingdom residents seem to favor Luxembourg¹⁴ and Malta, which are European havens. North America has a slightly more diversified distribution of haven use, incorporating a roughly similar number of companies in the British Virgin Islands, Luxembourg, Bermuda, Malta and other havens. Geographical proximity and historical and cultural links thus seem to play an important role in explaining these heterogeneous tax haven preferences.

In addition to providing an insight into the heterogeneity of tax haven use across the globe, we use the leaks to identify the real owners of a sub-sample of companies buying properties in the UK.

Register of Overseas Entities. To identify the owners of companies buying properties in the UK, we also exploit the Register of Overseas Entities (ROE). The ROE is the result of a UK law passed in the Spring of 2022 requiring all foreign entities holding UK properties to report the identity of their beneficial owners before February 2023. The data is held and published by Companies House, which is also responsible for maintaining the UK companies register. Registered companies report information on their beneficial owners. When the beneficial owners are individuals, it includes their name, date of birth, nationality and correspondence address. When beneficial owners are companies, the name, address and country of incorporation of the company are reported.

As of August 2023, almost 29,000 companies have registered some information with Companies House, covering more than 140,000 properties held at that date. Despite the fact that the ROE still leaves major gaps open to keep the identity of the real property owners hidden (Advani et al., 2023), this data source allows us to identify the ultimate owner of a significant share of UK properties in the Land Registry dataset.¹⁵

¹³We group countries by region of the world. We draw on the groups defined in Badarınza and Ramadorai (2018), who study foreign real estate investments in London, and we create finer sub-groups to reflect the importance of the countries we identify as buying UK properties in section 3.3.2. We therefore create 11 groups: the United Kingdom alone, the Arabian Peninsula, the rest of the Middle East, North America and the English world (including South Africa), Africa (excluding South Africa), Asia-Pacific, Western Europe (excluding the UK), South Asia, Southern Europe, Eastern Europe and South and Central America.

¹⁴As we have access to administrative data on companies incorporated in Luxembourg, but only to a sub-sample of companies incorporated in other havens with the leaks, it is likely that the share of Luxembourg for each region of the world is overestimated. However, the differential use of havens by individuals from different regions of the world is not affected by this bias.

¹⁵Some of the most important gaps of the ROE include the 25% threshold to qualify as a beneficial owner, which makes it easy to divide ownership between e.g. several members of a family to not be reported; and the fact that trust structures information is not being made available to the public. Moreover, a substantial portion of foreign companies did not comply with their reporting obligations: about 10% of properties held in the UK in August 2023 are owned by companies that failed to register their beneficial owners. Advani et al. (2023) estimate that because of both design flaws and non-compliance, essential beneficial ownership information for more than 71% of properties owned by foreign companies are still missing or publicly unavailable. In addition, the ROE does not provide a random picture of the offshore world. It only includes companies owning properties

3 The UK offshore real estate market

This section proposes a descriptive analysis of the UK offshore real estate market. We first compare the offshore property market to the overall UK real estate market and discuss its magnitude. Next, we compare our estimates of the UK offshore real estate market size to similar estimates for other countries to give a sense of its global relevance. Then, we provide more information on the identity of those owning offshore real estate in the UK, directly and indirectly. We also identify the country of residence of a sub-sample of individuals channeling their investments through foreign companies and distinguish between owners from CRS-adopting countries and other countries.

3.1 Size of the offshore real estate market in the UK

We study the stock of foreign-owned properties as of January 2018 to provide a comprehensive picture of the UK offshore real estate market. We provide estimates of the stock of real estate in dollars, for comparability with other studies.

We combine the stock of properties held by foreign companies with the stock of properties held by individuals with a foreign correspondence address. Foreign companies held 93,332 properties in the UK in 2018. The characteristics of these properties are comparable to that of the average property purchased over the period 2000-2020 (see Appendix table A.2 for a comparison). Individuals with a foreign address held more than 150,000 titles in the UK in January 2018.

We supplement these two datasets with UK national balance sheet estimates to provide descriptive statistics on the UK's onshore and offshore real estate markets. We construct estimates of total real estate wealth owned by corporations and households in the UK based on UK national balance sheet estimates published by the Office for National Statistics (ONS). These national accounts statistics provide information on the net worth of assets owned in the UK by type of asset and category of owner. We construct estimates of total residential and commercial real estate owned both directly and through corporations in England and Wales in 2018.¹⁶ We describe our methodology in more detail in Appendix section C.

Figure III shows the size of the real estate market in England and Wales by type of ownership (corporate, direct and total) and provides estimates of how much is owned from abroad. In total, we estimate that around \$6,000 billion of real estate wealth is owned in 2018 in England and Wales. 75% of this wealth is held directly —i.e. by households— while the rest is held by corporations. We estimate that almost \$240 billion worth of real estate in England and Wales is owned from abroad in January 2018, around 4% of the stock of properties at this time. About 60% of this real estate wealth is held through foreign corporations, while 40% is held directly by individuals with a foreign correspondence

and the UK, and it is not a random event; registration of foreign entities has been discussed in the UK for several years, and individuals who wished to avoid being reported on could have taken steps to restructure their offshore affairs. As such, we do not use the ROE to gather information on the heterogeneous use of tax havens by country. We only use it to identify a sub-sample of companies appearing in the OCOD dataset.

¹⁶This means that we only focus on real estate assets owned by households and private corporations (both financial and non-financial).

address. The share of foreign ownership is also significantly higher for corporate ownership than for properties held by households (9% vs. 2%).

Appendix figure A.2 breaks down real estate wealth in England and Wales by type of property. We estimate that the total residential real estate market is approximately worth \$5,270 billion in 2018, with 15% of the properties being held by foreign and domestic corporations. The commercial real estate market is much smaller, totaling \$730 billion, with 96% of it owned by corporations.

To the best of our knowledge, our study is the first to propose an estimation of the overall size of the offshore real estate market in the UK, taking into account both residential and commercial real estate, and both direct and corporate ownership. Two studies provide estimates of the value of residential properties (excluding commercial properties) held by foreign corporation in England and Wales (Johannesen et al. (2022) for December 2019) and in London only (Bourne et al. (2022) for January 2022). In Appendix figure A.3, we compare their estimates with our own, focusing on the same segment of the real estate market.¹⁷ Overall, the figures are of the same order of magnitude across all three studies. On the one hand, our estimates for England and Wales are higher than those of Johannesen et al. (2022) (\$90 billion vs. \$70 billion). The difference is mostly due to methodological choices when classifying properties as residential, as Johannesen et al. (2022) use a more conservative classification rule.¹⁸ On the other hand, when focusing on housing wealth held in London, we get lower estimates than Bourne et al. (2022). The two studies thus provide bounds on our estimates of offshore real estate and confirm that the order of magnitude we provide is likely to be correct.

3.2 Comparison with other offshore real estate markets

Figure IV compares our estimates of the size of the offshore real estate market in England and Wales to other studies on foreign-owned real estate in other locations. In the global cross-border real estate market, the United Kingdom, as proxied by England and Wales, stands out due to its substantial size. Morel and Uri (2021) focus on France and find that \$140 billion of French domestic real estate was owned from abroad at the end of 2019, directly and through corporations. This represents less than 60% of our estimates for England and Wales in January 2018. Note however that our figures include both residential and commercial real estate while estimates for France only capture residential real estate.¹⁹ Alstadsæter and Økland (2022) combine for the first time residential and commercial real estate and find that at the end of 2017, about \$20 billion worth of Norwegian real estate was owned by non-residents. Exploiting data on 800,000 properties in Dubai, Alstadsæter et al. (2022) show that around \$150 billion in wealth is invested in the

¹⁷We express all estimates in January 2018 dollars.

¹⁸We describe in Appendix section B the method used to classify the properties by type. Johannesen et al. (2022) consider as residential the properties from the OCOD register that match a transaction in the Price Paid Data (on address and timing of the transaction), which is more restrictive than our method.

¹⁹Data on direct ownership of UK properties do not provide information on the type of property, so we don't have estimates of how much residential real estate wealth is owned from abroad in aggregate. However, when focusing on corporate ownership, about 62% of total foreign-owned UK real estate is residential.

Dubai property market by non-nationals, which is comparable to our estimates for offshore real estate held in London.

3.3 Who owns offshore real estate in the UK?

3.3.1 Immediate ownership

We first provide evidence of total ownership of UK real estate by country of *immediate ownership* by adding total direct ownership by residence country of the buyer and total indirect ownership by country of incorporation of the corporate vehicle. Figure V shows estimates for the top 20 buyers, providing insights into the structure of the offshore real estate market. First, ownership is very concentrated. The top 5 countries own about 50% of total foreign-owned real estate wealth, the top 10 countries own 65% and the top 20 own more than 80%. Second, the seven most important owners are tax haven countries.²⁰ In total, more than 74% of foreign investment in UK real estate goes through tax havens when including both properties purchased by individuals with an address in a tax haven, or by corporations incorporated in a tax haven.

Appendix figure A.4 breaks down these estimates. Panel A provides estimates of the value of properties held by individuals with a foreign correspondence address. Two-thirds of the foreign directly owned real estate wealth is linked to a dozen countries, and more than 80% is owned by the top 20 countries. The main countries of residence are located in South-East Asia, including two tax havens, Hong Kong and Singapore. Other important countries are the United States, countries from the Arabian Peninsula (United Arab Emirates, Kuwait, Saudi Arabia), some European countries (Ireland, France, Italy) and Australia.

Panel B of Appendix figure A.4 provides estimates of ownership by country for corporate owners only (transactions in the OCOD). Indirect ownership is even more concentrated than direct ownership. Indeed, the top 5 owners hold almost 80% of indirectly owned UK real estate in 2018. Ownership is also more concentrated among tax havens, as more than 94% of the properties in the register are owned through companies incorporated in a tax haven. Except for Luxembourg, the five most important buyers also have particularly strong links with the UK: the Channel Islands (Jersey and Guernsey), the British Virgin Islands and the Isle of Man. Companies incorporated in Jersey are the most popular vehicle of investment, with more than 25% of the value of all foreign corporately owned properties are held via Jersey. Ownership from non-havens is mainly coming from the Netherlands, the United States, and Germany.²¹

3.3.2 Identifying the ultimate beneficial owners of UK properties

Immediate vs. ultimate ownership. If the majority of foreign-owned UK real estate is held from tax havens, it is unlikely that the ultimate beneficial owners of these properties are effectively residents of these countries. Indeed,

²⁰There is no consensus on which countries should be considered as tax havens. We use the list of tax havens of Menkhoff and Miethe (2019), which is obtained by combining the lists of Gravelle (2009) and Johannesen and Zucman (2014). They classify 58 countries as tax havens, which are listed in Appendix table A.5.

²¹Note that in the list of tax havens we follow (Menkhoff and Miethe, 2019), the Netherlands is not considered as a tax haven.

the population of places such as Jersey, the British Virgin Islands or Guernsey barely exceeds 100,000 inhabitants.²² Therefore, these territories are probably used as conduits for offshore real estate investments, with the real ultimate owners residing in another country.

This additional layer between an investor and their ultimate assets makes it harder to determine who owns real estate in the UK. To formalize the problem, consider first that the total value of UK real estate wealth ultimately owned from country c through foreign companies is equal to I^c . Define Ω^O as the set of firms appearing in the OCOD and Ω_h^O a subset of Ω^O composed of all firms incorporated in jurisdiction h . Then, let us define as P_i the total value of UK properties owned by company i and as ϕ_i^c the share of owners of company i who are residents from country c . I^c is equal to $\sum_{i \in \Omega^O} \phi_i^c \cdot P_i$. If the jurisdiction of incorporation of company i and the country of residence of its owner are the same, namely if $c = h$, ϕ_i^c is equal to 1 for $c = h$ and 0 otherwise and I^c is simply equal to $\sum_{i \in \Omega_c^O} P_i$ which is directly observable in the OCOD. However, c and h do not coincide in most cases, in particular when h is a tax haven. Thus, we lack information on ϕ_i^c . Indeed, the OCOD provides information on the companies buying UK real estate but not on the owners of the companies.

Method. To learn more about the beneficial owners of companies purchasing UK properties, we complement our data on corporate real estate ownership with the Leaks data and the Luxembourg companies register, which provides information on more than 1 million of shell companies and their owners, and with the Register of Overseas Entities, giving information on overseas companies owning real estate in the UK. This allows us to observe ϕ_i^c for a subset of companies and thus help us estimate the geographical distribution of UK real estate ultimate ownership. We match the OCOD with the beneficial ownership data based on the company name and its country of incorporation. Our matching algorithm is described in more detail in Appendix section C.2.

Table II displays the outcome of the matching by data source, while Appendix table C.1 presents the source of the matches. In total, we identify roughly 60,000 owners involved in 49% of the property transactions in our sample. However, a lot of matched companies report beneficial owners who are companies themselves rather than individuals. After discarding them, the share of properties for which we identify the ultimate owners is reduced to 20%, totaling 21% of the properties value. The share of matched properties for London only is comparable, with the individual ultimate owners identified for 22% of the properties (Appendix table C.2).

Appendix figure A.5 presents the results of our matching process, for the main countries of ultimate owners buying in the UK. Panel A shows the geographic distribution of all identified owners. A significant share of properties are matched to owners linked to a tax haven, with Hong Kong, Switzerland, the Isle of Man, Jersey and Cyprus among the top ten countries of origin of identified buyers. Two elements can explain this result. First, in the Register of Overseas Entities, individuals have to provide a correspondence address and not a residence address. Some individuals

²²The number of inhabitants in Jersey in 2018 was less than 110,000, the British Virgin Islands and Guernsey have even fewer inhabitants.

likely delegate the management of their company to corporate service providers, some of them are based in tax havens. Second, in the Offshore Leaks data, individuals are linked to countries based on all information available, including business addresses and addresses of intermediaries, likely resulting in individuals being linked to tax havens even though they live elsewhere.

To ensure that we identify the correct country of residence of owners in our matched sample, we allocate all owners linked to a tax haven to their most probable country of residence. We do so by exploiting information on individuals' nationality (Register of Overseas Entities) or by predicting individuals' residence country through a name analysis (Leaks data).²³ We provide details of this reassignment exercise in Appendix section C.3. Panel B of Appendix figure A.5 shows the results of this allocation. A large proportion of individuals linked to tax havens are reallocated to non-haven countries. The share of properties attributed to individuals from the United Kingdom and China increases significantly, likely reflecting the fact that they are among the main users of the British tax havens (for the UK) and Hong Kong (for China).

Results. The matching process reveals that residents of the United Kingdom constitute by far the main group of buyers through offshore companies. This is suggestive of a “home bias” in the individuals' investment decisions (Coeurdacier and Rey, 2013). This is also coherent with the fact that tax planning considerations could motivate UK buyers to own their property through a shell company instead of in their own name. Non-domiciled UK taxpayers, in particular, could benefit from tax rebates, as they do not have to pay taxes on their foreign assets as long as they do not repatriate them to the UK. Before 2017, this rule allowed them for example not to pay inheritance tax on UK properties held through foreign companies. Other tax rules favor properties held through companies rather than directly. For example, buying the shares of a company owning a property rather than the property directly allows one to avoid the stamp duty tax, and therefore to save a lot in tax payments, particularly for expensive properties. Holding UK properties through foreign companies is much less profitable now than it used to be, but the benefits can remain non-negligible in some cases.²⁴ Investors from the Middle East, and in particular the United Arab Emirates, account for another important share of the identified ultimate investors. As residents from these countries face very low, and sometimes even zero, effective tax rates on their income and wealth, this suggests that these investors are using a shell company to channel their purchases for secrecy motives rather than to lower their tax liabilities²⁵. Other main countries of origin of

²³We reallocate tax haven owners to their most likely country of residence using two different methods. For owners identified through the Register of Overseas Entities, we use the country of citizenship, as information on individuals' nationality is available. However, this information is not available in the Offshore Leaks data. To circumvent this issue, we conduct a name analysis of individuals identified through the Leaks data using a software able to predict the most probable country of residence given the name of a person. Appendix Figure A.6 presents the results of this allocation separately for the Register of Overseas Entities and for the Offshore leaks matches.

²⁴From 2013, an annual tax –ATED– was introduced on the ownership of such “enveloped” properties, but this affects only residential properties and not commercial ones. However, the rate is not high enough to offset the advantage of avoiding the stamp duty for many property owners, especially for very expensive properties (Dan Neidle, “ATED – the obscure tax that nobody was supposed to pay” (2022)). See Appendix section D for more details on the legislation.

²⁵Indeed, for high net-worth individuals living in a politically unstable country, avoidance of political reprisals is sometimes pointed as a major consideration in the use of tax havens (Harrington, 2016)

real estate investments include the United States and South Africa, which maintain strong relationships with the United Kingdom, and China.

We do not identify many properties ultimately owned by Russians, even though London has sometimes been nicknamed “Londongrad” and the implementation of the Register of Overseas Entities (ROE) was ultimately triggered by the need to identify Russian assets in the broader context of sanctions against oligarchs in 2022 (Collin et al., 2023). However, the true proportion of real estate assets held by Russian individuals is likely to be higher than what our matching results indicate. Indeed, most of our matches come from the ROE, and as some Russian individuals already had their assets frozen or were under the threat of sanctions from the UK at this point, it is likely that they either did not comply with their reporting requirements or structured their assets in a way their name and nationality would not appear in the public register. Moreover, we identify a lot of properties as held by Cypriot owners (Appendix Figure A.5). A share of these properties is likely to be ultimately held by Russian individuals, as Cyprus has been shown to be a place of choice for Russian people to park their money in (Chang et al., 2023).²⁶

3.3.3 Geographical distribution of the offshore real estate market in the UK

The goal of this subsection is to show the geographical distribution of total foreign ownership of UK real estate, including both direct and corporate ownership. To do so, we first allocate the unmatched properties in the OCOD to their most probable owners’ residence country based on what we observe in the sample of matched properties from section 3.3.2. Combining the matched and unmatched properties, we obtain an estimate of the geographical distribution of cross-border corporate ownership of properties in the UK. Second, we correct the ownership country of UK properties held by individuals when the correspondence address is in a tax haven by predicting the most probable true residence country of the owner. We describe our methodology in detail in appendix section C.4.

Combining indirect and direct ownership, we present in figure VI our estimates of the geographic distribution of UK real estate ownership, by ultimate owners’ country of residence. Appendix figure A.7 shows these estimates separately for indirectly and directly held properties. The geographic distribution of ownership is very similar to what is presented in Appendix figure A.5. The United Kingdom remains the largest group of buyers through foreign vehicles, followed by countries from the Arabic Peninsula. South East Asia countries also make up for a large share of real estate owners, which is driven by their importance in directly held properties. Appendix Figure A.8 shows how our estimates vary depending on how we treat owners from tax havens, corporate owners or how we compute the residence country. Appendix Figure A.9 proposes an alternative way to allocate the unidentified sample of properties, using the leaks data.²⁸ All specifications produce similar results, with a comparable amount of wealth allocated to the main countries

²⁶Cypriot finance and banking professionals have also been found to help Russian clients avoid sanctions, sometimes by finding Cypriot individuals to formally hold Russian assets instead of their true beneficial owners.²⁷ The recent “Cyprus Confidential” leaks confirm the central role of Cyprus for the Russian offshore industry.

²⁸This alternative allocation uses the full information available in the leaks data, including both the country of incorporation of the company and the country of residence of the owner. We provide more details in Appendix section C.4.

of residence. In particular, the United Kingdom is always the most important investor.

Overall, this section has shown that the UK offshore property market i) is large, ii) attracts a large number of buyers from the UK, the Arabian Peninsula and South-East Asian, and iii) usually involves intermediaries located in tax havens. The nature of this market makes it a destination likely to attract investors willing to keep their offshore wealth hidden. In the next section, we study the extent to which taxpayers responded to the Common Reporting Standard by reallocating their offshore wealth towards UK properties.

4 The effect of the CRS on the demand for offshore real estate

The main goal of this section is to test the assumption that taxpayers responded to the Common Reporting Standard by rebalancing their offshore portfolio towards real estate assets. To do so, we build a measure of exposure to the CRS at the jurisdiction level and show that companies incorporated in the most exposed jurisdictions increased their investments in UK properties significantly more than their less exposed counterparts.

4.1 The Common Reporting Standard

In September 2013, G20 leaders endorsed automatic exchange of information as the new global standard of cross-border information exchange. This standard requires financial institutions of participating countries to collect information on the identity of their clients. Then, when the ultimate account holder is a taxpayer from a country that adopted the standard, information about their identity and their financial holdings is shared with the account holder's tax authorities.²⁹ A few months later on March 19, 2014, forty-four jurisdictions signed a joint statement announcing they would implement the OECD Common Reporting Standard by the end of 2015. On May 6, 2014, the OECD adopted its Declaration on Automatic Exchange of Information in Tax Matters, in which forty-seven countries also committed to implement a single global standard of Automatic Exchange of Information. In our analysis, we pool the March and May 2014 event together as they are published almost at the same time and are signed by a very similar set of countries, and refer to them as the 'Joint Announcement'.

In total, sixty-seven countries committed to the CRS either in 2013 or 2014. We call these countries the "early adopters", as opposed to countries that will adopt the CRS later, or never. The lists of participating countries for each of these events are presented in table A.3. Main early-adopters include European Union countries, Canada, and China while non-adopters include the United States (which implements its own unilateral exchange of information policy), the United Arab Emirates and Qatar.³⁰

²⁹That is, if a taxpayer from country A owns an account in country B, the tax administration of country B will automatically and annually report to the tax authorities of country A the information linked to this account and its owner.

³⁰More precisely, the relevant timing of treatment is the date at which both the residence country of the ultimate owner and the country where offshore assets are held enter the CRS. This is because participating countries will start to exchange information with each other when they both enter the agreement. As some of the most important offshore financial centers are among the early-adopters (e.g. Switzerland), the date at which residence

These major steps towards global tax transparency change the trade-off faced by non-compliant taxpayers. It leads to an important increase in the expected cost of *financial evasion*, defined as owning undeclared financial assets and receiving unreported income, by substantially increasing the probability of detection. Available evidence suggests that owners of offshore accounts responded to this transparency shock by removing financial assets from tax havens. Indeed, the event of September 2013, together with the announcement of the US-Swiss Bank Program (August 29, 2013) has been shown to be concomitant to a sudden decrease in foreign-owned deposits in Switzerland, while the “Joint Announcement” led to a 11% reduction in offshore deposits owned by residents from all committing countries (O’Reilly et al., 2021). However, there is uncertainty on whether this reduction in offshore deposits is due to enhanced tax compliance globally or the development of alternative evasion strategies. One major loophole of the CRS is that it only covers financial assets, leaving room for evaders to keep their wealth hidden by restructuring their offshore portfolios away from financial assets and towards real assets.

In the next section, we present our method to study this asset rebalancing response to the CRS, by focusing on investments in the UK offshore real estate market.

4.2 Empirical strategy

Data and sample restrictions. In this section, we exploit the OCOD (presented in section 2.1), which gathers comprehensive information on properties in the UK acquired through foreign companies. Given the prevalence of intermediaries situated in tax havens, this dataset is especially relevant for examining the tendencies of taxpayers seeking to evade compliance with the CRS. We exclude from the dataset the small set of properties acquired by companies incorporated in non-haven countries, and transactions made after 2017 as new jurisdictions commit to the CRS over time. We pool all purchases at the jurisdiction - quarter level. Our sample is a balanced panel of 50 tax havens over the period 2000-2017.

Constructing a measure of exposure to the CRS. One caveat of the OCOD is that it provides information on the country of incorporation of the buying companies but not on the country of residence of their owners. This limitation prevents us from estimating the response to the CRS by comparing directly the evolution of real estate investments from residents of countries adopting the CRS to purchases from residents of non-adopting countries. To circumvent this issue, we leverage the country-specific tax haven preferences documented in section 2.2. We compute estimates of the share of all company owners in jurisdiction h that are residents from country c , observed using the Leaks data. The estimates show the extent to which each tax haven is used by residents from different countries.

countries commit to the CRS provide a good approximation of the timing of treatment for individuals. Note that individuals from early-adopting countries will be affected by the CRS even if the offshore assets are held through a shell company incorporated in a non-participating country. In this case, the bank in which the assets are held will have to gather and exchange information on the ultimate owner of the assets, whether or not the shell company is incorporated in a country participating to the CRS.

Using these figures, we construct a measure of “CRS exposure”, which is equal to the proportion of company ultimate owners who are resident from early-adopting countries. This measure of CRS exposure is based on companies incorporated before or during the second quarter of 2013, so before the transparency shocks that we exploit.³¹ If a jurisdiction does not appear in the leaks or the OpenLux data as an incorporation country (or if less than 10 corporations are incorporated), we consider that exposure to the CRS is null.³² This continuous measure of exposure to the CRS goes from 0 (0% of company owners in that jurisdiction come from early adopter countries or that jurisdiction hosts less than 10 shell companies in the beneficial ownership data) to 1 (100% of company owners are residents from early-adopting countries).

Figure A.10 displays the distribution of CRS exposure among tax havens in our sample, focusing on the 30 most important buyers.³³ If investors willing to circumvent the CRS invest part of their offshore wealth in real estate by using shell companies incorporated in their preferred tax havens, we should see an increase in purchases channeled through the most exposed jurisdictions. On the other hand, we expect investments from less exposed tax havens to react less to the CRS as they are also used by a lot of individuals not affected by the policy. Hence, real estate investments channeled through these countries should be less affected by the CRS.

Difference-in-differences specification. We analyse the response to the CRS in a dynamic difference-in-differences setting using CRS exposure as the treatment. The treatment is thus defined at the jurisdiction level, and is continuous. Time trends stemming from the evolution of the dynamics of the UK real estate market as well as from tax changes are identified from tax havens with a CRS exposure of zero. We estimate the following equation:

$$Y_{hq} = \sum_{j \neq 2013q2} \beta^q \cdot Q_{j=q} \cdot Exposure_h + \gamma_h + \eta_q + v_{hq} \quad (1)$$

where Y_{hq} denotes the real estate investment outcome for tax haven h in quarter q (in 3-quarters moving average). We consider several outcome variables: investment in value (million Pounds), investment in value scaled by pre-CRS investment (2012q3-2013q2) and number of transactions. $Q_{j=q}$ is a quarter dummy equal to one when the quarterly date is equal to q . $Exposure_h$ is the continuous measure of exposure to the CRS, going from 0 (0% of company owners in that jurisdiction come from early adopter countries) to 1 (100% of company owners are residents from early adopter countries). We control for country-specific links to the UK real estate market by including country fixed effects γ_h , and for common changes in the UK real estate market by including quarter fixed effects η_q . v_{hq} is the error term. The treatment variable being continuous, a coefficient $\hat{\beta}^q$ of 100 in quarter q means that the average difference in real estate

³¹We show in section 5 that our results are robust to using CRS exposure measures computed using companies incorporated after 2013q2.

³²In total twenty-six havens have a CRS exposure of zero. We show later on that our results are robust to a sample restriction where we only keep tax havens for which we observe at least 10 corporations in the leaks or OpenLux.

³³Given how skewed the distribution of offshore real estate investment is, these 30 jurisdictions represent 99.6% of all investment from tax havens in our data. Some of the most exposed countries are tax havens investing a lot in the offshore real estate market, including Jersey, Guernsey, the Isle of Man and Luxembourg. The British Virgin Islands, however, the second investor overall both in value and volume of transactions, counts only 54% of owners residing in early-adopting countries and is therefore less exposed than the above-mentioned jurisdictions.

investment between fully exposed tax havens (exposure of one, meaning that the jurisdiction is only used by residents from early adopter countries) and non-exposed tax havens (exposure of zero) is higher by £100 million in quarter q compared to the reference period (second quarter of 2013).

4.3 Results

Highly exposed vs. the rest. As first descriptive evidence, Figure VII shows total real estate investments from firms incorporated in “highly-exposed” tax havens, defined as jurisdictions with a CRS exposure higher than 75%, and from firms incorporated in all other tax havens. The flows of investments follow each other closely during a very long period spanning from 2000 to mid-2013 before diverging sharply in the third quarter of 2013, right when the G20 countries endorse automatic exchange of information. From this period, we observe a large jump in real estate investments from highly-exposed jurisdictions, that is not matched by investments from the other tax havens.³⁴

Investment in value. Panel A of figure VIII displays the coefficients $\hat{\beta}_q$ estimated from equation 1. The figure confirms that there is no significant pre-trend between 2000 and the second quarter of 2013. Immediately after the G20 endorses automatic exchange of information, there is a sharp increase in the level of investments coming from tax havens with a higher exposure to the CRS. After September 2013, the difference in the value of purchases made by fully exposed havens and the havens with exposure of zero surges and reaches an average of about £150 million until the end of 2017.

Investment relative to pre-reform levels. A potential concern raised by outcome variables in level is that large variations from a small subset of countries can significantly affect the results. The fact that the outcome variable is often equal to zero in our estimation sample prevents us from dealing with this issue by using a log transformation. Instead, we scale the outcome variable for each jurisdiction by its value before the announcement of CRS. More specifically, we scale quarterly investment of country h by the average quarterly investment from country h between the third quarter of 2012 and the second quarter of 2013.³⁵ Results are shown in figure VIII, Panel B. As for the specification in levels, there is no significant difference in real estate investments before the CRS, but a sharp divergence in investments from September 2013 onward. A coefficient of 1 in quarter q means that the difference in investment between the fully exposed and 0% exposed group in quarter q exceeds the difference in investment in the reference period by 1 time the average pre-CRS investment in the exposed group. The average of the post-CRS coefficients is 1.1. This means that the country’s average increase in real estate investments due to the transparency policy is equivalent to more than the average amount invested one year before the policy change.

³⁴Investments from the other tax havens still increase more than in other quarters, which could reflect the fact that some individuals from early-adopting countries also use these havens to channel their investments to the UK.

³⁵The scaled values are winsorized at the top at the 95% level. The results are robust to other pre-CRS periods.

Number of transactions Next, we study how the total number of transactions evolves around the endorsement and commitment to the CRS. Appendix Figure A.11 shows the results of the estimation. The number of properties purchased from differently exposed tax havens was not following similar trends before the CRS. To investigate the channels behind the surge in investment, we study separately transactions for which the price is respectively below and above £1 million. Figure IX displays the estimates obtained from equation 1. Panel A shows the results for property transactions below £1 million. The number of less expensive purchases channeled through differently exposed havens is following the same pattern than all transactions, with a significantly increasing trend before 2013. Panel B shows the same specification for transactions above £1 million. In this case, the pattern in real estate investments is very close to what we observe when considering the value of transactions, with parallel trends before the CRS, diverging sharply after the G20 commitment to the policy. Taken together, these findings suggest that the increase in real estate investments channeled through tax havens that are mostly used by CRS-adopting countries is driven by the purchase of relatively expensive properties. This is what we would expect if the CRS was pushing offshore users to invest in the real estate market, as they are more likely to come from the top of the wealth and income distribution (Alstadsæter et al., 2019; Guyton et al., 2021; Londoño Vélez and Ávila-Mahecha, 2021).

5 Robustness checks

5.1 Tax reforms in the UK

We observe an increase in offshore investments in UK properties from the second quarter of 2013. However, interpreting this surge in investments as a rebalancing response to the Common Reporting Standard requires to ensure that it is not driven by factors unrelated to the CRS. Our difference-in-differences methodology, leveraging the different level of exposure to the CRS across jurisdictions, aims to address this concern. Moreover, the fact that our main outcome variables evolve similarly before the announcement of the policy regardless of the level of exposure of the tax haven indicates that the CRS caused the sharpened divergence observed in 2013. We further validate our interpretation by establishing that the tax reforms implemented in the UK during the period we consider do not confound our results.

Since 2012, several important tax reforms affecting the purchase and ownership of properties by companies have been introduced in the UK. These reforms include changes to Stamp Duty and Land Tax (notably in 2012 and 2014), the implementation of the Annual Tax on Enveloped Dwellings (ATED) targeting properties owned through corporate entities (introduced in 2013), an expansion of capital gains tax (implemented in 2015), and a broadening of the inheritance tax base (enacted in 2017). A detailed overview of these reforms can be found in Appendix section D. These tax reforms have altered the tax liabilities associated with offshore real estate investments, consequently affecting the after-tax return of these assets. As a result, the attractiveness of offshore real estate may have been impacted. As these

reforms take place over our period of analysis, they could influence our estimation results.

However, these tax changes primarily impact residential properties while the incentives to acquire or hold commercial properties remain largely unaffected.³⁶ We thus investigate the potential confounding effect of the tax reforms by comparing the purchases of residential properties (significantly affected by tax reforms) to commercial properties (largely unaffected by tax changes). More specifically, we estimate the following equation:

$$Y_{hqt} = \sum_{\substack{j \neq 2013q2, \\ t \in \{r, c\}}} \beta^{qt} \cdot Q_{j=q} \cdot Exposure_h \cdot Type_t + \gamma_h + \eta_q + Exposure_h \cdot Residential_t + Residential_t + v_{hqt} \quad (2)$$

where Y_{hqt} denotes the investment (in million Pounds) in property type t (residential (r) or commercial (c)) for country h in quarter q . $Exposure_h$ is the continuous measure of exposure to the CRS, $Type_t$ is a dummy equal to one when the property type is equal to t , $Residential_t$ is a dummy equal to 1 when the type of the property purchased is residential, $Q_{j=q}$ is a quarter dummy, γ_c is a country fixed effect, η_q a quarter fixed effect and v_{hqt} the error term.

Appendix Figure A.12 presents the results of the estimation of equation 2. First, we observe a similar investment pattern across both property classes, with a substantial surge in investment from tax havens exposed to the automatic exchange of information events after 2013. This result confirms that the tax changes implemented since 2012 are not likely to confound our estimated response to the CRS. Second, although the difference is not statistically significant, the increase in investment appears to be higher for commercial properties than for residential properties. This finding is not surprising, considering the substantial rise in tax liabilities for owners of residential properties. It suggests that the tax changes affecting offshore residential investment may explain partly the importance of foreign-owned commercial real estate in the UK today.³⁷

5.2 Alternative measures of exposure to the CRS.

Persistence of tax haven use by country of ultimate ownership. Our analysis relies on the key assumption that tax haven use by country of ultimate ownership is persistent, and in particular that it does not change as a consequence of the CRS. This assumption is plausible because while the CRS has been shown to have an impact on the location of offshore bank deposits (see Cusi et al. (2020)), it has not altered the incentives associated with the choice of tax haven for incorporating shell companies. To further validate this assumption, we construct an alternative exposure

³⁶Commercial properties are still affected by some of these tax changes. First, an important tax change for commercial properties happened in 2019, with the extension of the Non-Resident Capital Gains Tax to commercial properties. However, we stop our analysis of the responses to the CRS in 2018. Second, the business rates multipliers, which determine the business rates bill that the occupier of a non-domestic property will have to pay annually, are updated every year. The national multipliers go from 45.8p in 2012/13 to 49.7p in 2016/17. Special cases apply to the City of London, but we exclude this area in our sample (see Appendix section B). Third, the schedule of the Stamp Duty tax for non-residential properties changed in March 2016, increasing the tax liabilities for expensive properties (and reducing it for lower-value properties). However, this is a modest change relative to the evolution of the Stamp Duty tax schedule for residential properties.

³⁷In 2020, overseas capital is estimated to represent 51% of all investment in UK commercial real estate.

measure based on shell companies that were incorporated only after the implementation of the CRS. Figure A.13 shows the results of estimating equation 1 using this alternative measure of exposure. The results are qualitatively and quantitatively similar to when we compute tax haven preferences before the commitment to the policy.

Selection bias in the leaks. Our identification strategy also relies on our measure of heterogeneous use of tax havens across countries built from the beneficial ownership data, mostly composed of leaked datasets. The key strength of such leaks is that they can be seen as random events that offshore users could not predict, alleviating the concern of a strong selection bias. This randomness has allowed researchers to exploit leaked data in order to study the distribution of tax evasion [Alstadsæter et al. \(2019\)](#); [Londoño Vélez and Ávila-Mahecha \(2022\)](#); [Collin \(2021b\)](#). Yet, being based on individuals who used specific offshore service providers, the leaked data only provide a partial picture of the offshore world.³⁸

To address this concern, we construct an alternative measure of tax haven preferences based on data from the Bank for International Settlements (BIS). The BIS data offers insights into cross-border bank deposits on a bilateral basis for 48 reporting countries. Leveraging this data, we construct an alternative CRS exposure measure for each tax haven based on the jurisdiction's proportion of deposits held by residents from early-adopting countries. Appendix Figure A.14 presents the results obtained when estimating equation 1 using this alternative exposure measure. The findings closely align with those obtained from our main specification, ruling out the concern that a strong selection in the leaks would bias our measure of CRS exposure.

Another potential issue with the Leaks and OpenLux data is that we only observe companies incorporated in a selected sample of tax havens. When constructing the main measure of exposure to the CRS, we consider that tax havens that host less than 10 companies have an exposure of zero. While this assumption seems reasonable in practice, we could misclassify some tax havens that are actually used a lot by CRS-adopting countries, because of data limitations. We check that this is not driving our results by restricting our estimation sample to tax havens with at least 10 incorporated corporations. This refinement reduces our sample size by half, from 50 countries to 24. Appendix Figure A.15 presents the results obtained when estimating equation 1 using this restricted sample. Although the confidence intervals are wider due to the smaller sample size, the qualitative findings remain largely unchanged.

Register of Overseas Entities weights. Little is known about how the ownership structure of offshore assets is influenced by the type of asset held. In particular, financial offshore assets might be structured in a different way than real estate offshore assets. To address this potential issue, we exploit the Register of Overseas Entities, which provides information on foreign companies holding real estate assets specifically.³⁹ We use the complete Register data to build another exposure measure for each tax haven, based on the proportion of company owners resident in early-adopting

³⁸Note that this is not the case for the Bahamas Leaks, as they are a sample of files taken from the company register of the Bahamas.

³⁹These companies can hold other types of assets as well, but their holding portfolio always includes real estate assets.

countries. Results of the difference-in-differences estimation using these weights are presented in Appendix Figure A.16. Because United Kingdom company owners are probably over-represented in the Register (see section 2.2 and 3.3), there is less variation in treatment intensity across jurisdictions, leading to less precise estimates. However, the results are qualitatively similar to our main specification.

5.3 Identified sample of owners

We can match the OCOD with the Leaks, the OpenLux data and the Register of Overseas Entities over the whole period of analysis. As before, we discard matches where companies declare beneficial owners that are companies themselves. We match about 19% of the transactions over the period or 23% of the values of the properties bought. Appendix Figure A.17 presents the rate of matching over time and Appendix Figure A.18 shows the main country of residence of identified beneficial owners. The geographic distribution of clients is very similar to what we find for properties held in January 2018 only. We can exploit information on the country of residence of the owners to directly check whether the increase in real estate investments from 2013 is due to residents from CRS-adopting countries.

In order to investigate the effect of the CRS, we compare the total amount invested in real estate from 2005 to 2016 by residents of countries committing to the CRS in 2013 or 2014 and by residents of other countries. We stop the analysis in 2016 because the Panama Papers, one of the most important leak of the last decade, were published in April 2016.

While in the main analysis the measure of exposure to the CRS captures the probability that purchases are coming from CRS-adopting countries, in the identified sub-sample we directly know the residence country of the buyer.⁴⁰ We thus run a modification of equation 1 using a binary variable capturing whether a country commits to the CRS in 2013 or 2014 as the treatment variable. Due to a much smaller sample size, results are less precise than when using the full OCOD. We address this issue by using semesters rather than quarters as the unit of observation:

$$Y_{is} = \sum_{j \neq 2013s1} \beta^s \cdot S_{j=s} \cdot Treat_i + \gamma_i + \eta_s + v_{is} \quad (3)$$

where Y_{is} denotes the amount in million Pounds invested in real estate by country i at semester s , $Treat_i^g$ is a dummy equal to 1 when country i commits to the CRS in 2013 or 2014, γ_i is a country fixed effect, η_s a semester fixed effect and v_{is} an error term. The difference-in-differences coefficient β^s captures the effect of the CRS on real estate investments in semester s relative to the pre-reform semester, 2013s1.

Results are presented in Appendix Figure A.19. We confirm that the increase in property purchases documented in the previous section is actually driven by individuals affected by the CRS. Indeed, while there is no statistically significant difference in real estate investments between treated and control groups between 2005 and September 2013,

⁴⁰Provided that the owners of a given company have not changed between the transaction date and the date of the leak/release of the ROE.

there is an increase in real estate investments immediately after the endorsement of the CRS by G20 countries.⁴¹ The difference in investments between control and treatment group increases and becomes significant in the second semester of 2015.

Even if these results are to be interpreted with caution due to the small size of our identified sample, they indicate that individuals react to their country of residence committing to automatically exchange information by increasing their UK real estate investment through shell companies.

5.4 Further robustness checks

46% of the transaction prices in the estimation sample are inferred (see section 2.1). To address concerns about the potential impact of inferring prices on our results, we conduct additional tests presented in Appendix figure A.20. In Panel A, we estimate equation 1 using a restricted estimation sample that includes only transactions where the price is observed. We see that the results are very close to what is observed in figure VIII (Panel A). Then, we investigate in Panel B whether CRS exposure is correlated with the purchase of properties with (non-)missing prices, both before and after the implementation of AEOI. We find no significant correlation between CRS exposure and the likelihood of purchasing properties with missing prices. Taken together, these findings show that the inclusion of inferred transaction prices in the analysis does not substantially impact the robustness and validity of our main results.

Finally, a relatively small number of tax havens make up for the majority of purchases made on the real estate market: more than 60% of the transactions over the period we study are made by companies incorporated in Jersey, the British Virgin Islands and Guernsey. To check that purchases from these countries are not driving our results, we estimate equation 1 excluding these transactions from our sample. Appendix figure A.21 presents the results. When removing these countries from the sample, the estimated response to the CRS is smaller but our results stay qualitatively similar.

The last two sections have shown that the endorsement and commitment to the CRS triggered a significant increase in offshore real estate investments in the UK. Consistent with tax evaders responding to the transparency policy by rebalancing their offshore portfolio from financial to real estate assets, we find that the CRS caused a large increase in real estate purchases by companies based in tax havens that are the most exposed to the transparency policy.

6 Estimating the rebalancing response to the CRS

The aim of this section is to assess the extent of the rebalancing response to the CRS. To do so, we quantify the increase in real estate investment in the UK due to the CRS and compare it to the estimated effect of the policy on offshore financial wealth found in the existing literature. This exercise is a back-of-the-envelope calculation and these

⁴¹For visibility, our analysis period starts in 2005 but starting in 2000 does not change the results at all.

results should be interpreted with caution. However, it is useful to get a sense of the magnitude of the rebalancing response to the CRS.

6.1 Effect of the CRS on the offshore real estate market in the UK

We compute the total effect of the CRS on UK real estate investments by summing the estimated coefficients ($\hat{\beta}_{qt}$) from equation 2 over the post-CRS period (third quarter of 2013 to the fourth quarter of 2016) and multiplying the obtained quantity by the average exposure to the CRS (0.298) and the number of tax havens in the sample (50).⁴² The results are presented in Table III. According to our calculations, the commitment to the CRS in 2013 and 2014 resulted in an increase of approximately \$45 billion in investments in the UK real estate market through shell companies incorporated in tax havens. Commercial properties account for 55% of the total effect, and residential properties for 45%. This estimated effect is substantial and highlights the significant impact of the CRS on offshore real estate investments in the UK. When considering residential investment alone, the introduction of the CRS explains approximately 2% of the overall investment in the UK residential market over the same period, using transactions appearing in the Price Paid Data as a proxy for the denominator.

6.2 Effect of the CRS on offshore financial wealth

We quantify how much offshore financial wealth flowed out of tax havens in response to the CRS in two steps: i) we estimate the amount of offshore financial wealth owned by residents of early-adopting countries in 2013, and ii) we apply to this stock estimates from the literature on the impact of the CRS on offshore deposits.

First, in order to estimate the stock of financial wealth held by early-adopting countries in tax havens in 2013, we draw on country-by-country estimates of offshore wealth obtained by [Alstadsæter et al. \(2018\)](#). They update the global offshore wealth measure of [Zucman \(2013\)](#) and construct a time series of the evolution of offshore wealth from 2001 to 2015. Then, they allocate this amount to each of the world's countries, for the year 2007.⁴³ The amounts of offshore wealth held by residents of each country are very heterogeneous, amounting to the equivalent of 60% of GDP for countries like Russia, and to only a few percentage points for countries like Japan or Denmark. To get estimates for 2013, we allocate their 2013 estimates of global offshore wealth (\$7.7 trillion) to each country, according to the country-by-country shares of 2007. We thus assume that the geographical distribution of offshore wealth remained unchanged between 2007 and 2013. The total stock of offshore financial wealth owned by the early adopters in 2013 is obtained by simply adding the figures for all non-haven countries committing to the CRS in 2013 or 2014.⁴⁴ We find

⁴²Note that the tax havens investing the most in the UK property market have on average a higher exposure to the CRS (exposure of 0.55 for the top 10 buyers). Thus, the estimate obtained here should be considered as a lower bound.

⁴³The complete results of this allocation are available in appendix table A.3 of their paper.

⁴⁴Out of the 67 countries from the G20 (excluding the US and Russia) or participating either to the Joint Announcement or the OECD Declaration on Tax Matters, we only keep the 42 non-haven countries (using the list from [Menkhoff and Miethe \(2019\)](#)). We do not keep Greenland and Faroe Islands either, as we do not have information on the amount of offshore financial wealth they own in [Alstadsæter et al. \(2018\)](#). Thus, we are left with

that these countries were holding more than \$ 4.5 trillion in tax havens that year.

Second, we build on O' Reilly et al. (2021), who estimate that the Joint Announcement of March 2014 led to a reduction of approximately 11% in bank deposits held in tax havens. This figure is based on a two-period difference-in-differences model using Bank for International Settlements (BIS) cross-border deposits data. While other papers studied the effect of the CRS on offshore bank deposits (Menkhoff and Miethe, 2019; Cusi et al., 2020; Beer et al., 2019), O' Reilly et al. (2021) is to the best of our knowledge the only paper studying specifically the effect of the Joint Announcement, which is the event we exploit in our paper together with the previous commitment to the CRS of G20 countries. By applying the 11% effect of the CRS to the stock of offshore financial wealth held by residents of early-adopting countries in tax havens in 2013, we obtain an estimate of the effect the policy had on offshore financial wealth. We find that the Joint Announcement led to a decrease of about \$506 billion in the amount of offshore financial wealth held by residents from participating countries.

6.3 Magnitude of the rebalancing response to the CRS

Comparing effect of the CRS on real estate investments in the UK to its effect on offshore financial wealth, we find that the equivalent of about 9% of the observed reduction in offshore financial assets caused by the Joint Announcement was ultimately invested in UK properties. Results are presented in Table IV. This finding suggests that around 9% of the financial assets held in tax havens before the implementation of exchange of information were shifted to UK real estate assets as a means to evade the reporting requirements of the transparency policy. Investing in real estate assets would thus constitute a significant margin of evasion for the CRS.

Even though there is uncertainty on the exact share held by the UK in the global cross-border real estate market, anecdotal evidence suggests that the UK is one of the most popular destinations globally, as it would be the first destination for cross-border real estate in Europe and the second one at a global scale, behind the United States.⁴⁵ It is likely that similar asset-shifting responses to the CRS also occurred in other countries' real estate markets, albeit potentially to a lesser extent, given the special role played by London in high-net-worth individuals locations' choices (DeVerteuil and Manley, 2017). The estimated 9% can thus be considered as a lower bound of the portfolio rebalancing response to the CRS at the global level.

This result should be interpreted with caution due to at least two elements. First, there is no unanimity about how the CRS affected offshore deposits. Estimates of the magnitude of the response to the CRS vary substantially depending on the study. Available papers focus on various sets of countries and sample periods and find estimates ranging from 11.5% to 31.8%. In appendix table A.4, we show that our estimate of the rebalancing response ranges

40 countries.

⁴⁵Estimates for Europe are based on BNP Paribas Real Estate analysis of 2019, <https://www.realestate.bnpparibas.com/sites/default/files/2019-10/AAG%20Capital%20Flows%20Q2%202019.pdf>, retrieved on the 27/08/21. Estimates for the United States come from Tranio analysis, professional cross-border real estate broker, https://tranio.com/articles/investment-flows-around-the-globe-cross-border-property-transactions-in-2016_5321/, retrieved on the 27/08/21.

from 3% to 9% depending on the estimate we consider. Second, previous studies focused on the effect of the CRS on bank deposits rather than total financial wealth. If the effect of the CRS varies across different types of financial assets, aggregate rebalancing responses to the CRS may differ from the estimates based on bank deposits alone.

Overall, despite some uncertainties, our estimated rebalancing response to the CRS of 9% is likely to represent a conservative lower bound at the global level. This result suggests that a significant portion of assets targeted by the CRS were shifted into real estate globally to be shielded from reporting obligations.

7 Conclusion

The Common Reporting Standard constitutes a credible threat for offshore users who do not declare their assets to the tax authorities. Its multilateral nature makes it challenging for individuals to relocate their financial assets to non-cooperative tax havens, as more than 100 jurisdictions are now automatically exchanging information with each other. As such, it has decreased the amount of bank deposits that were held in tax havens. However, by only targeting financial assets, this policy leaves the door open for new evasion strategies to develop. In its current form, it creates incentives for non-compliant taxpayers to restructure their offshore portfolios away from financial assets and toward properties.

In this paper, we show that this new international transparency initiative played an important role in the growth of the offshore real estate market in the UK over the last decade. We show that it led to an inflow of investments of around \$45 billion over the 2013-2016 period, suggesting that substantial investments have been directed towards properties to circumvent the new reporting obligations. By documenting a substantial reshuffling response to the CRS, our results imply that excluding real estate assets from the reporting requirements is an important design flaw of this policy.

This paper demonstrates the necessity of implementing more comprehensive and ambitious automatic exchange of information agreements. To effectively combat tax evasion, a global information exchange treaty that covers all types of assets, including non-financial ones, is necessary. By expanding the scope of the reporting requirements to encompass a wider range of assets, policymakers can aim to minimize the potential for new evasion strategies to emerge and ensure a more robust and effective framework for international tax transparency.

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Tables and Figures

TABLE I: CHARACTERISTICS OF THE PROPERTIES PURCHASED

Variable	Total	Commercial	Residential
All transaction prices (GBP)	1,123,968	2,039,686	851,483
Observed transaction price (GBP)	1,768,254	4,372,988	1,243,283
London	48%	43%	49%
Freehold	52%	69%	47%
Share of indicated price	54%	39%	58%
Number of transactions	103,848	23,815	80,033
		(23%)	(77%)

Notes: This table displays the characteristics of the properties of the Overseas Companies Ownership Dataset we use in our analysis. The first column presents the characteristics of our full sample, while columns “Commercial” and “Residential” show the characteristics of respectively commercial and residential properties. The row “All transaction prices (GBP)” gives the average price (observed + predicted) in the sample. The row “Observed transaction price (GBP)” presents the observed price, either directly (OCOD data) or indirectly (PPD data). The row “London” displays the share of transactions taking place in the Greater London area. The row “Freehold” shows the percentage of freehold properties in each categories, the rest of the transactions corresponding to Leasehold titles. The last row “Share of indicated price” presents the percentage of transactions for which we recover a purchase price, either through the OCOD or the PPD. The rest of the prices are estimated as described in Appendix section B.1.

TABLE II: PERCENTAGE OF COMPANIES MATCHED AND IDENTIFIED IN THE OCOD SAMPLE

Source	Number of Transactions	Amount Invested (£billion)	Fraction of Total Transactions	Fraction of Total Amount Invested
Full Dataset	93,332	102	100%	100%
Matched	45,409	44.500	48.7%	43.6%
Identified	19,056	21.400	20.4%	21.0%

Notes: This table shows the number of properties held through foreign companies in January 2018 we manage to link with their ultimate beneficial owners' using the Bahamas Leaks, the Offshore Leaks, the Paradise Papers, the Panama Papers, the Pandora Papers, OpenLux, CNBIOM data and the Register of Overseas Entities. Columns 2 and 3 show the raw number of transactions and their value in the full and matched samples, while columns 4 and 5 show the corresponding percentages the matched transactions represent. The "Matched" row shows these figures for all companies we find in one of these datasets. The "Identified" row shows these figures excluding companies whose beneficial owners are other companies we do not know the owners of.

TABLE III: EFFECT OF THE CRS ON THE UK REAL ESTATE MARKET - 2013-2016

Property type	In billion GBP	In billion USD
Residential	13	20
Commercial	16	25
All	29	45

Notes: This table shows the estimates obtained when computing the effect of the CRS based on the estimated $\hat{\beta}^{qt}$ from equation 2, average exposure to the CRS (0.298) and number of tax havens in the sample (50). These estimates are for the period 2013q3-2016q4. The GBP - USD exchange rate used is from January 2015.

TABLE IV: ESTIMATION OF ASSET SHIFTING RESPONSES TO THE CRS

	(1)	(2)	(3)
Paper	Estimates	Wealth decrease (USD billion)	Asset shifting
O'Reilly et al. (2019)	11%	506	9%

Notes: This table compares the effect of the CRS on real estate investments we estimate in our paper to the amount of offshore financial wealth that left tax havens due to the transparency shock as estimated in [O' Reilly et al. \(2021\)](#). Column “Estimates” presents estimates of the reduction in offshore financial wealth caused by the CRS, as computed in [O' Reilly et al. \(2021\)](#). Column “Wealth decrease” presents a computation of the stock of offshore financial wealth that flew out of tax havens from 2013 because of the CRS. It is computed using estimates of the stock of offshore financial wealth held by CRS early-adopting countries in tax havens measured in [Alstadsæter et al. \(2018\)](#), multiplied by the estimated CRS financial effect of column (1). Column “Asset shifting” computes the ratio of our real estate effect for the UK (\$45 billion) over the offshore wealth decrease of column (2).

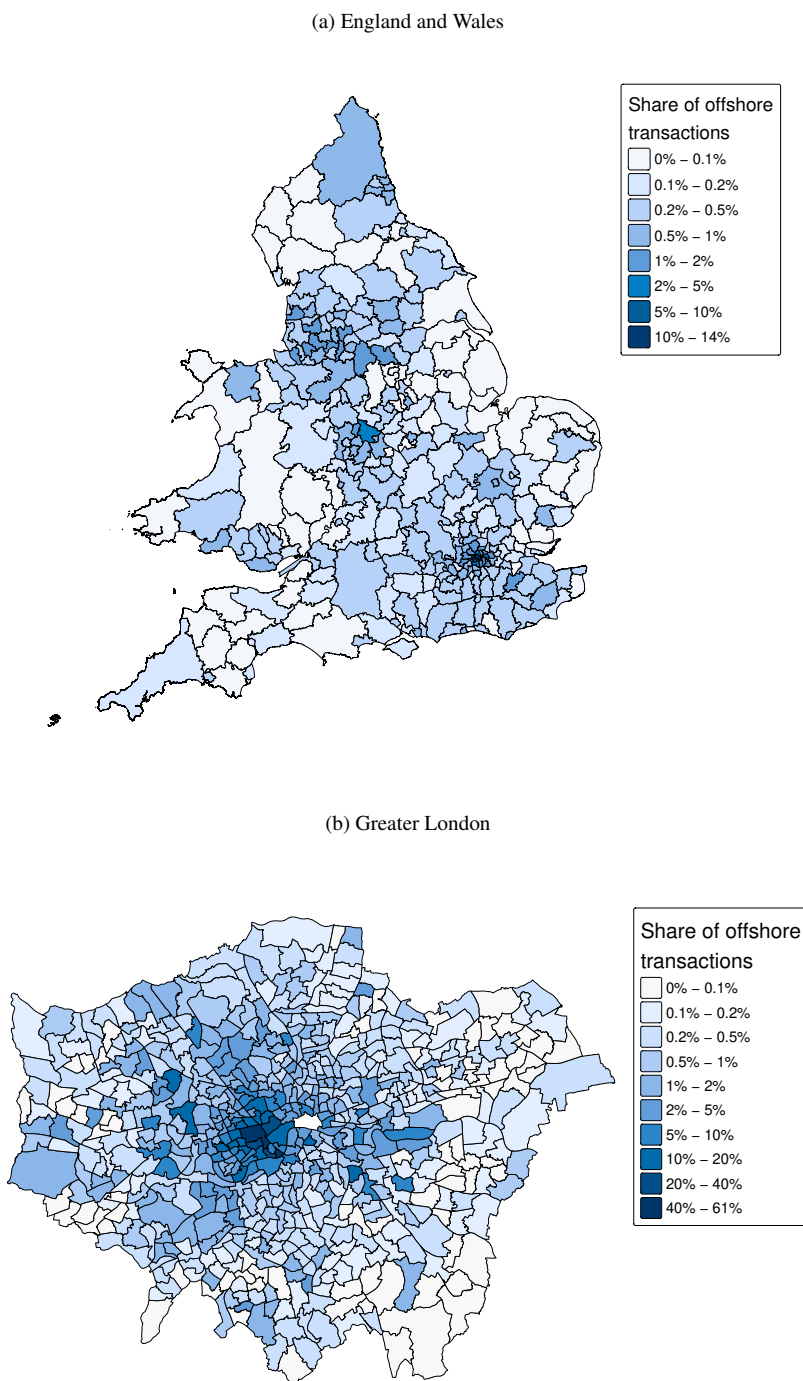


FIGURE I: SHARE OF RESIDENTIAL PURCHASES FROM FOREIGN COMPANIES IN TOTAL RESIDENTIAL PURCHASES

Notes: This figure shows the ratio of residential real estate purchases made by foreign companies in England and Wales over all residential property purchases. Panel A shows this ratio over England and Wales by local authority, and Panel B in Greater London only, by ward. This share is computed over the 2000-2019 period. Residential real estate purchases made by foreign companies are identified through the OCOD, while residential purchases are computed using the Price Paid Data. Commercial real estate purchases made by foreign companies are excluded from the sample. The City of London local government is excluded because the Price Paid Data does not provides information on purchases made in the City.

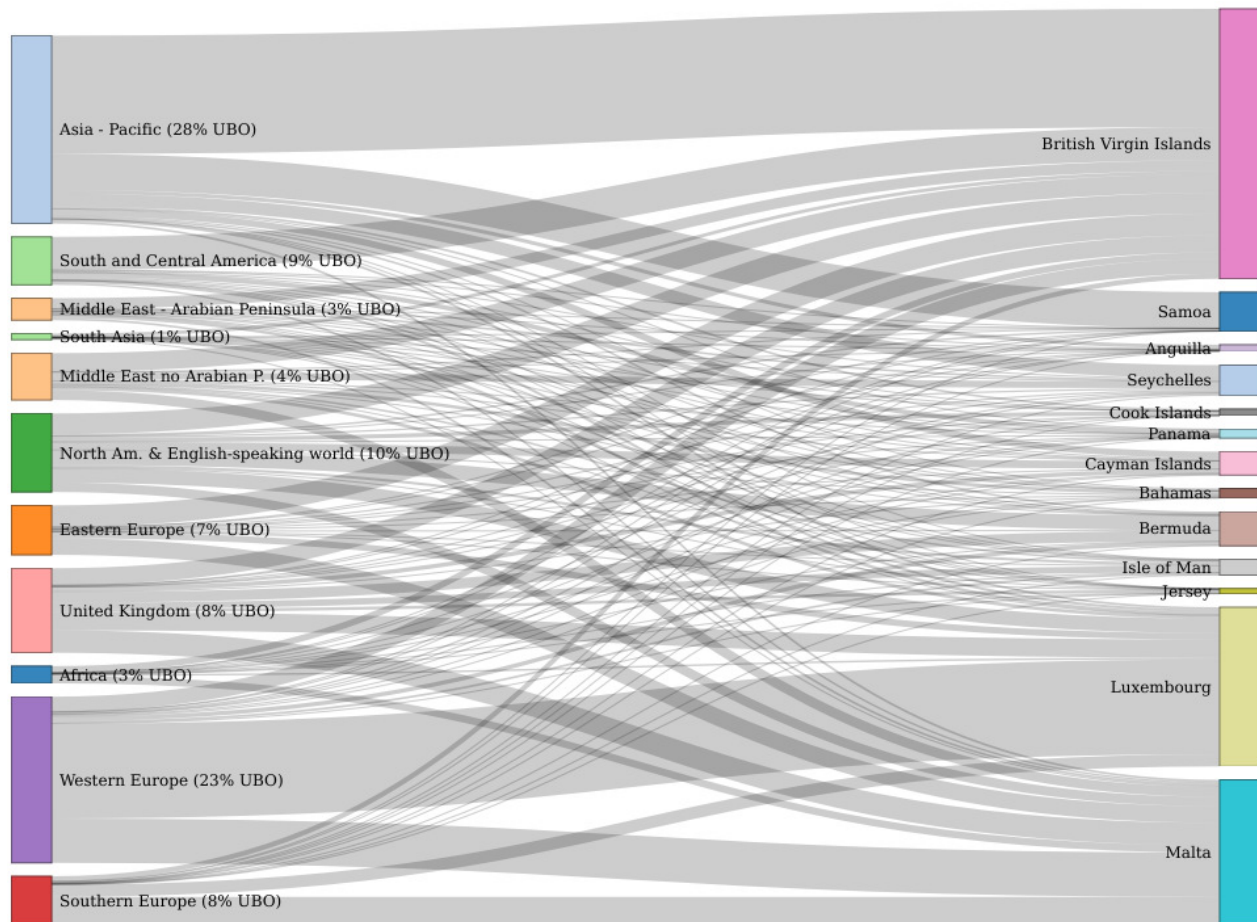


FIGURE II: MOST FREQUENT TAX HAVENS HOSTING COMPANIES BY WORLD REGION OF ULTIMATE BENEFICIAL OWNER

Notes: This figure shows the most frequent tax havens used to incorporate companies, by region of the ultimate beneficial owner(s) (UBO). Beneficial owners are on the left-hand side of the figure, and countries they use to incorporate companies on the right-hand side. The figure is constructed using data from the Bahamas Leaks, the Offshore Leaks, the Panama Papers, the Pandora Papers, the Paradise Papers, the CNBIOM leaks and the OpenLux data. We display the percentages of owners from each world region, as they appear in these datasets. To compute the percentages, we remove beneficial owners who are companies and beneficial owners linked to tax havens. The total might not add up to 100% because of rounding.

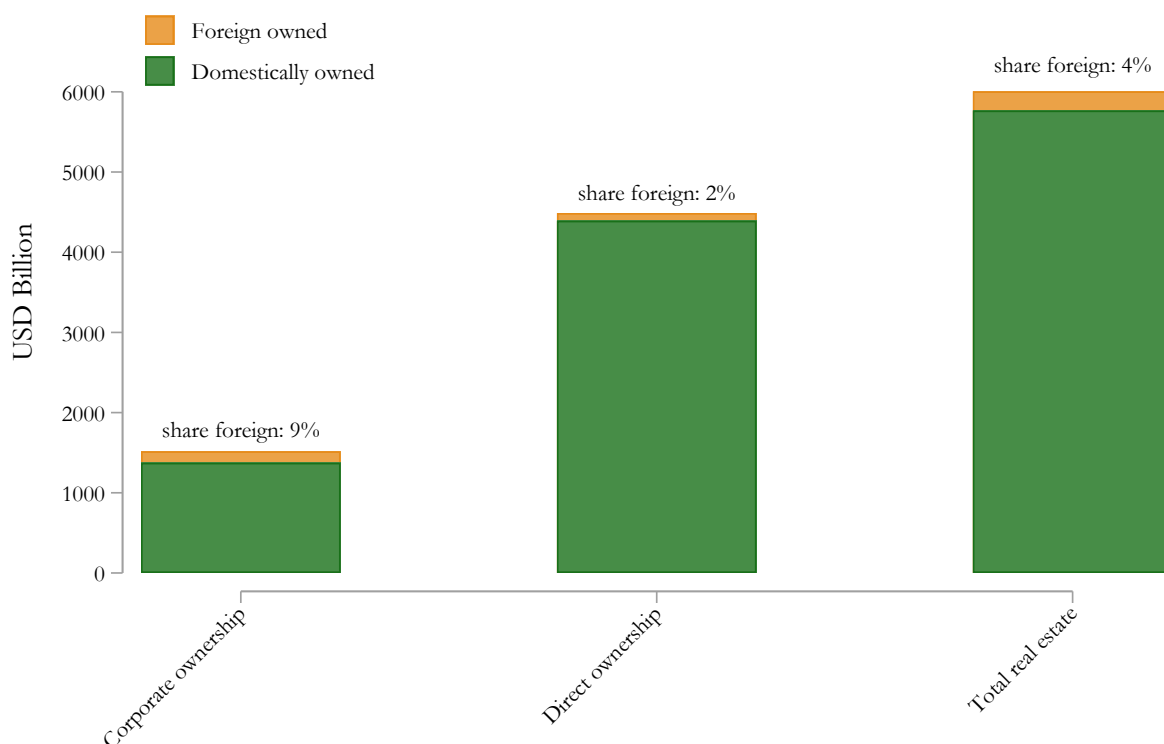


FIGURE III: SHARE OF FOREIGN OWNERSHIP IN REAL ESTATE - ENGLAND AND WALES

Notes: This graph shows the importance of real estate wealth owned in England and Wales by type of ownership and provides estimates of how much is owned from abroad, in 2018. Total real estate wealth is estimated using balance sheet estimates. Our method is detailed in Appendix section C. Foreign real estate wealth is estimated using the Overseas Companies Ownership Dataset (for corporate ownership) and data from an FOI request made by the Centre for Public Data (for direct ownership). The property prices are expressed in 2018 value. The value of directly owned properties is computed using the number of properties held by individuals with foreign correspondence addresses, multiplied by the average price in the first quarter of 2018 of the district where they are located. Average house prices by local authorities are published by the ONS and available [here](#). The 2018 value of corporately owned properties are displayed in table A.2 and are estimated from the district specific house price evolution between the date of transaction and the first quarter of 2018. All values are expressed in dollars (January 2018's exchange rate).

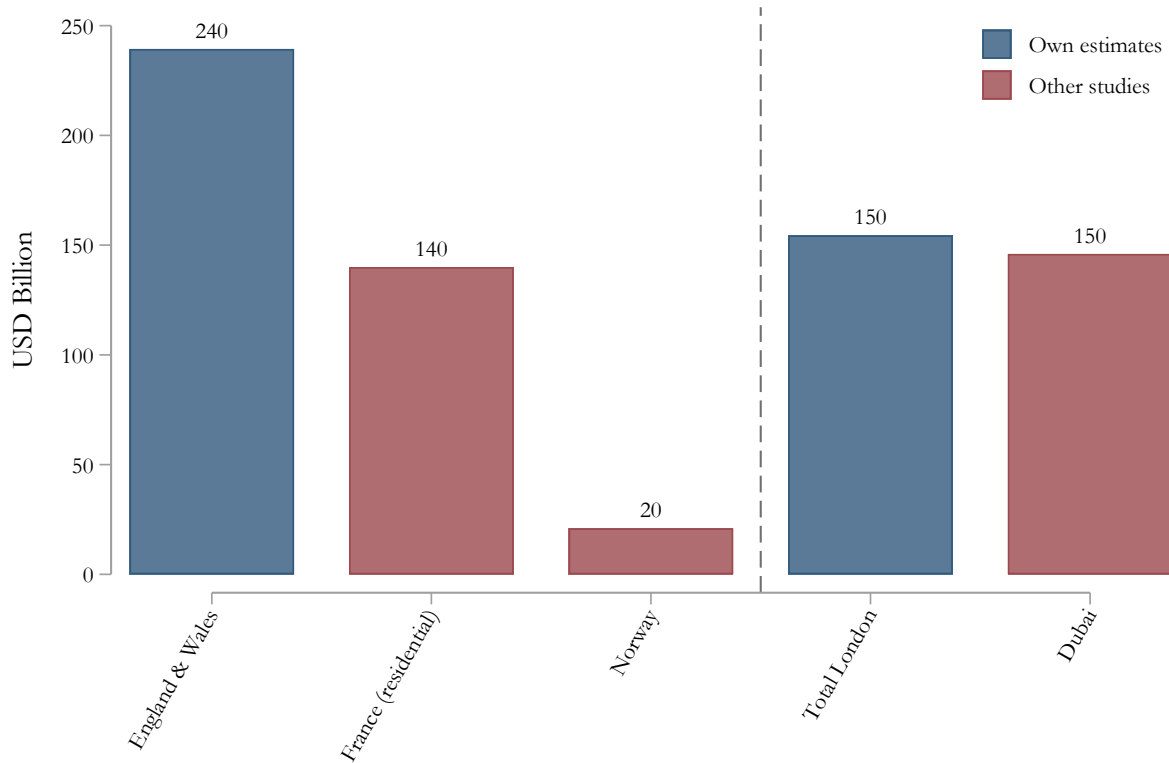


FIGURE IV: ESTIMATES OF OFFSHORE REAL ESTATE WEALTH IN OTHER COUNTRIES

Notes: This figure compares our estimates of real estate wealth owned from abroad to offshore real estate wealth estimates for other countries. We express our estimates in January 2018 dollars. Estimates for France, Norway and Dubai are from [Morel and Uri \(2021\)](#), [Alstadsæter and Økland \(2022\)](#) and [Alstadsæter et al. \(2022\)](#) respectively. For France, we use the same dollar estimates as [Alstadsæter et al. \(2022\)](#). We keep the other studies' estimates, which are already expressed in dollars.

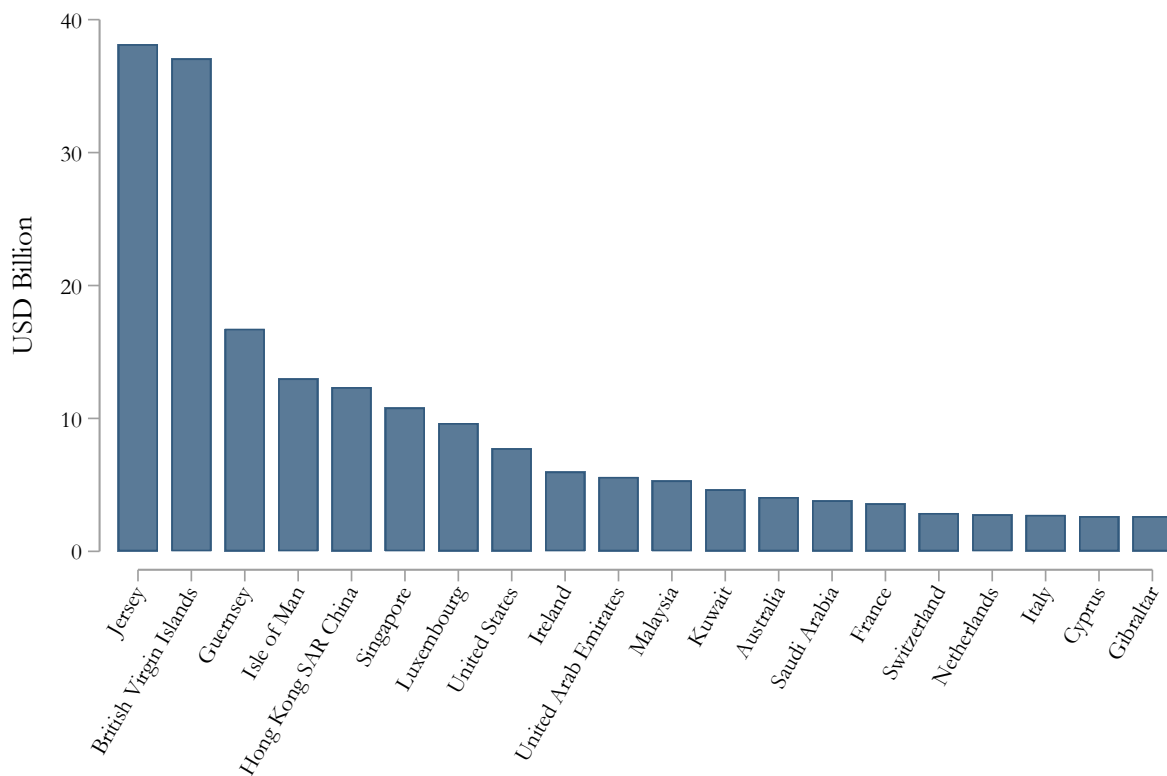


FIGURE V: COUNTRY OF UK REAL ESTATE OWNERS, BY IMMEDIATE OWNERSHIP

Notes: This figure displays the size of foreign owned real estate wealth by country of immediate ownership for the top 20 owners. Immediate ownership country is defined as the jurisdiction of incorporation for corporate ownership and as the country of correspondence address for direct ownership. Estimates are computed using the Overseas Companies Ownership Dataset (corporate ownership) and data on properties held by overseas individuals from the Centre for Public Data (direct ownership). All values are expressed in dollars (January 2018's exchange rate).

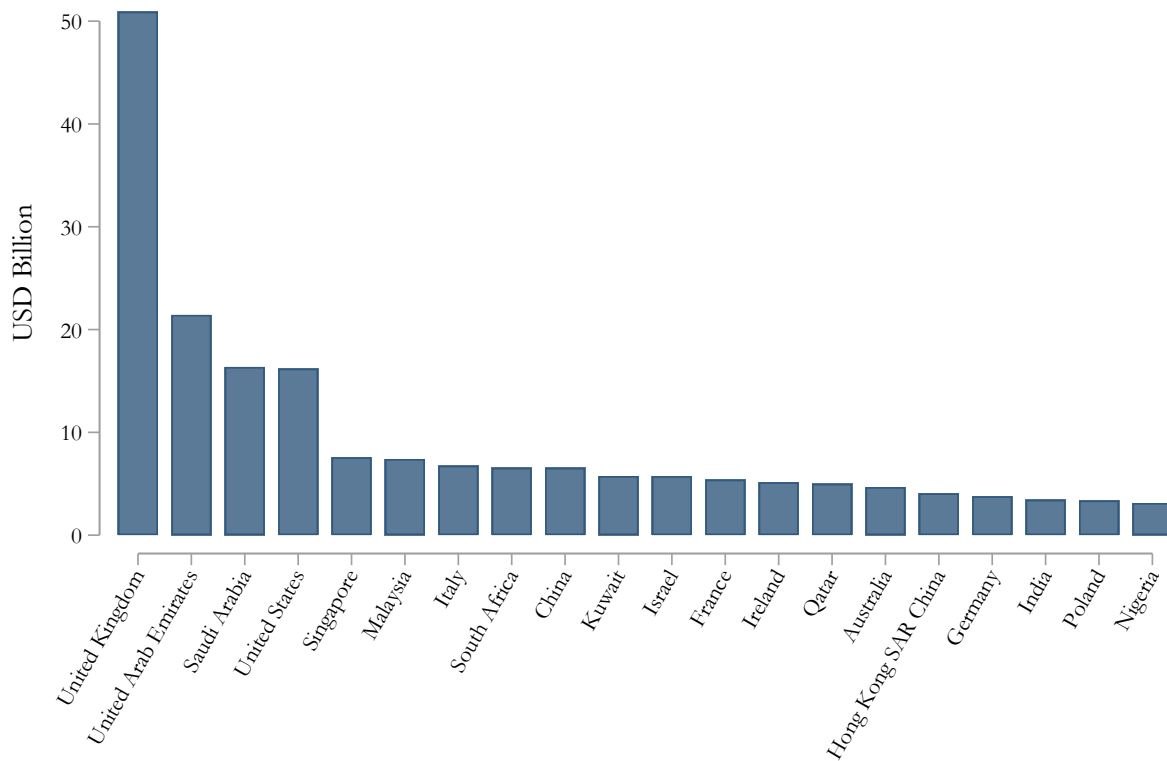


FIGURE VI: COUNTRY OF UK REAL ESTATE OWNERS, BY ULTIMATE OWNERSHIP

Notes: This figure displays estimates of the size of foreign owned real estate wealth in England and Wales by country of ultimate ownership for the top 20 owners. We compute these estimates using the Overseas Companies Ownership Dataset (for properties held by foreign companies) and data from an FOI request made by the Centre for Public Data (for properties held by individuals with foreign correspondence addresses). The estimates are for January 2018. Ultimate ownership country refers to the property ultimate owner’s country of residence. It includes both properties held through foreign companies, and properties held by individuals with a foreign correspondence address. Ultimate ownership of properties held through foreign companies is estimated by matching the Overseas Companies Ownership Dataset with the Register of Overseas Entities and the beneficial ownership data (Panama Papers and other leaks, and OpenLux data). The distribution of real estate wealth non-matched to these datasets is estimated using the distribution of properties matched to the beneficial ownership data (i.e. excluding the matches from the Register of Overseas Entities). Ultimate ownership of properties held by individuals with foreign correspondence addresses is estimated by allocating ownership from individuals with addresses i tax havens to the most probable “true” residence country of the owners using Register of Overseas Entities data. All values are expressed in dollars (January 2018’s exchange rate).

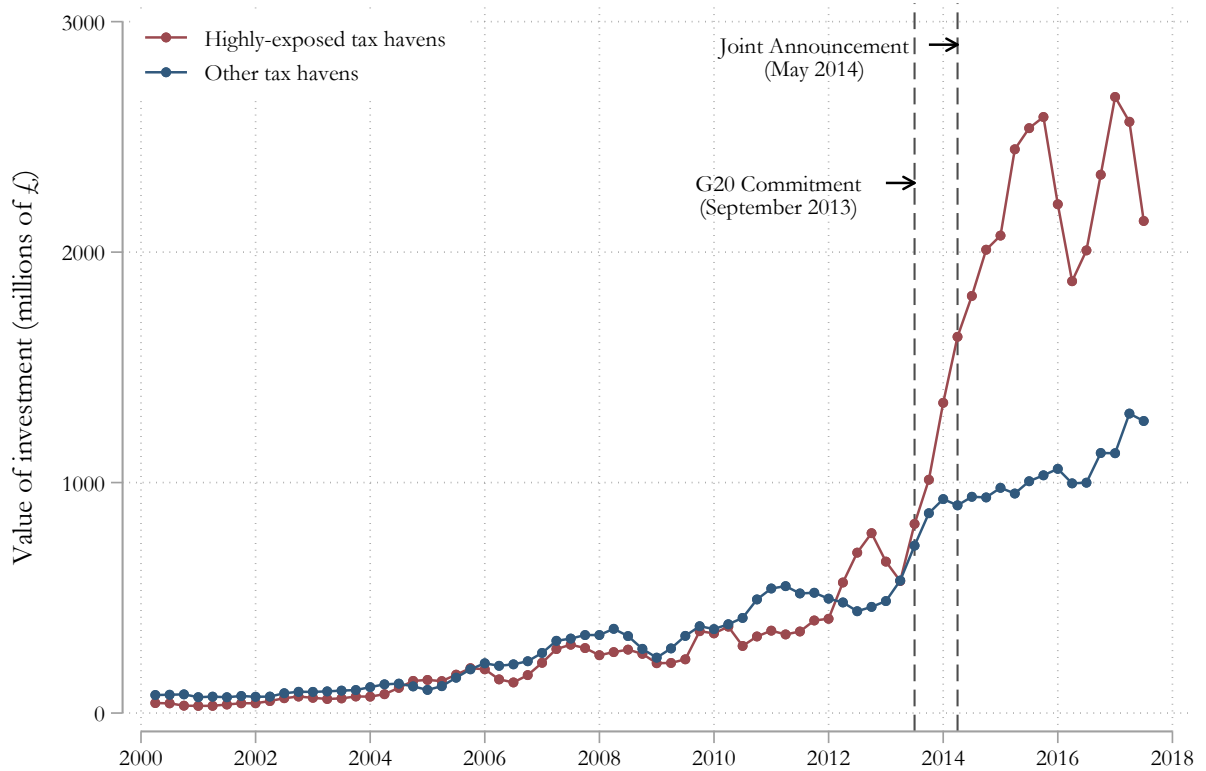


FIGURE VII: TOTAL VALUE OF TRANSACTIONS FROM COMPANIES INCORPORATED IN “HIGHLY-EXPOSED” TAX HAVENS VS OTHER HAVENS

Notes: This figure presents the aggregated amounts invested in England and Wales by companies incorporated in “highly-exposed” tax havens vs. companies incorporated in the other havens, over the period 2000-2017. It is based on the Overseas Companies Ownership Dataset. The measure of exposure to the CRS is computed using the beneficial ownership dataset (leaks data and OpenLux data). For each jurisdiction of incorporation h , we compute the proportion of company beneficial owners coming from countries adopting the CRS in 2013 or 2014. Highly-exposed havens are defined as havens with more than 75% of company owners coming from early-adopting countries. Other havens include all havens with less than 75% of company owners coming from early-adopting jurisdictions, or havens that host less than 10 shell companies in the beneficial ownership data. Investment is expressed in 3-quarters moving averages. Number of “highly exposed” tax havens: 9. Number of other tax havens: 41.

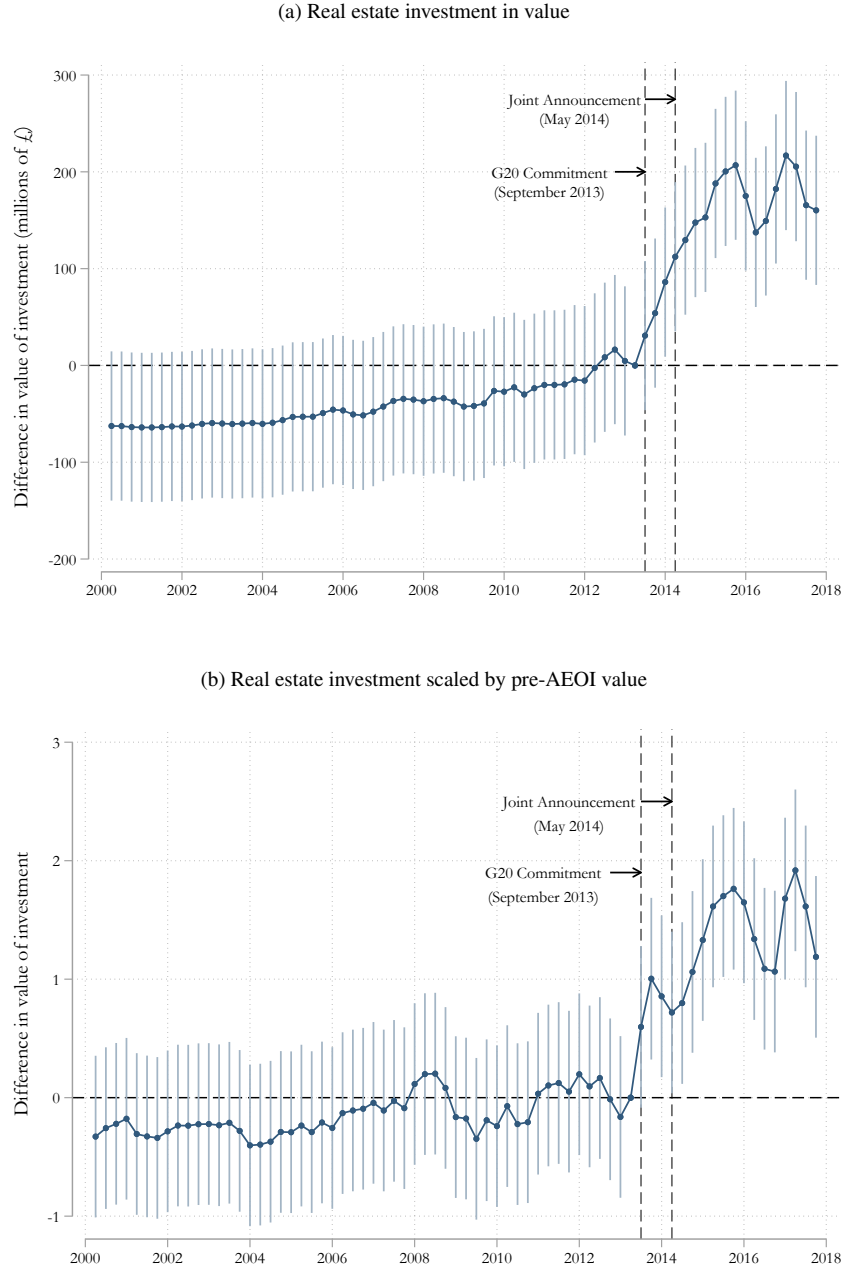


FIGURE VIII: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - AMOUNT INVESTED

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. The outcome variables –expressed in 3-quarters moving average– are investment in million Pounds (Panel A) and investment scaled by country average investment one year before AEOI endorsement i.e. the period between the 3rd quarter of 2012 and the 2nd quarter of 2013 (Panel B). The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

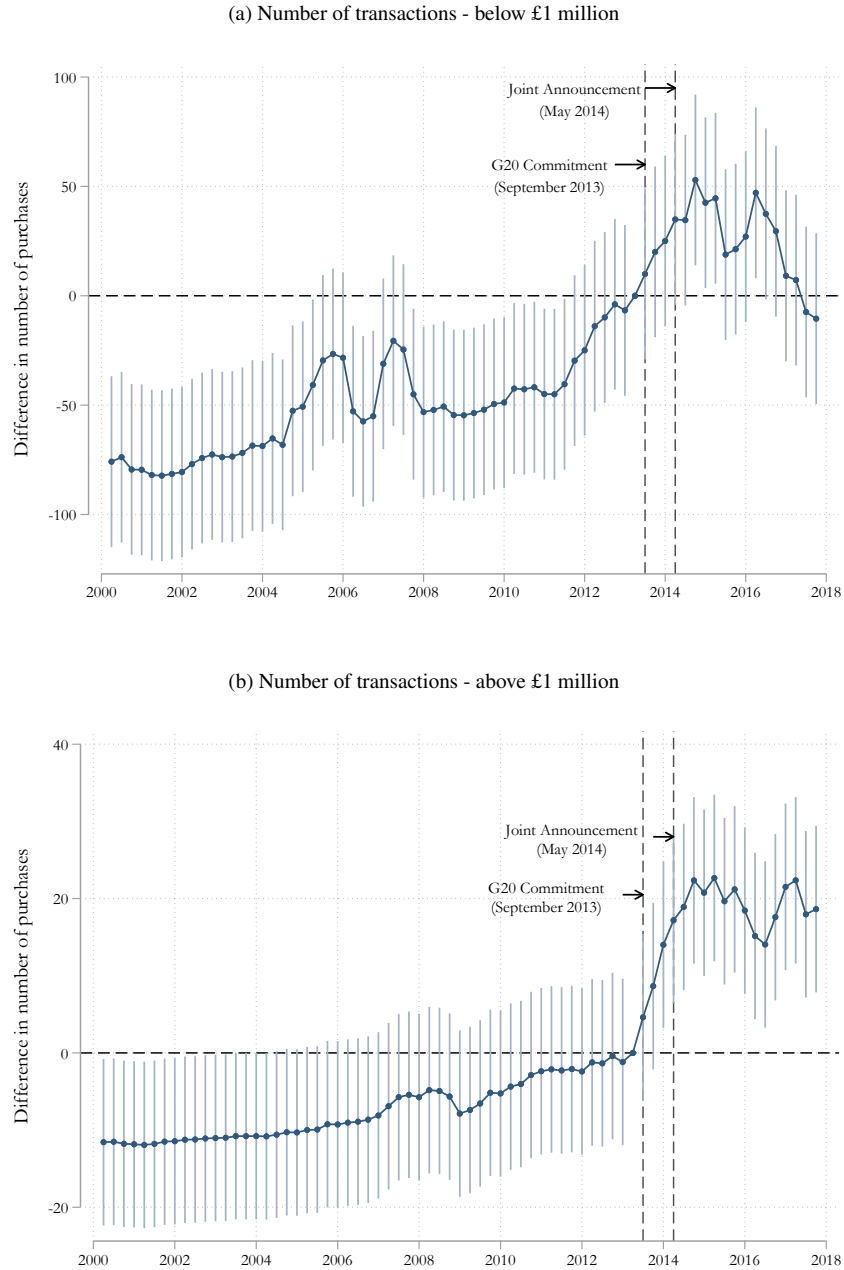


FIGURE IX: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - NUMBER OF TRANSACTIONS

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. The outcome variables –expressed in 3-quarters moving average– are number of properties costing less than £1 million (Panel A) and number of properties costing more than £1 million (Panel B). The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

Appendices

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A Additional Figures and Tables

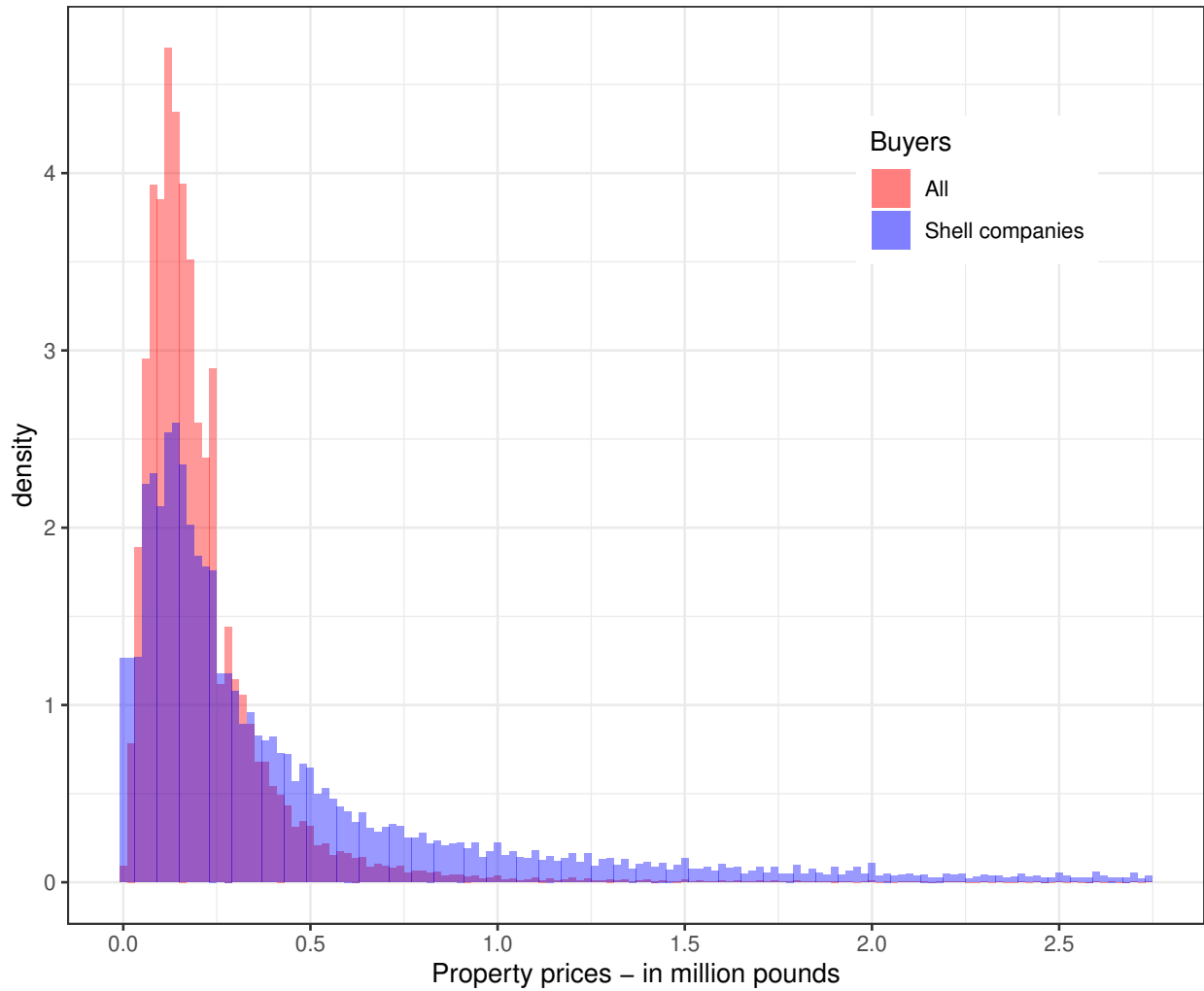


FIGURE A.1: DISTRIBUTION OF PRICES OF RESIDENTIAL PROPERTIES BOUGHT THROUGH SHELL COMPANIES, AND OF ALL RESIDENTIAL PROPERTIES IN THE UK REAL ESTATE MARKET.

Notes: This figure compares the distribution of prices of English and Welsh properties bought through foreign companies (in blue), and the distribution of prices of all residential properties bought in England and Wales (in red) over the 2000-2020 period. It is constructed using the Overseas Companies Ownership Dataset, keeping only transactions for which the transaction price is observed, and the Price Paid Data. Both datasets come from the UK Land Registry. For better visibility, the prices are capped at 99.8% of the price distribution of the Price Paid Data.

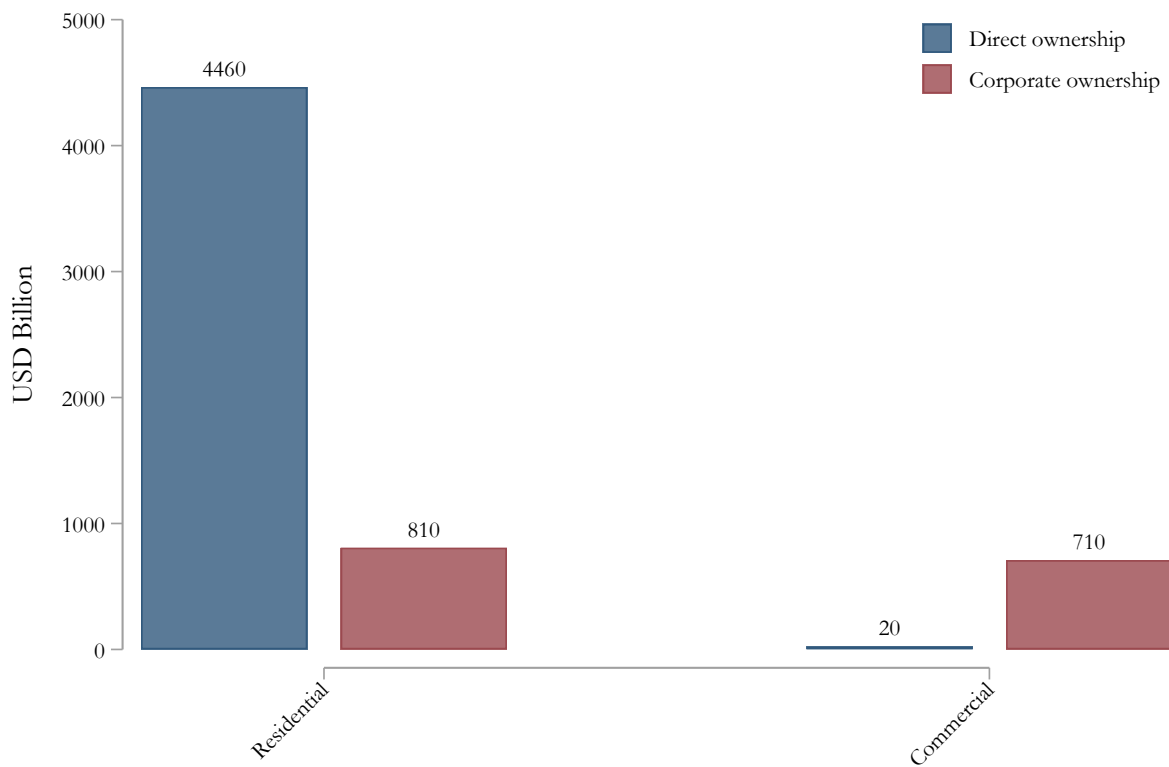


FIGURE A.2: RESIDENTIAL AND COMMERCIAL REAL ESTATE WEALTH BY OWNERSHIP CATEGORY IN ENGLAND AND WALES

Notes: This figure shows the size of residential and commercial real estate wealth owned in England and Wales, by type of ownership. The estimates are based on UK national balance sheet for 2018. Residential real estate is composed of dwellings and land underlying them. Commercial real estate is defined as other buildings and land underlying them. Direct ownership refers to properties owned by households. Corporate ownership refers to properties owned by private corporations. We provide details on how the figures are computed in Appendix section C. All values are expressed in dollars (January 2018's exchange rate).

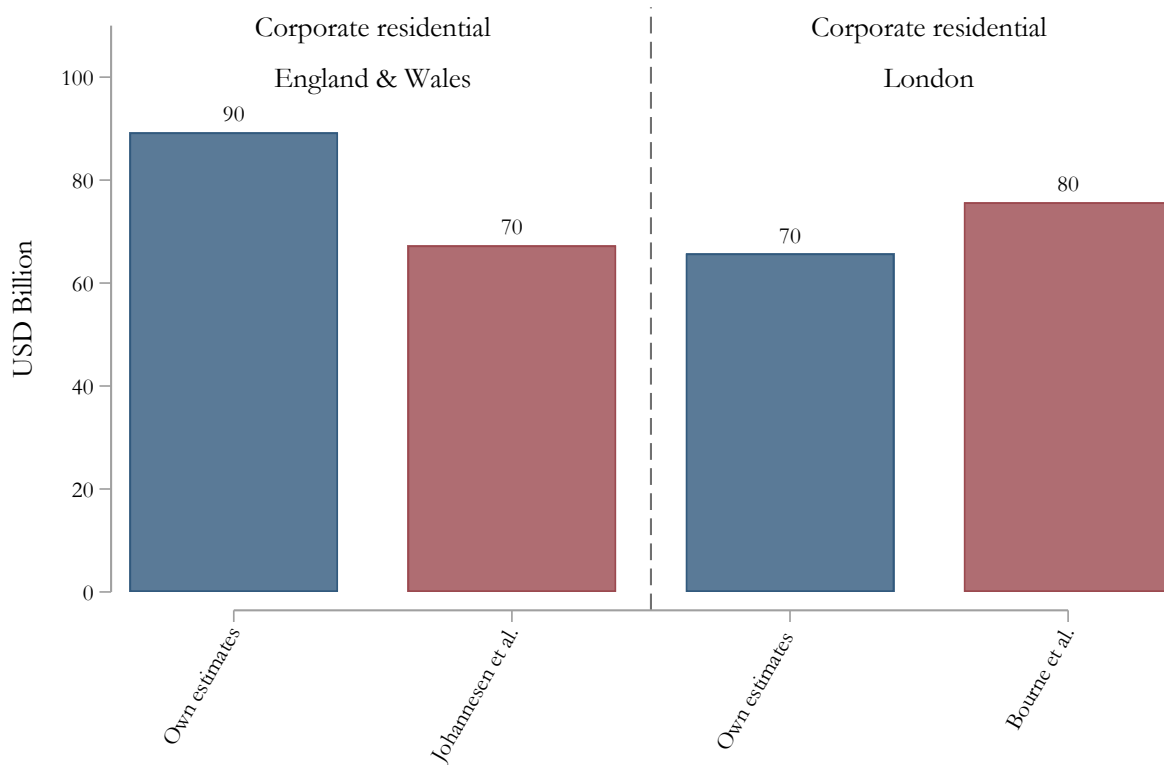


FIGURE A.3: COMPARISON WITH OTHER STUDIES ON OFFSHORE REAL ESTATE IN ENGLAND AND WALES.

Notes: This figure compares our estimates of real estate wealth owned by foreign corporations based in the Overseas Companies Ownership Dataset to what is found in other studies. We restrict our estimates to the same segment of the real estate market to make the comparisons meaningful. For the comparison with [Johannesen et al. \(2022\)](#), we focus on residential properties only and for the comparison with [Bourne et al. \(2022\)](#), we focus on residential properties held in London only. Our estimates are for January 2018, while [Johannesen et al. \(2022\)](#)'s figure is estimated for December 2019 and [Bourne et al. \(2022\)](#)'s for January 2022. All values are expressed in dollars (January 2018's exchange rate).

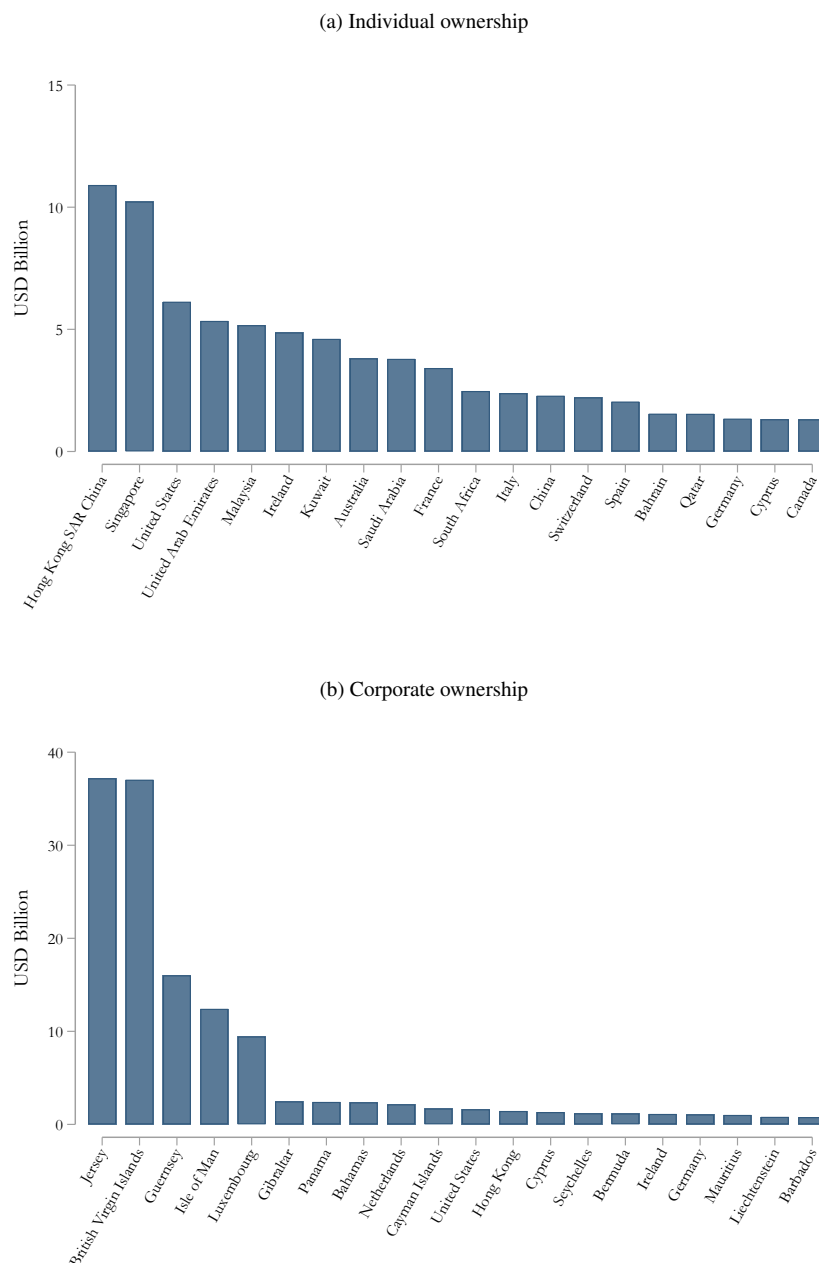


FIGURE A.4: FOREIGN REAL ESTATE OWNERSHIP IN THE UK, BY COUNTRY OF THE IMMEDIATE OWNER

Notes: This figure displays estimates of the size of foreign owned real estate wealth by country of immediate ownership for the top 20 owners. Foreign owned real estate includes both properties held by individuals with foreign correspondence addresses and by foreign companies. In Panel A, individual (or direct) ownership estimates are computed using a Freedom of Information Request established by the Centre for Public Data. The country of immediate ownership is computed using the country of correspondence address of individuals. In Panel B, corporate (or indirect) ownership estimates are computed using the Overseas Companies Ownership Dataset. Immediate ownership country is defined as the jurisdiction of incorporation of the purchasing companies. Estimates are for January 2018 and all values are expressed in dollars (January 2018's exchange rate).

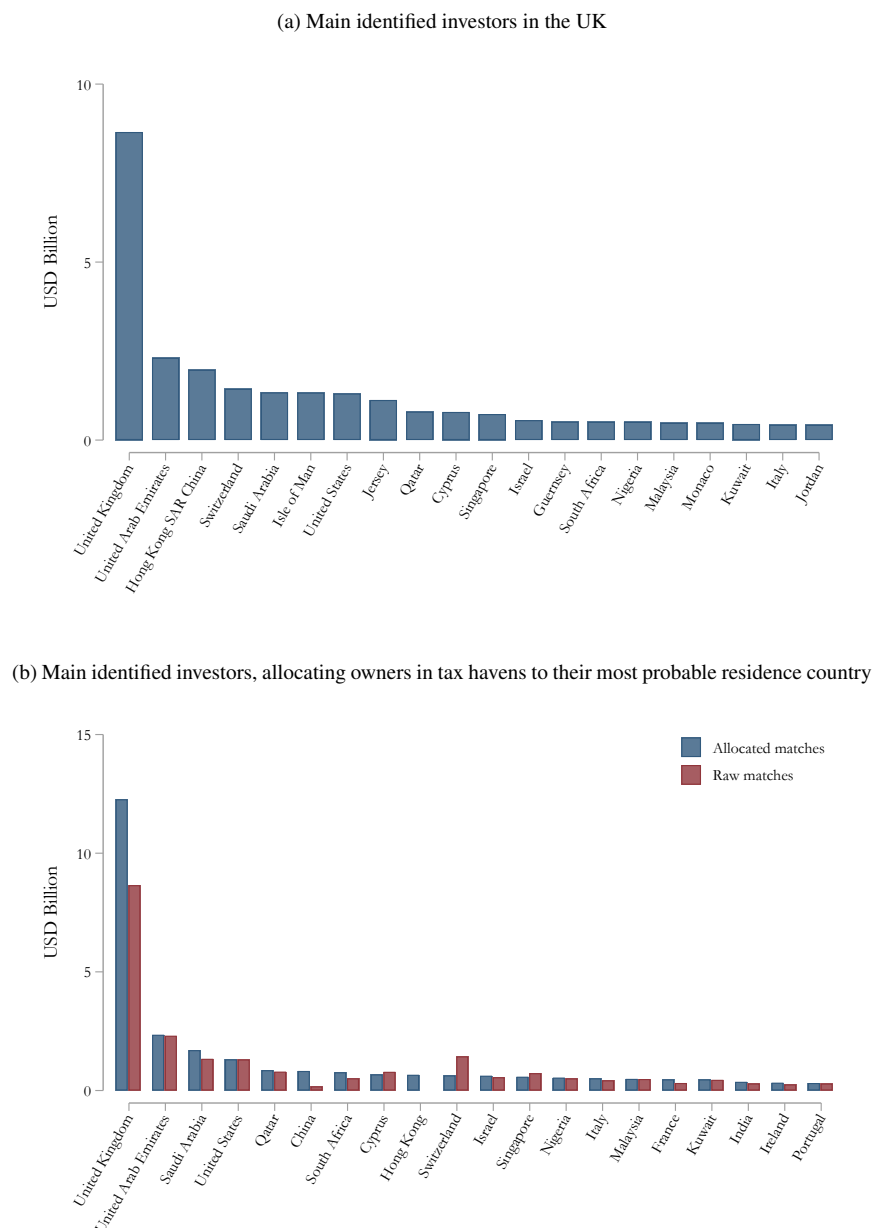


FIGURE A.5: COUNTRY OF ORIGIN OF IDENTIFIED OWNERS BUYING PROPERTIES IN THE UK THROUGH A FOREIGN COMPANY, STOCK OF PROPERTIES IN JANUARY 2018

Notes: This figure shows the distribution of country of residence of identified beneficial owners of companies holding real estate in England and Wales in January 2018. Data from companies holding properties in England and Wales comes from the Overseas Companies Ownership Dataset (OCOD) maintained by the Land Registry. We identify beneficial owners of these companies matching the OCOD with the Leaks Data (Bahamas Leaks, Offshore Leaks, Panama Papers, Pandora Papers, Paradise Papers, and OpenxLux and CNBIOM Leaks) and the Register of Overseas Entities (ROE). We match companies based on their name, country of incorporation and date of incorporation. In Panel A, we show the raw results we obtain after the matching process, using the address country as declared in the ROE and country linked to the beneficial owner as per the Offshore Leaks. In Panel B, we allocate all owners from tax havens we found in Panel A to their most probable country of residence. For matches with the ROE, we do so by selecting the country of citizenship of the individual instead of the address country. This information, however, is not available in the Leaks data. As a result, we use a software predicting the most likely nationality according to the name to assign a country of residence to owners linked to a tax haven. The matching is based on the property stock of the UK as of January 2018, and all values are expressed in dollars (January 2018's exchange rate).

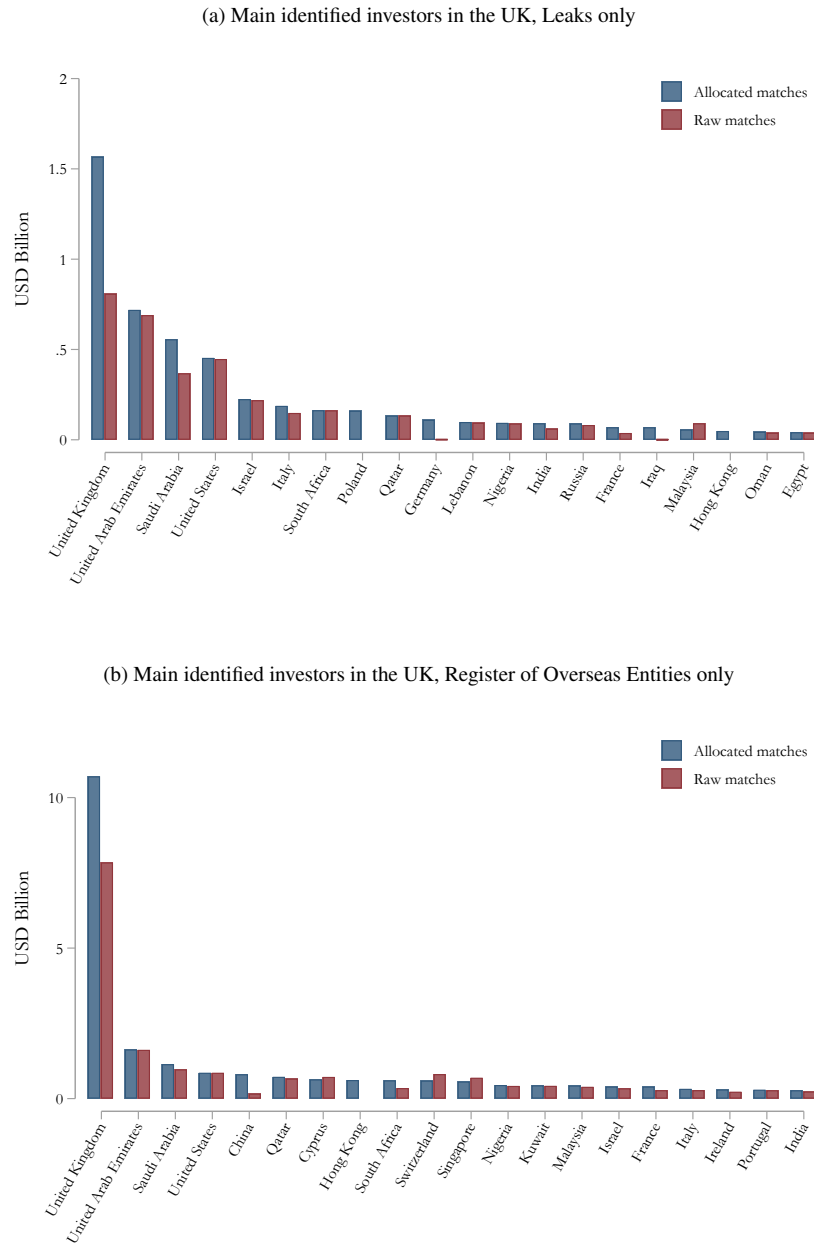


FIGURE A.6: COUNTRY OF ORIGIN OF IDENTIFIED OWNERS BUYING PROPERTIES IN THE UK

Notes: This figure shows the distribution of country of residence of identified beneficial owners of companies holding real estate in England and Wales in January 2018. Data from companies holding properties in England and Wales comes from the Overseas Companies Ownership Dataset (OCOD) maintained by the Land Registry. We identify beneficial owners of these companies matching the OCOD with the Leaks Data (Bahamas Leaks, Offshore Leaks, Panama Papers, Pandora Papers, Paradise Papers, and OpenLux and CNBIOM Leaks) and the Register of Overseas Entities (ROE). We match companies based on their name and country of incorporation. In Panel A, we show results of the matching process with the Leaks data only (which include Bahamas Leaks, CNBIOM data, Offshore Leaks, OpenLux data, Panama Papers, Pandora Papers, Paradise Papers). Raw results are presented in red. We also present a specification in which we use a software predicting the most likely nationality according to the name to assign owners from tax havens to their most likely country of origin. These estimates are presented in blue. In Panel B, we present the results of the matching process with the ROE data only. Raw results are presented in red. Because the ROE data provide more information than the Leaks, we reallocate owners from tax havens using their country of citizenship. The matching is based on the property stock of the UK as of January 2018, and all values are expressed in dollars (January 2018's exchange rate).

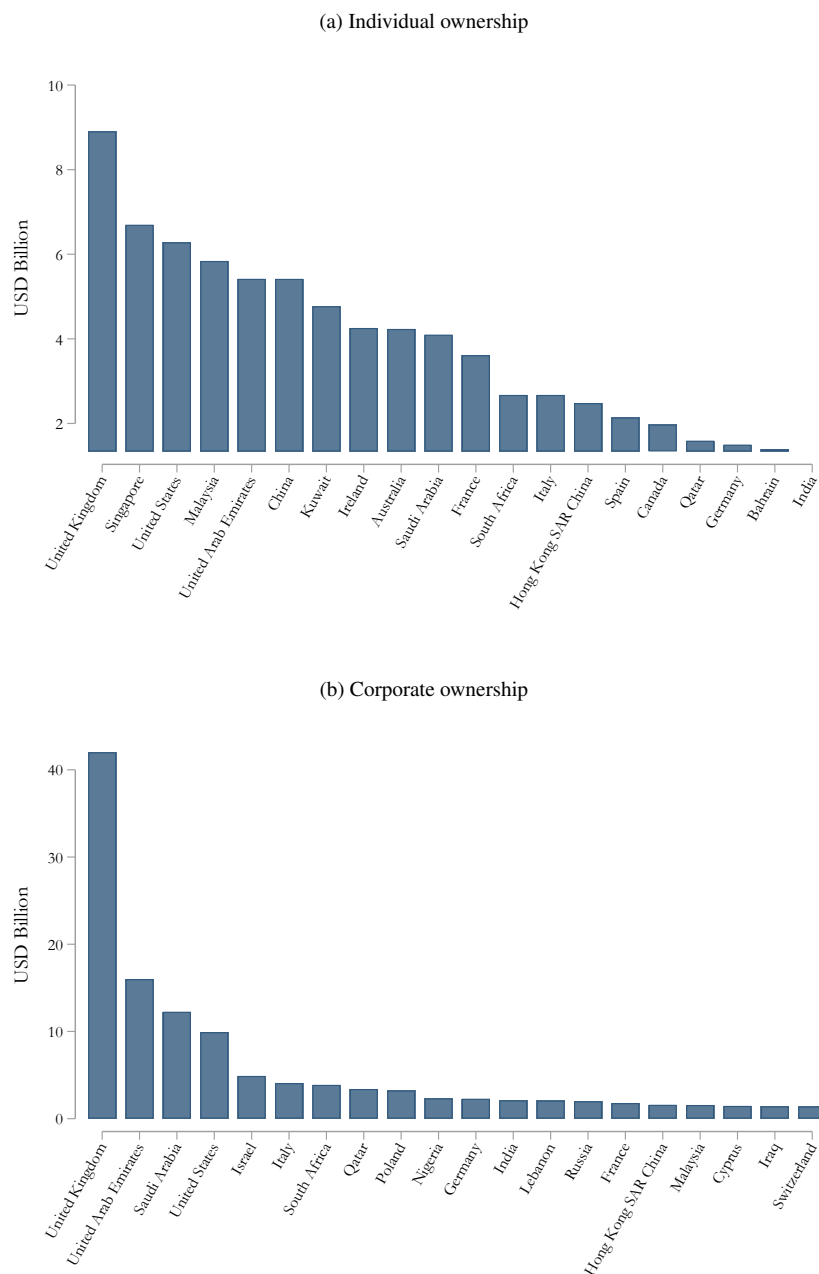


FIGURE A.7: FOREIGN REAL ESTATE OWNERSHIP IN THE UK, BY RESIDENCE COUNTRY OF THE ULTIMATE OWNER

Notes: This figure displays estimates of the size of foreign owned real estate wealth in England and Wales by country of ultimate ownership for the top 20 owners. We compute these estimates using the Overseas Companies Ownership Dataset (for properties held by foreign companies) and data from an FOI request made by the Centre for Public Data (for properties held by individuals with foreign correspondence addresses). The estimates are for January 2018. Panel A presents estimates of ultimate ownership for properties held by individual with foreign correspondence addresses. We reallocate addresses in tax havens to the most probable “true” residence country of the owners using Register of Overseas Entities data. The reallocation procedure is detailed in the text. Panel B shows estimates of ultimate ownership of properties held through foreign companies. It is estimated by matching the OCOD with the Register of Overseas Entities and the Leaks data. we allocate the companies we did not find in the ROE or the Leaks according to the geographic distribution of UK real estate ownership estimated through matches with the Leaks data. Estimates are for January 2018 and all values are expressed in dollars (January 2018’s exchange rate).

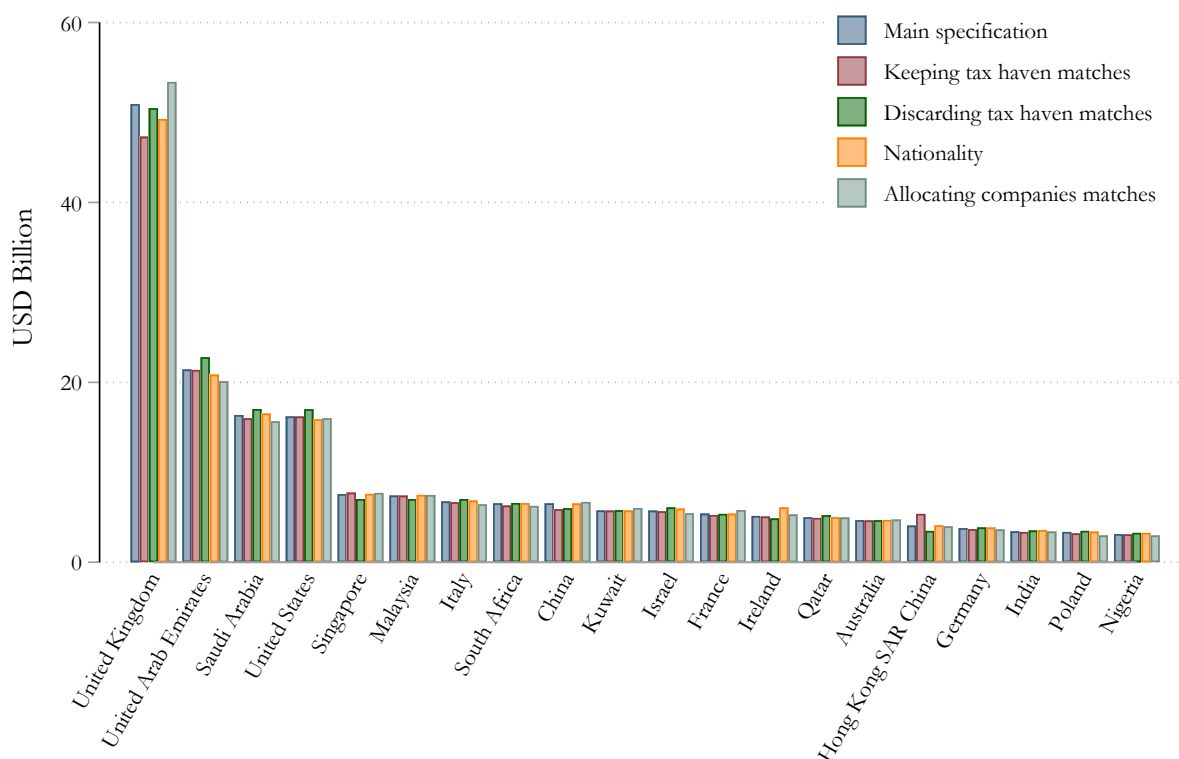


FIGURE A.8: COUNTRY OF UK REAL ESTATE OWNERS, BY ULTIMATE OWNERSHIP - ROBUSTNESS TO RESIDENCE COMPUTATION

Notes: This figure displays different estimates of the size of foreign owned real estate wealth by country of ultimate ownership for the top 20 owners. We compute these estimates using the Overseas Companies Ownership Dataset (for properties held by foreign companies) and data from an FOI request made by the Centre for Public Data (for properties held by individuals with foreign correspondence addresses). Ultimate ownership country refers to the property ultimate owner's country of residence. It includes both properties held through foreign companies, and properties held by individuals with a foreign correspondence address. The main specification (in blue) is showed in Figure VI. It is based on the distribution of owners identified through the Register of Overseas Entities and the beneficial ownership data, where individuals linked to tax havens are allocated to their most probably country of residence based on either their country of citizenship or their name. The first alternative specification (in red) keep these tax havens as the ultimate country of residence of the owners, instead of predicting the residence. The second specification (in green) discard these matches altogether. The third specification (in yellow) takes the country of citizenship as the country of origin of company owners (or the most probably country of citizenship based on names, for matches with the leaks data). The last specification (in grey) includes matches in the OCOD where the ultimate beneficial owner is a company. To distribute the value of companies matched to corporate owners, we proceed in the following way. We leverage the full Register of Overseas Entities data, which provides information on the country of residence of a significant share of foreign companies holding properties in the United Kingdom. For each country of incorporation h of these companies, we compute the distribution of owners residing in each country c . Then, we allocate these weights to the corporate beneficial owners identified in the matching process. For example, in the Register, we find that 18.9% of individuals owning a company incorporated in Hong Kong are residents of China. Then, when the matching process shows that the beneficial owner of a company is another company incorporated in Hong Kong, we allocate 18.9% of the value of the corresponding property to China. Estimates are for January 2018 and all values are expressed in dollars (January 2018's exchange rate).

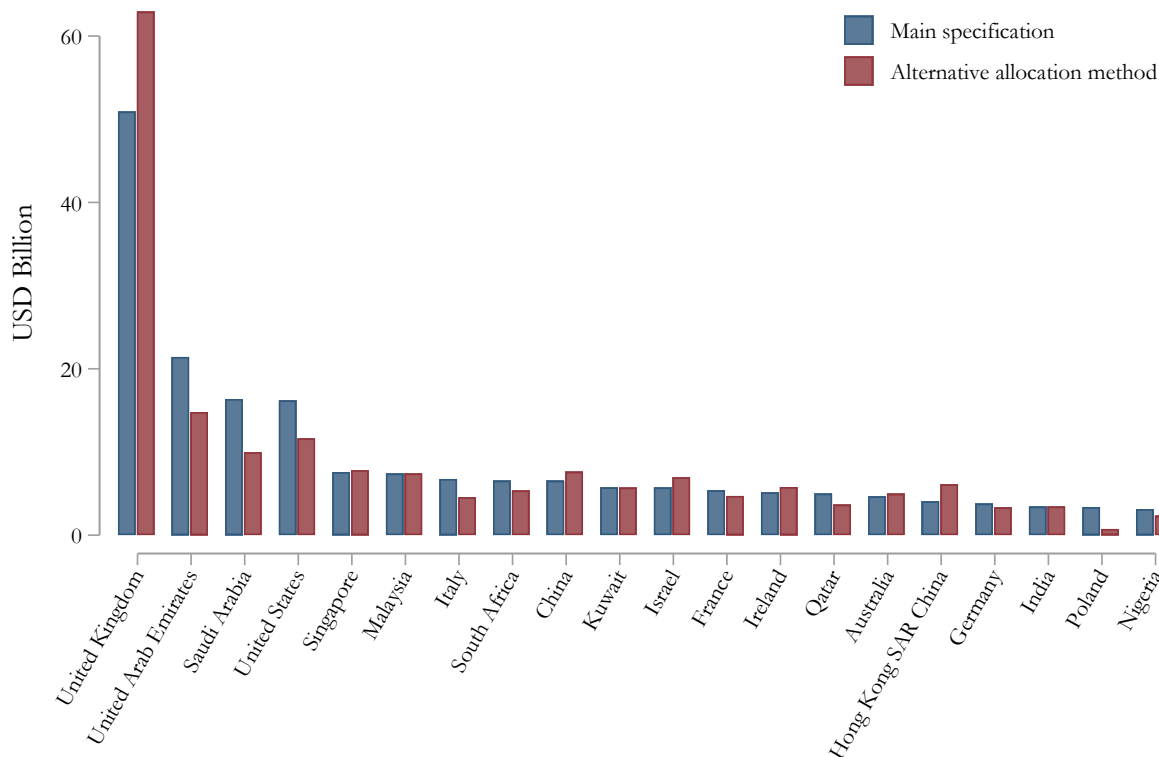


FIGURE A.9: COUNTRY OF UK REAL ESTATE OWNERS, BY ULTIMATE OWNERSHIP - ROBUSTNESS TO ALLOCATION METHOD COMPUTATION

Notes: This figure displays estimates of the size of foreign owned real estate wealth by country of ultimate ownership for the top 20 owners. We compute these estimates using the Overseas Companies Ownership Dataset (for properties held by foreign companies) and data from an FOI request made by the Centre for Public Data (for properties held by individuals with foreign correspondence addresses). Ultimate ownership country refers to the property ultimate owner's country of residence. It includes both properties held through foreign companies, and properties held by individuals with a foreign correspondence address. The main specification (in blue) is showed in Figure VI. It is based on allocating the unmatched companies using the country distribution found in the subsample identified thanks to the Leaks data. This means that as in the sample of OCOD companies matched to the leaks, 26.7% of the real estate wealth is owned by the United Kingdom, we allocate 26.7% of the unidentified wealth to the United Kingdom. In the alternative allocation method (in red), we exploit information on country distribution identified through the leaks for each country of incorporation of companies. This means that as according to matches between the OCOD and the Leaks e.g. 16% of owners of companies incorporated in the Bahamas are from the United Kingdom, we allocate 16% of the unidentified sample of Bahamas companies to the United Kingdom. We repeat this exercise for each country of incorporation. Estimates are for January 2018 and all values are expressed in dollars (January 2018's exchange rate).

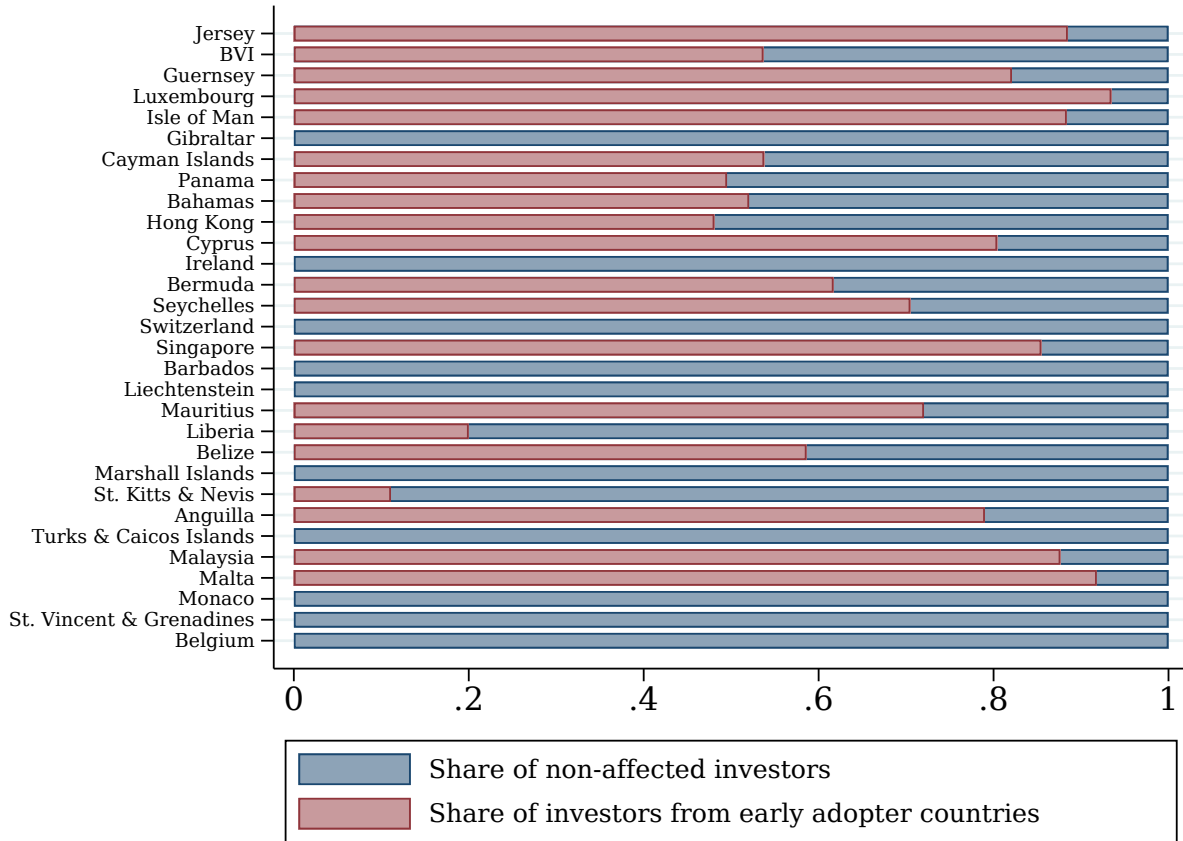


FIGURE A.10: DISTRIBUTION OF CRS EXPOSURE AMONGST THE TOP 30 TAX HAVENS.

Notes: This figure shows the distribution of exposure to the Common Reporting Standard for the 30 tax havens that channel the most UK real estate investment. Exposure to the CRS is defined as the share of investors affected by the CRS. It is computed using the Leaks data and the OpenLux data. We build it by computing, for each tax haven, the total share of individuals owning companies in that haven who reside in a country committing to the CRS early. A treatment intensity of e.g. 0.5 indicates that 50% of all company owners in the tax haven reside in countries participating to the Joint Announcement. Tax havens are ordered according to their importance in terms of flows of investments in the English and Welsh real estate market (e.g. Jersey is the most important buyer and it comes first).

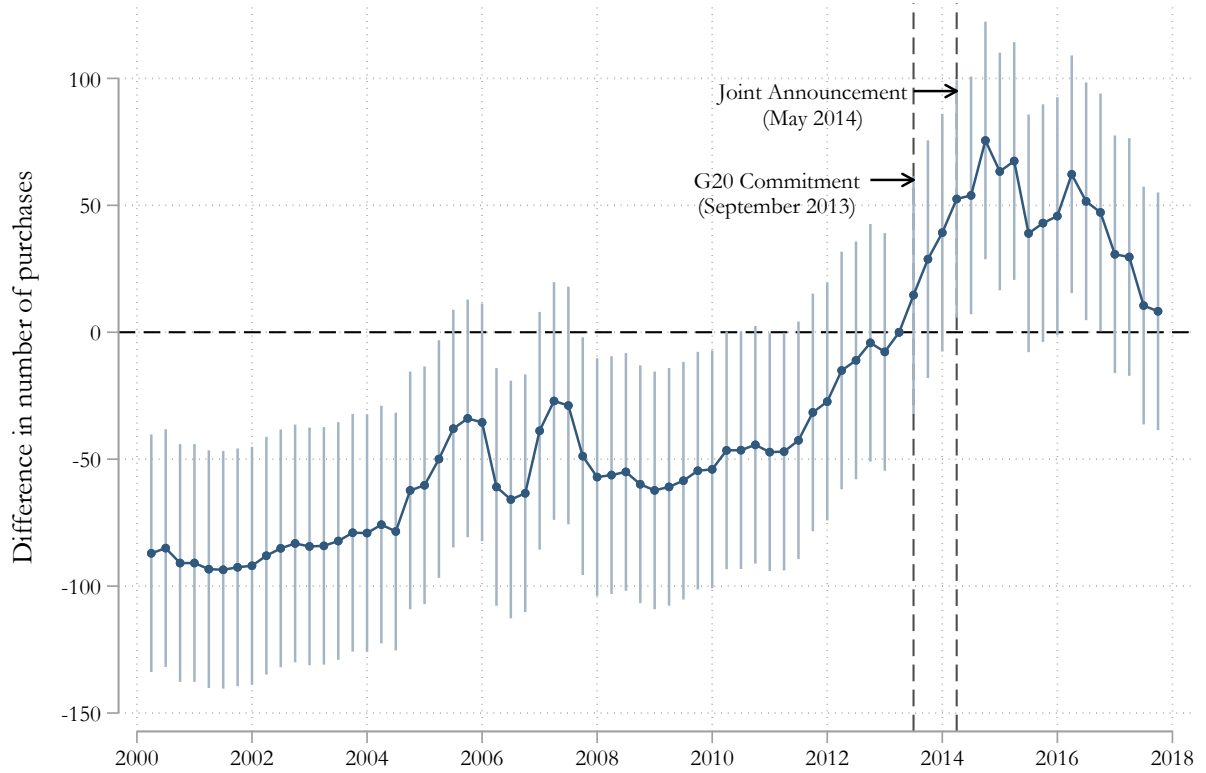


FIGURE A.11: IFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - NUMBER OF TRANSACTIONS

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. The outcome variable –expressed in 3-quarters moving average– is the number of properties purchased each quarter. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

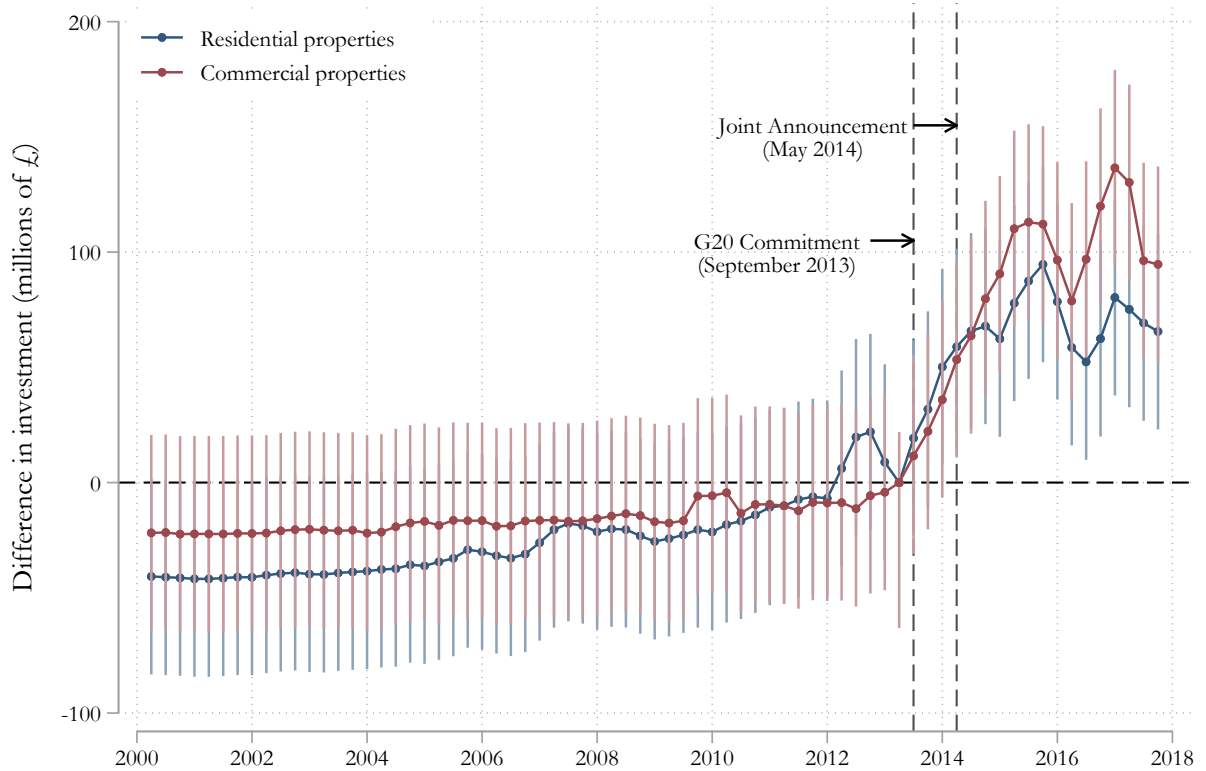


FIGURE A.12: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - BY CLASS OF PROPERTY

Notes: This figure plots the path of estimated β^{qt} and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 2. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. The outcome variable is investment in million Pounds, expressed in 3-quarters moving averages. Estimates are shown separately for residential and for commercial properties. The pre-CRS coefficients β^{2013q2} are normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

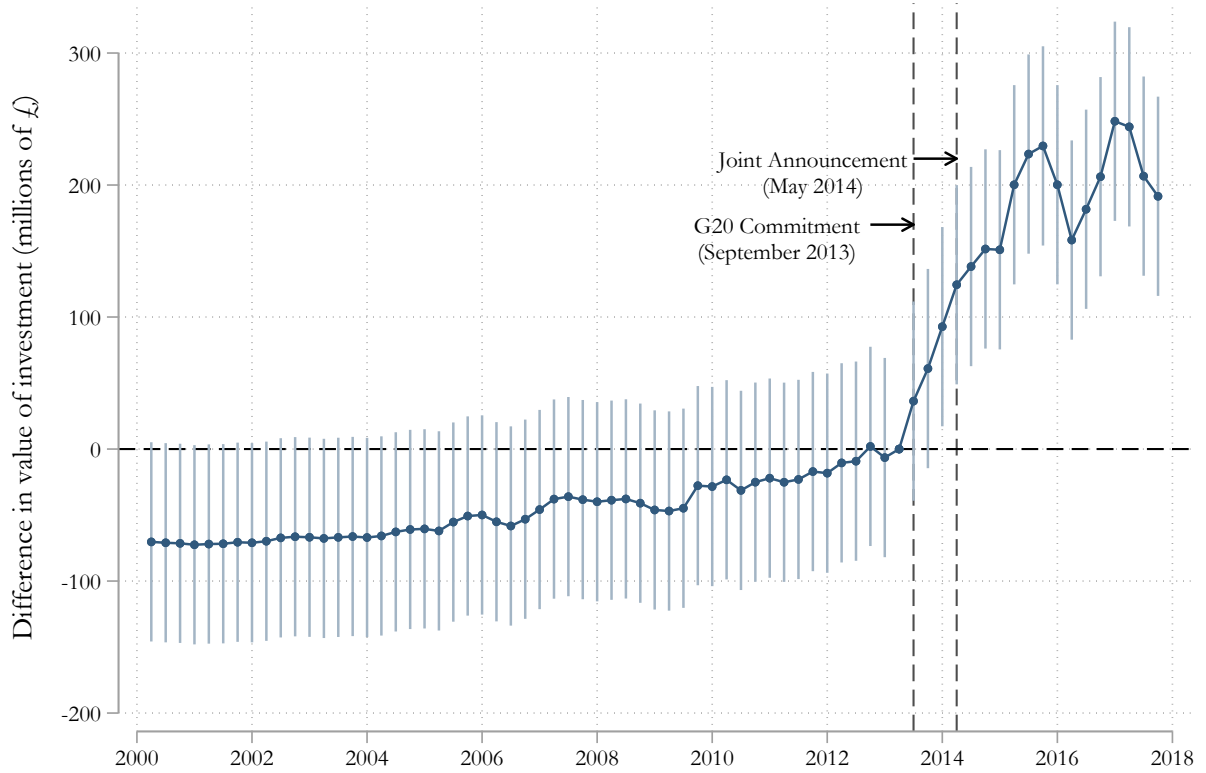


FIGURE A.13: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - EXPOSURE BASED ON POST-AEOI SHELL COMPANIES

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. Measure of exposure to the CRS is computed based on shell companies incorporated only after the second quarter of 2013. The outcome variables is investment in million pounds, expressed in 3-quarters moving average. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

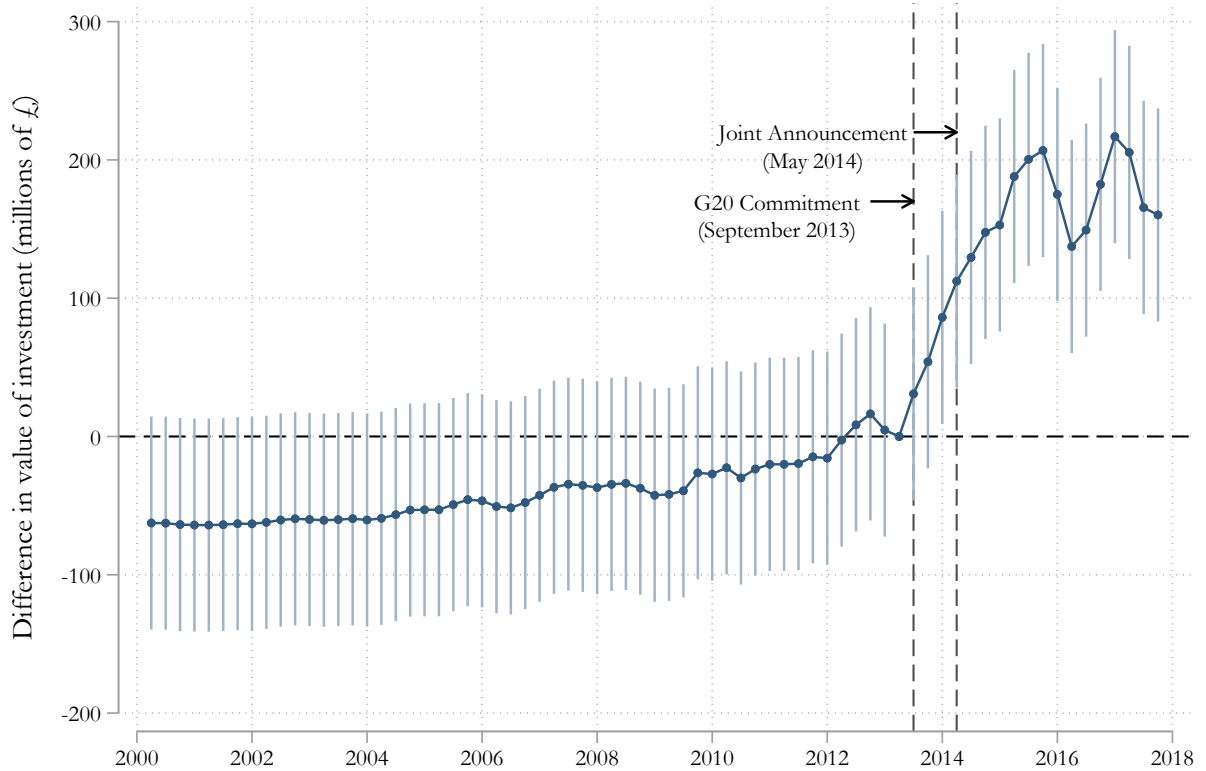


FIGURE A.14: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - EXPOSURE BASED ON BIS DATA

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. Measure of exposure to the CRS is computed based on cross-border bank deposits record by the Bank for International Settlements (BIS). The outcome variables is investment in million pounds, expressed in 3-quarters moving average. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

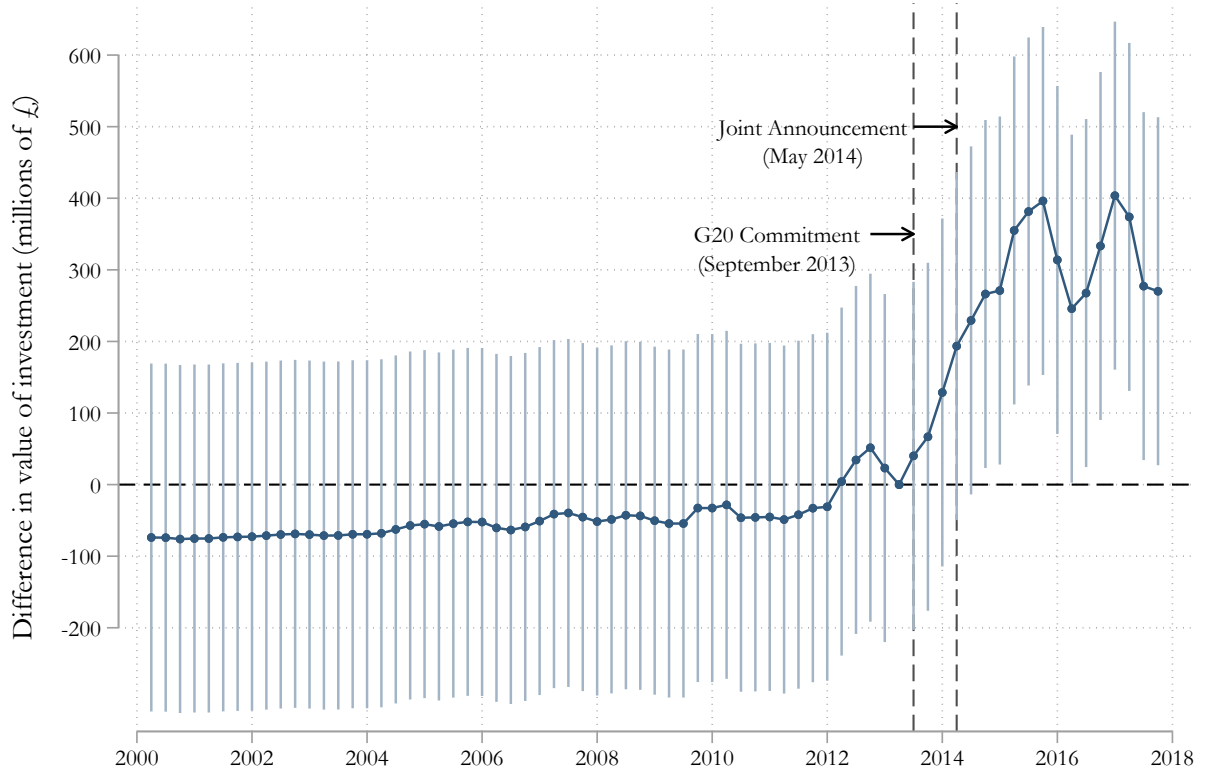


FIGURE A.15: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - LARGE TAX HAVENS ONLY

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. We restrict the sample to large tax havens i.e. tax havens in which more than 10 corporations have been incorporated before the third quarter of 2013. The outcome variables is investment in million pounds, expressed in 3-quarters moving average. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 24.

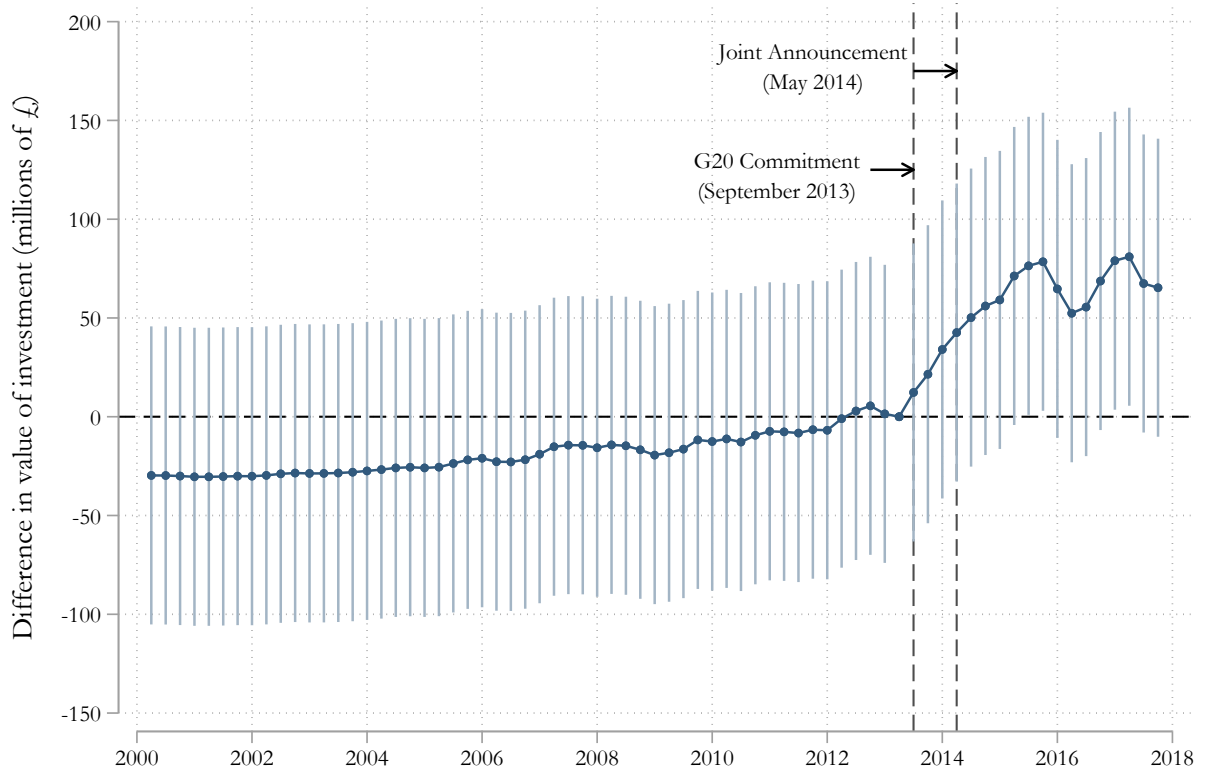


FIGURE A.16: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - EXPOSURE BASED ON REGISTER OF OVERSEAS ENTITIES DATA

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. Measure of exposure to the CRS is computed based on company ownership data from the Register of Overseas Entities. The outcome variable is investment in million pounds, expressed in 3-quarters moving average. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

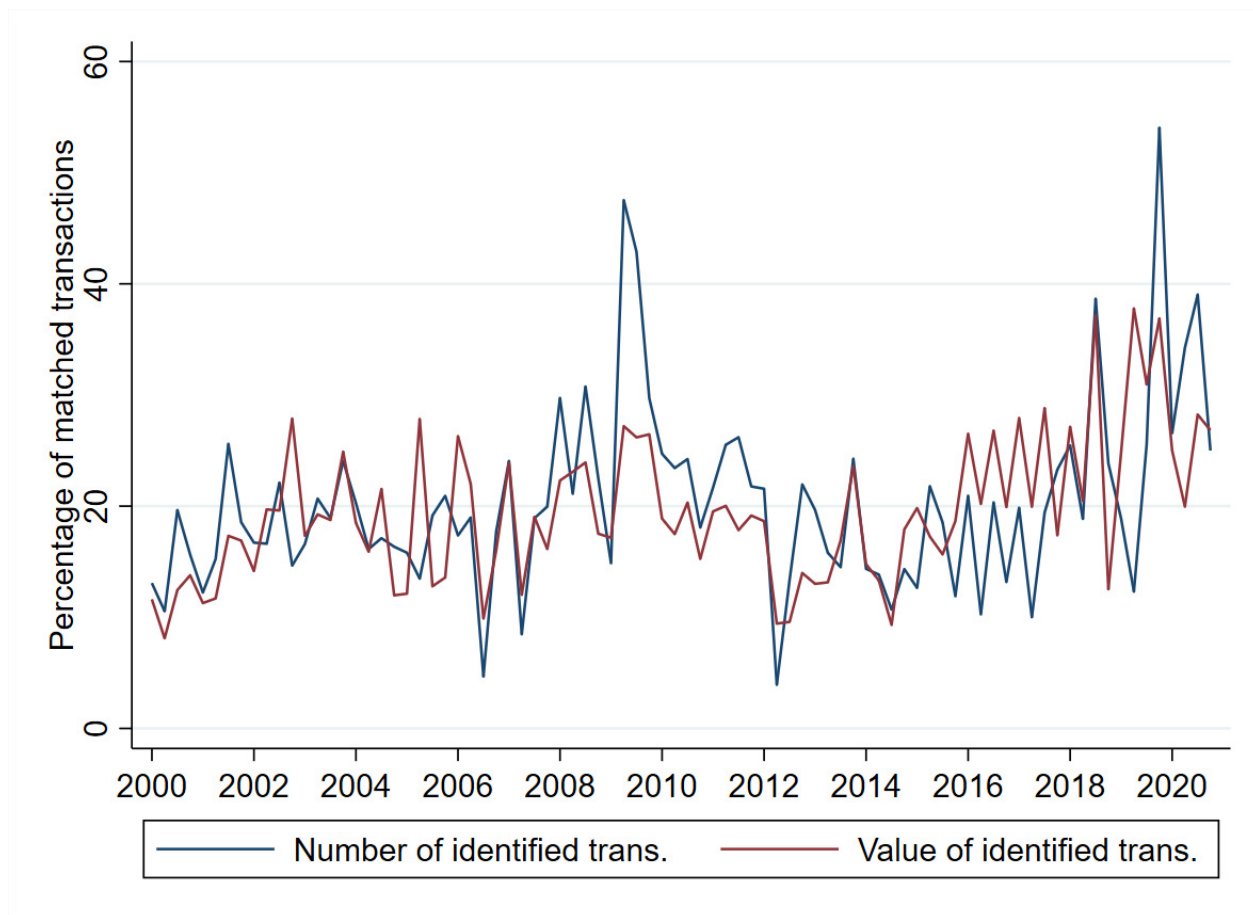
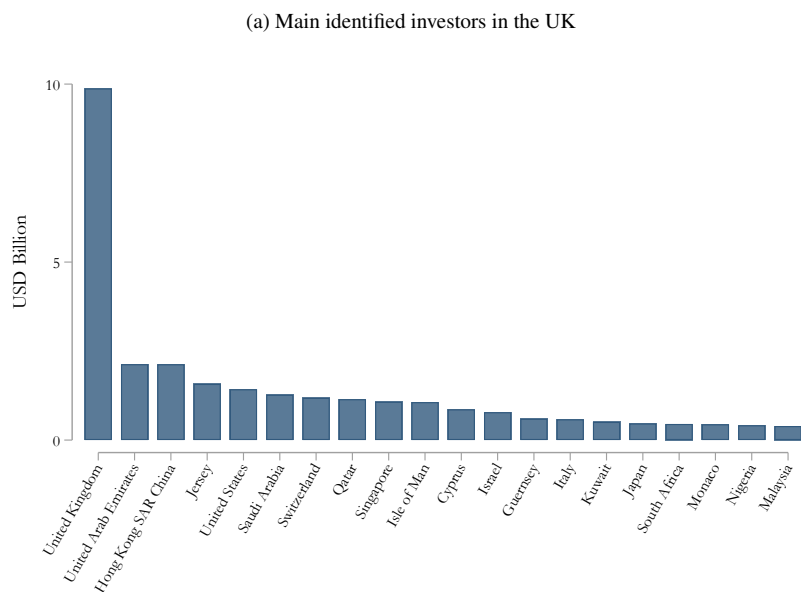


FIGURE A.17: RATE OF MATCHING WITH THE LEAKS AND THE ROE OVER TIME

Notes: This figure presents the matching rates of the Overseas Companies Ownership Dataset with the Leaks, the OpenLux and the Register of Overseas Entities over time. We match companies based on their name, their country of incorporation, and their date of creation. We discard matched companies whose owner is another company. We present the matching rate both for the number of transactions per period, and for the value of these transactions.



(b) Main identified investors, allocating owners in tax havens to their most probable residence country

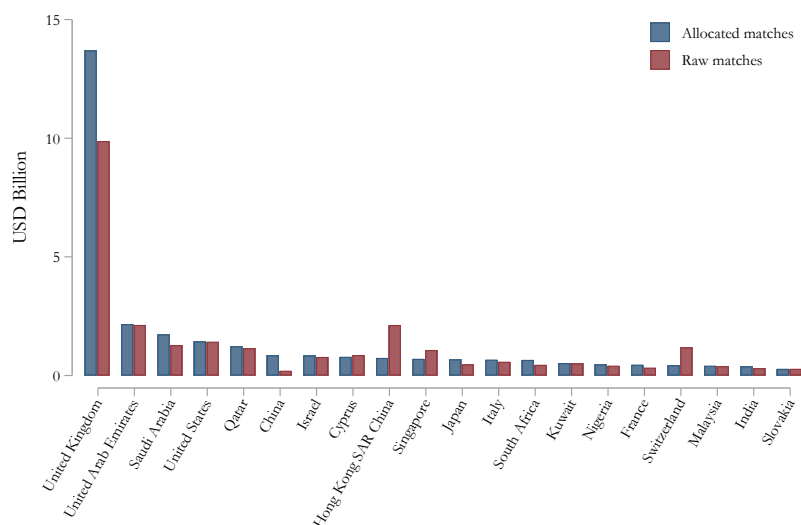


FIGURE A.18: COUNTRY OF ORIGIN OF IDENTIFIED OWNERS BUYING PROPERTIES IN THE UK, 2000-2020 PURCHASES

Notes: This figure shows the distribution of country of residence of identified beneficial owners of companies holding real estate in England and Wales in January 2018. Data from companies holding properties in England and Wales comes from the Overseas Companies Ownership Dataset (OCOD) maintained by the Land Registry. We identify beneficial owners of these companies matching the OCOD with the Leaks Data (Bahamas Leaks, Offshore Leaks, Panama Papers, Pandora Papers, Paradise Papers, and OpenxLux and CNBIOM Leaks) and the Register of Overseas Entities (ROE). We match companies based on their name, country of incorporation and date of incorporation. In Panel A, we show the raw results we obtain after the matching process, using the address country as declared in the ROE and country linked to the beneficial owner as per the Offshore Leaks. In Panel B, we allocate all owners from tax havens we found in Panel A to their most probable country of residence. For matches with the ROE, we do so by selecting the country of citizenship of the individual instead of the address country. This information, however, is not available in the Leaks data. As a result, we use a software predicting the most likely nationality according to the name to assign a country of residence to owners linked to a tax haven. The matching is based on all properties bought over the 2000-2020 period and all values are expressed in dollars (January 2018's exchange rate).

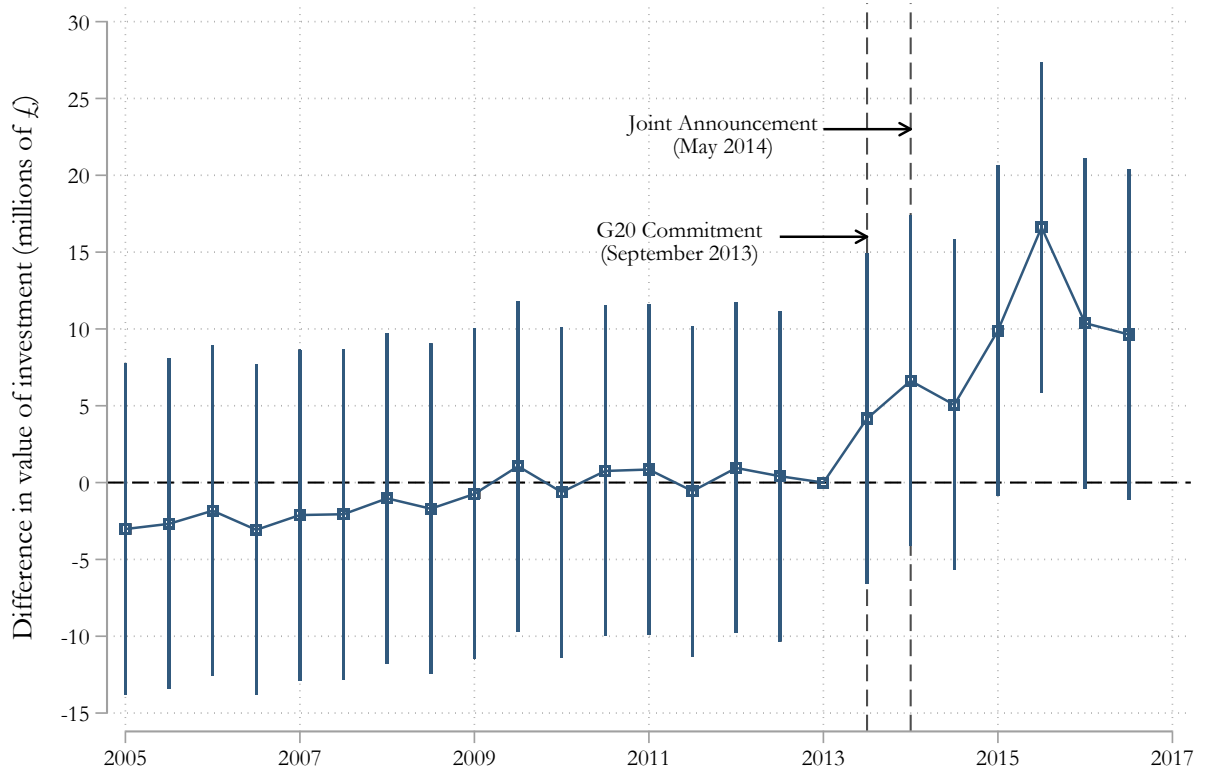


FIGURE A.19: DIFFERENCE-IN-DIFFERENCE WITH BINARY TREATMENT VARIABLE WHEN THE OWNER IS IDENTIFIED

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 3. In this version of the equation, we use a binary treatment variable taking 1 if a country commits to the Common Reporting Standard in 2013 or in 2014. The control group is composed of countries adopting the CRS later, or never adopting it. The outcome variable is investment in million Pounds. The pre-CRS coefficient β^{2013h1} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. The graphs start in 2005 for visibility, but the results are unchanged when including the 2000-2005 period.

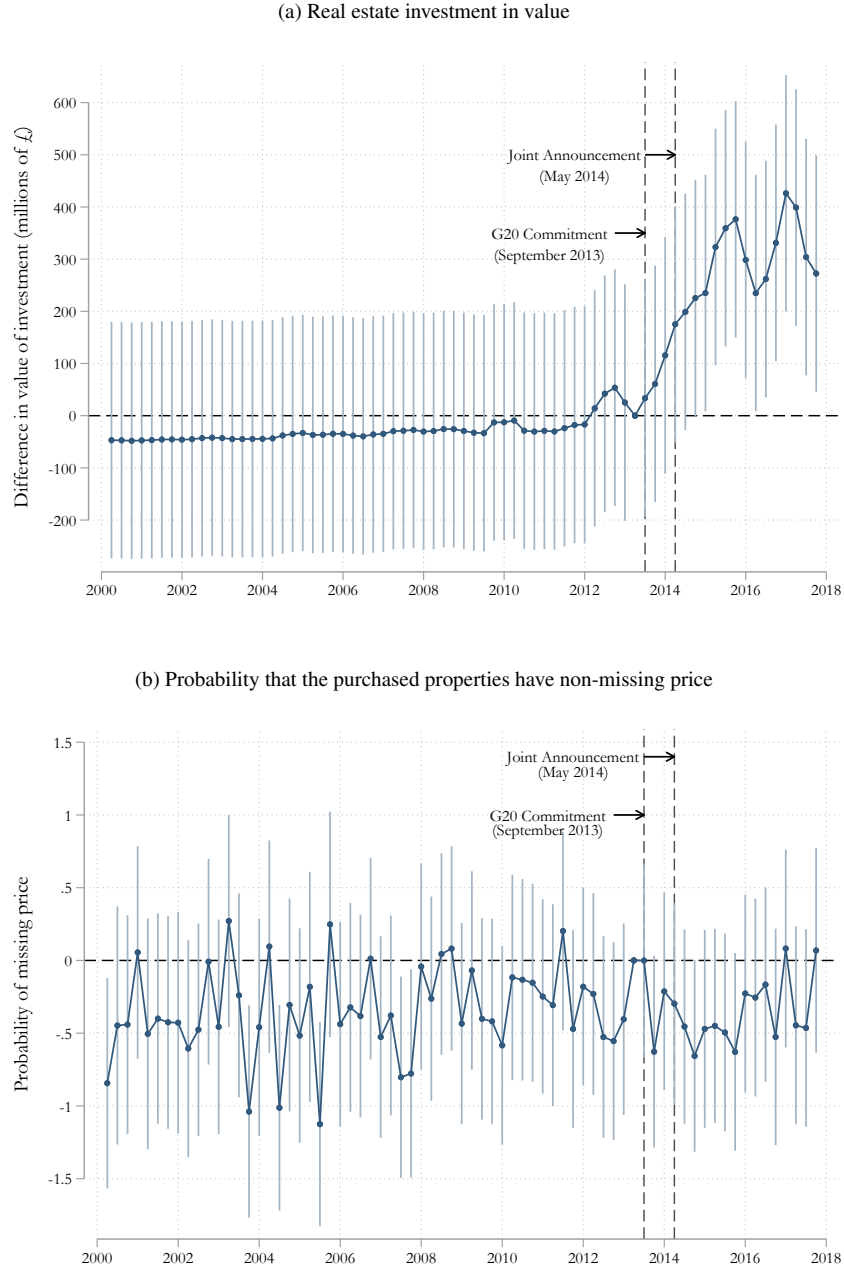


FIGURE A.20: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - ROBUSTNESS OF PRICE INFERENCE

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. Panel A shows real estate investment –expressed in 3-quarters moving average – and restricts the estimation sample to transactions with non-missing price information. Panel B shows estimates from a specification of equation 1 where the outcome variable is the probability that a transaction price is missing in our full OCOD estimation sample. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 50.

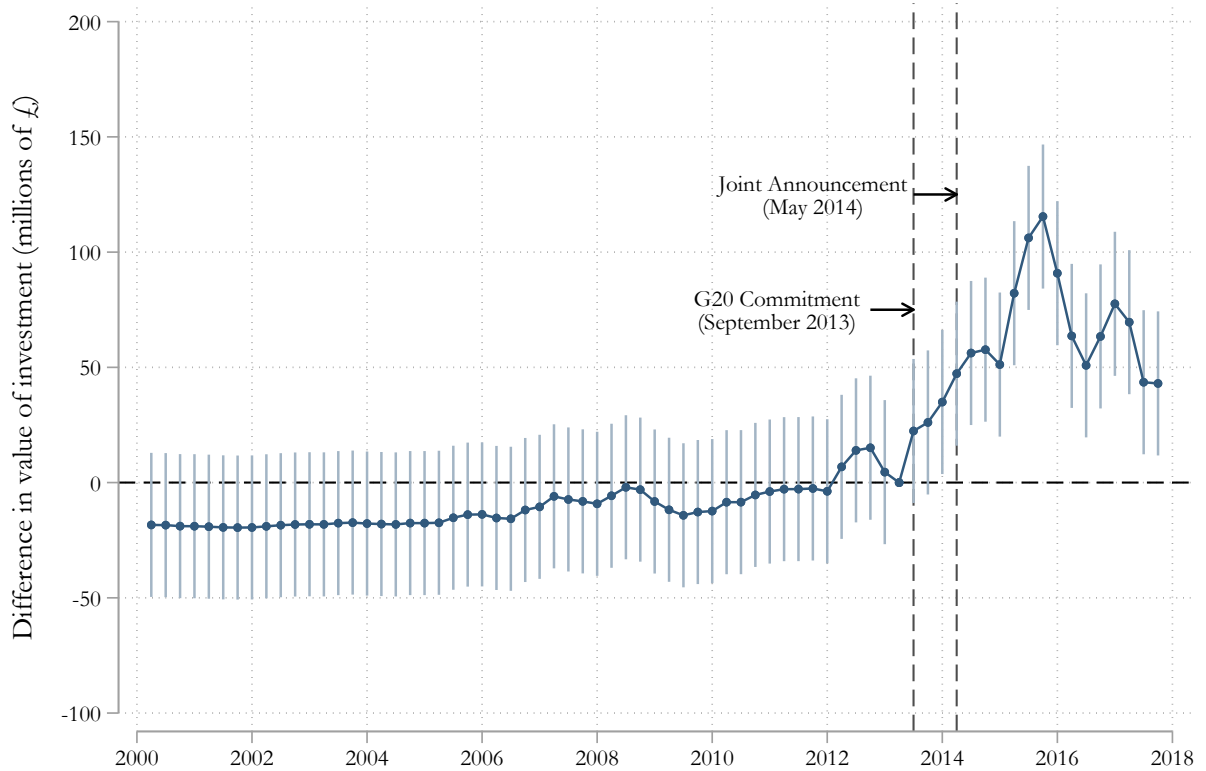


FIGURE A.21: DIFFERENCE-IN-DIFFERENCES WHERE TREATMENT IS CONTINUOUS - EXCLUDING JERSEY, THE BRITISH VIRGIN ISLANDS AND GUERNSEY

Notes: This figure plots the path of estimated β^q and their 95 percent confidence interval band from the difference-in-differences model summarized by equation 1. In this equation, the treatment is continuous and captures the exposure of each tax haven to the Common Reporting Standard. We exclude from the sample purchases made by companies incorporated in Jersey, the British Virgin Islands or Guernsey i.e. the three largest buyers in our sample. The outcome variables is investment in million pounds, expressed in 3-quarters moving average. The pre-CRS coefficient β^{2013q2} is normalized to zero such that estimates can be interpreted relative to before commitment to the CRS. The estimation is based on investments made by tax havens in England and Wales, as captured in the Overseas Companies Ownership Dataset. Number of tax havens: 24.

TABLE A.1: NUMBER OF COMPANIES AND ULTIMATE BENEFICIAL OWNERS BY SOURCE IN THE BENEFICIAL OWNERSHIP DATASET

Data source	Period covered	Number of companies	Number of unique beneficial owners
Bahamas Leaks	1919-2016	175,888	6
CNBIOM	2007-2019	1,406	927
OpenLux	1907-2020	261,249	70,795
Offshore Leaks	1918-2010	105,516	75,948
Panama Papers	1936-2015	213,634	238,055
Pandora Papers	1905-2019	29,221	41,745
Paradise Papers	1865-2017	290,086	133,555
Total		1,077,000	561,031

Notes: This table details the characteristics of the seven corporate ownership datasets we exploit in our analysis. For each of the datasets, it presents the period covered, the number of companies and the number of unique beneficial owners we get information on.

TABLE A.2: CHARACTERISTICS OF THE PROPERTIES PURCHASED

Variable	Descriptive sample			Estimation sample		
	Full	Commercial	Residential	Full	Commercial	Residential
All transaction prices (GBP)	894,310	1,904,632	649,753	1,123,968	2,039,686	851,483
Observed transaction price (GBP)	1,446,028	4,046,359	990,966	1,768,254	4,372,988	1,243,283
London	47%	48%	47%	48%	43%	49%
Freehold	56%	68%	53%	52%	69%	47%
Share of residential	81%	-	-	77%	-	-
Share of indicated price	44%	34%	47%	54%	39%	58%
Number of transactions	93,332	18,189	75,143	103,848	23,815	80,033

Notes: This table displays the characteristics of the properties of the Overseas Companies Ownership Dataset we use to describe the offshore real estate market in the UK (Descriptive sample) and to analyze the effect of the CRS on this market (Analysis sample). The first column presents the characteristics of our full sample, while columns “Commercial” and “Residential” show the characteristics of respectively commercial and residential properties. The row “All transaction prices (GBP)” gives the average price (observed + predicted) in the sample. The row “Observed transaction price (GBP)” presents the observed price, either directly (OCOD data) or indirectly (PPD data). The row “London” displays the share of transactions taking place in the Greater London area. The row “Freehold” shows the percentage of freehold properties in each categories, the rest of the transactions corresponding to Leasehold titles. The row “Share of residential” displays the percentage of residential transactions in the two samples. The last row “Share of indicated price” presents the percentage of transactions for which we recover a purchase price, either through the OCOD or the PPD. The rest of the prices are estimated as described in Appendix section B.1.

TABLE A.3: LIST OF CRS EARLY ADOPTERS

Country	Declaration on Automatic Exchange of Information	Joint Announcement
Anguilla		X
Argentina	X	X
Australia	X	
Austria	X	
Belgium	X	X
Bermuda		X
Brazil	X	
British Virgin Islands		X
Bulgaria		X
Canada	X	
Cayman Islands		X
China	X	
Chile	X	
Colombia	X	X
Costa Rica	X	
Croatia		X
Cyprus		X
Czech Republic	X	X
Denmark	X	X
Estonia	X	X
Finland	X	X
Faroe Islands		X
France	X	X
Germany	X	X
Gibraltar		X
Greece	X	X
Greenland		X
Guernsey		X
Hungary	X	X
Iceland	X	X
India	X	X
Indonesia	X	
Ireland	X	X
Isle of Man		X
Israel	X	
Italy	X	X
Japan	X	
Jersey		X
Korea	X	X
<i>Continued on next page</i>		

Continued from previous page

Country	Declaration on Automatic Exchange of Information	Joint Announcement
Latvia	x	x
Liechtenstein		x
Lithuania	x	x
Luxembourg	x	
Malaysia	x	
Malta		x
Mauritius		x
Mexico	x	x
Montserrat		x
Netherlands	x	x
New Zealand	x	
Norway	x	x
Poland	x	x
Portugal	x	x
Romania		x
San Marino		x
Saudi Arabia	x	
Seychelles		x
Singapore	x	
Slovak Republic	x	x
Slovenia	x	x
South Africa	x	x
Spain	x	x
Sweden	x	x
Switzerland	x	
Turkey	x	
Turks and Caicos Islands		x
United Kingdom	x	x

Notes: This table shows the list of countries participating to the Declaration on Automatic Exchange of Information (May 2014) and to the Joint Announcement (March 2014). Note that the United States signed the May 2014 Declaration but do not participate to the Common Reporting Standard, and are therefore excluded from this table. We add The Faroe Islands, Greenland, Korea, Mauritius, San Marino and the Seychelles to the list of countries signing the Joint Announcement as they appear in the [Joint Statement by the early adopter group](#) in October 2014.

TABLE A.4: ESTIMATION OF ASSET SHIFTING RESPONSES TO THE CRS

	(1)	(2)	(3)
Paper	Estimates	Wealth decrease (USD billion)	Asset shifting
Casi et al. (2020)	11.5%	534	8%
Menkhoff and Miethe (2019)	31.8%	1463	3%
Beer et al. (2019)	29.6%	1362	3%
O'Reilly et al. (2019)	11%	506	9%

Notes: This table compares the effect of the CRS on real estate investments we estimate in our paper to the amount of offshore financial wealth that left tax havens due to the transparency shock as estimated in the literature. Column “Estimates” presents estimates of the reduction in offshore financial wealth caused by the CRS. The estimate for [Casi et al. \(2020\)](#) comes from column 1, table 4 from their paper, for [Menkhoff and Miethe \(2019\)](#), from column 2, table 5 and for [Beer et al. \(2019\)](#), from column “Model 4”, table 3. Column “Wealth decrease” presents a computation of the stock of offshore financial wealth that flew out of tax havens from 2013 because of the CRS. It is computed using estimates of the stock of offshore financial wealth held by CRS early-adopting countries in tax havens measured in [Alstadsæter et al. \(2018\)](#), multiplied by the estimated CRS financial effect of column (1). Column “Asset shifting” computes the ratio of our real estate effect for the UK (\$45 billion) over the offshore wealth decrease of column (2).

TABLE A.5: LISTS OF TAX HAVENS USED IN THE ANALYSIS.

Country	Menkhoff and Miethe (2019)	Hines and Rice (1994)	Consensus list
Andorra	x	x	
Anguilla	x	x	
Antigua and Barbuda	x	x	x
Aruba	x		
Austria	x		
Bahamas	x	x	x
Bahrain	x	x	x
Barbados	x	x	x
Belgium	x		
Belize	x	x	x
Bermuda	x	x	x
British Virgin Islands	x	x	x
Cayman Islands	x	x	x
Chile	x		
Cook Islands	x	x	x
Costa Rica	x		
Cyprus	x	x	x
Dominica	x	x	
Gibraltar	x	x	x
Grenada	x	x	x
Guernsey	x	x	x
Hong Kong SAR China	x	x	x
Ireland	x	x	
Isle of Man	x	x	x
Jersey	x	x	x
Jordan	x	x	
Lebanon	x	x	
Liberia	x	x	x
Liechtenstein	x	x	x
Luxembourg	x	x	x
Macao SAR China	x	x	
Malaysia	x		
Maldives	x	x	
Malta	x	x	
Marshall Islands	x	x	
Mauritius	x		
Monaco	x	x	
Montserrat	x	x	x
Nauru	x		
Netherlands Antilles	x	x	x
Niue	x		
Panama	x	x	x
Samoa	x		
San Marino	x		
Seychelles	x		
Singapore	x	x	x
St. Kitts and Nevis	x	x	x
St. Lucia	x	x	
St. Vincent and Grenadines	x	x	x
Switzerland	x	x	x
Tonga	x		
Trinidad and Tobago	x		
Turks and Caicos Islands	x	x	x
U.S. Virgin Islands	x		
Uruguay	x		
Vanuatu	x	x	x

Notes: This table shows the list of tax havens we use in our analysis (column (1) [Menkhoff and Miethe \(2019\)](#)) and compares it to the list used by [Hines and Rice \(1994\)](#), which excludes 18 countries categorized as tax havens by [Menkhoff and Miethe \(2019\)](#). Column (3) shows a “consensus” list of tax havens. This list is compiled by [Menkhoff and Miethe \(2019\)](#) by choosing the 29 countries that all appear in most recent studies on tax evasion (see [Menkhoff and Miethe \(2019\)](#), appendix A.2).

B Data construction

B.1 Overseas Companies Ownership Dataset

The primary data source in our paper is the Overseas Companies Ownership Dataset (OCOD). Making use of this dataset requires a heavy data processing work that we detail here.

Obtaining a dataset at the transaction level

First, the raw OCOD data is at the stock level, meaning that if a property is bought by a company, for example, *Cyprus Investment Ltd* in April 2013 and sold in September 2019, it will appear as owned by *Cyprus Investment Ltd* in all registers between November 2015 and August 2019. If the property is sold to a UK company or a private individual, it will not appear in the subsequent snapshots. Conversely, if the property is sold to another foreign company, the new entity becomes the designated owner in the OCOD registers from September 2019 onward. While registration is mandatory, there may be a lag between the actual transaction date and its inclusion in the OCOD register because of registration delays. We build a dataset at the transaction level by combining all the stock-level registers.

Parsing properties addresses

If a single buyer purchases multiple properties, the register may list all the properties in the same entry, resulting in nested properties where the address information refers to a group of properties, such as "113, 115, 117 Morning Lane, London (E9 6LH)". To address this, we use an algorithm developed by [Bourne et al. \(2022\)](#) designed to parse and clean the OCOD dataset. This algorithm transforms the raw data into an enhanced dataset where each observation corresponds to a single property purchased on a specific date. Table B.1 provides an example of the output from this method for highly nested addresses. The table shows how the method enables us to convert a row that corresponds to the purchase of a bundle of properties with multiple nested addresses into multiple rows that each represent a unique address. For instance, in the OCOD data, one of the address entries reads as "flats 1-56, counter house, park street, london (sw6 2fl), flats 1-84, compass house, park street, london (sw6 2fb) and flats 1-143, doulton house, 11 park street, london (sw6 2ft)". Using the algorithm developed by [Bourne et al. \(2022\)](#), this single address entry can be transformed into a 283 distinct addresses, each corresponding to a flat ID located in either Counter House, Compass House, or Doulton House on Park Street, London. However, some addresses in the OCOD data are incomplete, and remain so after the data parsing. Thus, we only keep properties where the street name and street number are non-missing, or when the street name and building name is non-missing.

TABLE B.1: EXAMPLE OF AN ENHANCED ADDRESS USING [BOURNE ET AL. \(2022\)](#) ALGORITHM

Unit id	unit type	building name	street number	street name	postcode	city
1		counter house		park street	SW6 2FL	london
...		counter house		park street	SW6 2FL	london
56		counter house		park street	SW6 2FL	london
1		compass house		park street	SW6 2FB	london
...		compass house		park street	SW6 2FB	london
84		compass house		park street	SW6 2FB	london
1		doulton house	11	park street	SW6 2FT	london
...		doulton house	11	park street	SW6 2FT	london
143		doulton house	11	park street	SW6 2FT	london

Notes: This table displays the outcome of the address parsing process developed by [Bourne et al. \(2022\)](#) and adapted for our dataset. The parsing process converts a single, composite address: "flats 1-56, counter house, park street, london (sw6 2fl), flats 1-84, compass house, park street, london (sw6 2fb) and flats 1-143, doulton house, 11 park street, london (sw6 2ft)" into 283 unique addresses.

Importantly, we exclude from our analysis instances where a single proprietor purchases more than 100 properties on the same date, removing about 39,000 transactions from our sample. This is because although it is mandatory for buyers to register their purchases with the Land Registry, there is no formal time limit to do so. Therefore, we suspect that some highly-active buyers register their purchases in bulk, even though they happened on different dates. Excluding these transactions enhances the reliability of our final dataset, but our results are robust to this sample restriction.

Classifying properties

The OCOD register does not contain information on the type of properties owned, such as whether a property is residential, commercial, or a piece of land. We use the same property type classification method as the one developed in [Bourne et al. \(2022\)](#). Their method is described in their [appendix](#), and can be summarized as follows: a property will be classified as “business” if the address contains keywords related to business (such as “sho” or “restaurant”), if the property is matched (on street number, street name and district) to the Valuation Office Agency (VOA) rating list (which is used by local councils to charge business rates),⁴⁶ or if there is a business located in the same building. A property is classified as “residential” if its address contains keywords related to domestic properties (such as “flat”

⁴⁶Business rates are a tax charged on most non-domestic properties. This typically includes shops, offices, pubs, warehouses, factories, holiday rental homes or guest houses.

or “apartment”), if there are no businesses in the same geographic area (Output Area or Lower Layer Super Output Area), or if there is a match with the Price Paid Data. This classification method is hierarchical, which means that if two different rules leading to conflicting property types are satisfied by a property, the rule that comes first takes precedence and the property is classified accordingly.

Using this method, the OCOD properties are classified into five categories: land (10%), parking spaces (2%), air spaces (1%), residential (71%) and commercial (15%). 1% of the property types remain unknown. To focus our analysis on the effects of tax transparency on real estate markets, we only keep in our sample residential and commercial properties.

Inferring missing prices

Approximately 70% of the transactions in the dataset do not include information on the price paid. This can happen for several reasons. First, HM Land Registry only started collecting price data for transactions involving non-private buyers from October 2013.⁴⁷ Second, companies often purchase portfolios of properties, and if the price paid for each individual property is not specified in the transaction document,⁴⁸ the Land Registry excludes the price information from the register. To recover missing price information, we follow (Johannesen et al., 2022) and make use of another dataset managed by the Land Registry, the Price Paid Data (PPD). We predict the price of the remaining 46% of the purchases using the sample of transactions where the price is available.

Complementing the OCOD dataset with the PPD dataset The PPD provides details on the majority of residential and some non-residential property purchases in the UK since 1995, including price, address, and transaction date. After standardizing address information in both datasets, we match the OCOD to the PPD using the property’s address and postcode.

Several attempts to match the Price Paid Data to the Overseas Companies Ownership Dataset have been made before us (see Bourne et al. (2022); Johannesen et al. (2022)). Our own attempt is similar to what was done in Johannesen et al. (2022). Specifically, we focus on properties for which postcode information is available. We first match on simple addresses (street number, street name, and postcode). Then, we add properties where some information on the unit type (typically “flat”) and unit number is available. It is important to note that transactions involving a corporate buyer do not appear in the Price Paid Data before October 2013. Therefore, for all OCOD transactions that occurred before that date, a match is possible only if the same property was traded either before by a private individual or later by any kind of buyer. In this case, the date of the transaction appearing in both datasets will be very different. Contrary

⁴⁷For some transactions happening before that date the price information can be non-missing if the applications were lodged for registration to the Land Registry after 2013.

⁴⁸Form TR5: Transfer of a portfolio of titles (whole or part).

to Johannesen et al. (2022), we decide to keep these matches⁴⁹ and to use the price information from the Price Paid Data, and adjust for within-district house price inflation between the two dates of purchase. We were able to match 49% of the OCOD transactions to a Price Paid Data transaction when restricting the sample to properties classified as residential and where the postcode information was not missing.

Several reasons can explain why the match between the PPD and OCOD is not 100%. First, before Oct. 2013, if the buyer is a firm, the sale does not appear in the Price Paid Data. Thus, if a property is purchased by a company before Oct. 2013, the property appears in the Price Paid Data only if it has been purchased before⁵⁰ by a private individual or if there is a new transaction for this property later on. This condition therefore excludes potentially many properties we observe in OCOD from the Price Paid Data.⁵¹ Then, the Price Paid Data gathers information mostly on residential properties, even though some non-residential properties also appear in the data set. Therefore, most commercial properties appearing in the OCOD are likely to have no match in the PPD. Finally, even after 2013, there are a number of exclusions from the price paid data set, including, among others, some cases where a portfolio of properties is transferred.⁵²

Prediction of the remaining missing prices

46% of the OCOD estimation sample has missing price information, either because the information is missing in the OCOD dataset or because the property does not match to any property in the Price Paid Data. We predict missing prices using the sample of transactions where the price is available. Properties situated in the city of London are excluded as we lack any precise price information for this area. Let us denote Z_i a set of properties' characteristics that we observe in our sample. We can express the properties' (log) prices as

$$p_i = \beta Z_i + \epsilon_i \quad (4)$$

Where ϵ_i is the price component not captured by the set of predictors that we assume to be orthogonal to Z_i . Prices are winsorized at the 97% level, which is the level where the root mean square error is minimized. We estimate β from equation (4) using the subsample where the price is indicated, and the missing prices are then predicted using the resulting $\hat{\beta}$ as

$$\hat{p}_i = \hat{\beta} Z_i \quad (5)$$

⁴⁹ Johannesen et al. (2022) exclude matches when the transaction date as registered in OCOD and in PPD is more than four months apart.

⁵⁰but after 1995, when the Price Paid Data start.

⁵¹For transactions happening before Oct. 2013 that still match a PPD transaction (because the same property is traded later or before by a private individual), we still use the price information appearing in the Price Paid Data. In this case, the date of transaction recorded in the OCOD and Price Paid Data will be different and so we adjust for within district house price inflation between the two dates of purchase.

⁵²“Vesting Deeds Transmissions or Assents of more than one property”.

The prediction model is estimated by OLS, using 5-fold cross-validation. The set of predictors includes the property tenure (leasehold, freehold), a postcode fixed effect,⁵³ a quarter fixed effect and a dummy variable for residential. Using the Price Paid Data, we also include as predictors for each property the number of sales that occurred in the same postcode area, and the average price.⁵⁴ The $\hat{\beta}$ are estimated on a training sample composed of 80% of the transactions, and the quality of the predictions is evaluated with a test sample built with the 20% remaining observations. Table B.2 displays information on our out-of-sample fit (computed with our test sample). The adjusted R^2 is 0.69, the root mean square error (RMSE) is 0.86 and the mean absolute error (MAE) is 0.52.

TABLE B.2: PRICE PREDICTION - ALL

RMSE	Rsquared	MAE
0.86	0.69	0.52

Notes: This table describes the quality of our price inference and gives the value of the root mean squared error (RMSE), the R^2 and the mean absolute error (MAE) obtained when we regress p_i on Z_i (equation 4) using the full OCOD sample.

In figure B.2, we plot the predicted against the observed price in log for each transaction from the test sample. The smaller the distance between a point and the 45 degree line, the better the prediction. The figure shows that while some points are well above or below the 45 degree line, there is no evidence of systematic over- or underestimation of the true prices.

⁵³When we observe less than 6 transactions in a given postcode, we use a higher level of aggregation, until at least 6 transactions are observed in the considered area. We start by "output area", then use "middle layer super output area" and finally "district".

⁵⁴In case the postcode x quarter of a given transaction in the OCOD data does not match to any transactions in the Price Paid data, we use information on average price by Ward, which is a greater level of aggregation.

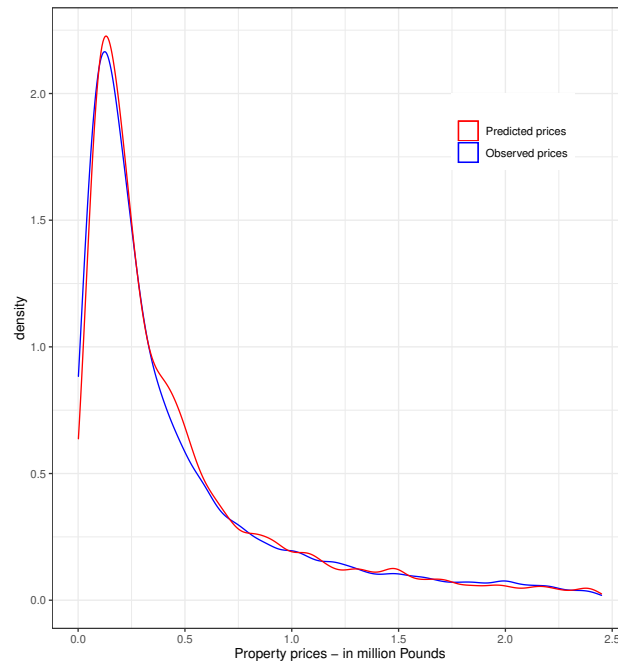


FIGURE B.1: DISTRIBUTION OF PREDICTED AND OBSERVED PRICES IN THE TEST SAMPLE

Notes: This figure displays the distribution of observed and predicted prices in the test sample. This sample is selected by randomly picking 20% of the transactions for which we have price information. It is not used to estimate equation (4) but only to test the quality of our price prediction. Because the price distribution is heavily right-skewed, we cut the distribution at the 90th percentile in order to see what is happening lower down in the price distribution, where most transactions happen.

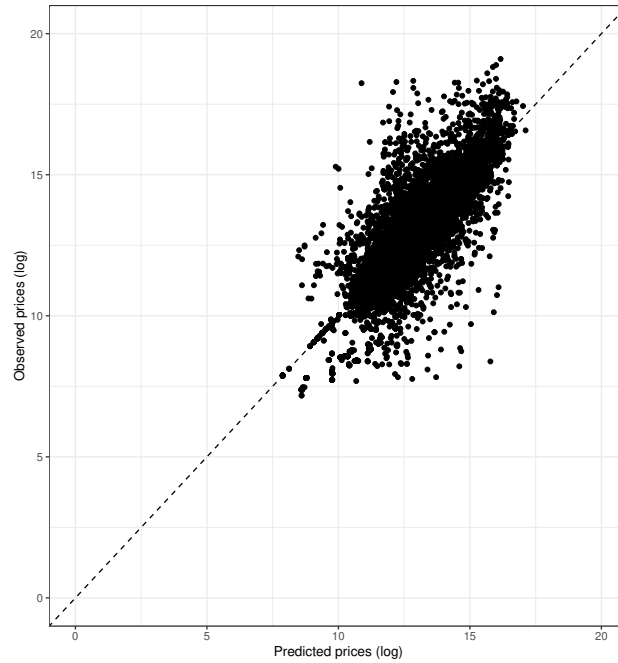


FIGURE B.2: DISTRIBUTION AGAINST OBSERVED PRICES IN THE TEST SAMPLE

Notes: This scatterplot displays the predicted against observed prices (in log) in the test sample, depending on whether we have information on the property size or not. The test sample is selected by randomly picking 20% of the transactions for which we have price information. It is not used to estimate (4) but only to test the quality of our price prediction. If a point lies on the 45 degrees line, this indicates that the predicted and observed prices correspond exactly.

C Offshore real estate wealth

C.1 Estimates of real estate wealth owned in England and Wales

Estimates presented in section 3 combine data on foreign ownership of real estate with estimates of the UK national balance sheet in 2018.⁵⁵ We compute total real estate wealth in England and Wales as the sum of residential and commercial real estate owned by (domestic + foreign) households and corporations.⁵⁶ We proceed in three steps. First, we compute the value of total commercial and residential wealth for the UK as a whole. Residential real estate wealth is simply the sum of the value of "dwellings" and "land underlying dwellings".⁵⁷ Commercial real estate is approximated in our paper by the category "other buildings" from the National Accounts.⁵⁸ Other buildings consist in fact of all buildings that are not dwellings and include commercial buildings, industrial buildings and buildings used to provide public services.⁵⁹ Since we ultimately discard properties owned by the government, the latter category is likely to be marginal in our estimates. However, our definition of commercial real estate will include some industrial properties (such as warehouses) and should therefore be considered as an upper bound for commercial real estate wealth. The ONS provides estimates of land underlying other buildings and structures. We allocate the value of land underlying other buildings proportionally to the value of other buildings relative to the total of other buildings and structures.⁶⁰

Then, following [Johannessen et al. \(2022\)](#), we scale our estimates by the land share of England and Wales (61.7%) in order to obtain estimates of the value of residential and commercial properties owned in England and Wales. Finally, the national accounts statistics provide information on who owns the assets (households, corporations, government), but the estimates for the value of the land owned are aggregated, thus the distinction between the value of the land underlying dwellings and underlying other buildings is not provided. In order to estimate how much is owned by households and (private) corporations only (thus discarding assets owned by the government or non-profit), we allocate the total value (buildings + land) obtained for the whole economy proportionally to the value of dwellings and other buildings owned by households and corporations in the total for all sectors, respectively.

C.2 Matching between OCOD dataset and beneficial ownership data

To learn more about the beneficial owners of UK properties, we match the OCOD data (2018 snapshot) to beneficial ownership data, namely we merge the January 2018 OCOD snapshot to the Leaks, the OpenLux and the CNBIOM data presented earlier. We conduct the matching in several steps:

⁵⁵National balance sheet estimates can be found [here](#).

⁵⁶Correspondence with the ONS taught us that UK real estate wealth owned by non-residents are included in the UK national balance sheet.

⁵⁷The ONS provides details on the value of land underlying dwellings and land underlying other buildings and structures [here](#).

⁵⁸System of National Accounts 2008 gives some example of other buildings: they can refer to "warehouses and industrial buildings, commercial buildings, buildings for public entertainment, hotels, restaurants, schools, hospitals, prisons etc."

⁵⁹Source: [here](#).

⁶⁰Note that by doing so, we are somewhat underestimating land underlying buildings (relative to structures) since buildings are likely located in areas where the land has a higher value.

TABLE C.1: NUMBER OF COMPANIES, TRANSACTIONS AND ULTIMATE BENEFICIAL OWNERS MATCHED TO LEAKS AND OPENLUX DATA

Source	Number of companies	Number of properties	Number of UBO
CNBIOM	29	66	36
OpenLux	151	288	197
Offshore Leaks	50	87	86
Panama Papers	995	1,905	1,636
Pandora Papers	334	554	466
Paradise Papers	146	424	530
Register	10,188	42,085	56,566
Total	11,893	45,409	59,517

Notes: This table shows by data source the number of companies, of transactions and of ultimate beneficial owners in our matched sample, obtained after matching the OCOD dataset to the beneficial ownership registry based on the name of the companies and country of incorporation. We do not identify any beneficial owners using the Bahamas Leaks, hence we do not show it in the data sources.

1. We use standardized companies names to match beneficial ownership data to the OCOD.⁶¹
2. We keep only companies for which the country of incorporation is the same in both datasets.
3. We keep companies that were active at the time of the purchase.⁶²
4. If a matched company appears to be owned by another company instead of a real person, we go one layer further and look for the owners of this second company in the leaks data. We repeat the operation four times in order to identify the “true” UBO of as many matched companies as possible.⁶³
5. We drop the “true” UBOs that still appear to be companies.⁶⁴
6. We allocate the shares of the companies identified. If a matched company is owned by n identified owners, we allocate $\frac{1}{n}$ share to each UBO.

C.3 Reallocating owners from tax havens to their most probable country of residence

The country information we have used comes from the beneficial ownership data. It encompasses the Offshore leaks and OpenLux databases, and the Register of Overseas Entities. However, information on country of residence could

⁶¹For example, we replace all the occurrences of “Ltd” by “Limited”, “Corp” by “Corporation” etc.

⁶²We have access to the company’s status over the years.

⁶³In some cases, the owners of a given company do not remain the same throughout the whole period. For cases when a person owns a company through different layers of companies, we impose the restriction that the owner owns each company at least over the period 2013-2015, which is the key period of interest for the CRS.

⁶⁴We exclude UBOs with the following words in their name: “COMPAGNIE”, “CORPORATION”, “COMPANY”, “INCORPORATED”, “TRUST”, “LTD”, “BUSINESS”, “LIMITED”, “LLC”, “FUND”, “INTERNATIONAL”, “EUROPE”, “FONDATION”, “FOUNDATION”, “INVESTMENT”, “CAPITAL”, “BANK”, “INC”, “LP”, “ACTION”, “ACTIVITY”, “HOLDING”, “GMBH”, “LLP”, “PLC”, “LTDA”, “GROUP”, “MANAGEMENT”, “SA”.

TABLE C.2: PERCENTAGE OF COMPANIES IDENTIFIED IN THE LONDON OCOD SAMPLE

Source	Number of transactions	Amount invested (extsterling billion)	Fraction of total transactions	Fraction of total amount invested
Full dataset	43,915	72.3	100%	100%
Matched	19,417	32.4	44.2%	44.7%
Identified	9,437	17.0	21.5%	23.5%

Notes: This table shows the number of OCOD transactions located in London we manage to link with their ultimate beneficial owners' using the Bahamas Leaks, the Offshore Leaks, the Paradise Papers, the Panama Papers, the Pandora Papers, OpenLux, CNBIOM data and the Register of Overseas Entities. Columns 2 and 3 show the raw number of transactions and their value in the full and matched samples, while columns 4 and 5 show the corresponding percentages the matched transactions represent. The "Matched" row shows these figures for all companies we find in one of these datasets. The "Identified" row shows these figures excluding companies whose beneficial owners are either i) from a tax haven ii) another company.

be erroneous for at least two reasons. First, in the Register of Overseas Entities, the country of residence variable is built using the correspondence address that beneficial owners are providing when the company is registered. They could choose to provide the address of their corporate service provider, or their wealth manager, services that are often provided in tax havens. For the Leaks, owners are linked to countries using all available information. This could lead to owners being wrongly associated to tax havens, if the only information available pertain to their offshore affairs and not to their true country of residence.

To gather the most accurate information on the residence of individuals purchasing real estate through companies in the United Kingdom, we reallocate company owners from tax havens to their most probable country of residence. We proceed in two different ways. For matches with the Register of Overseas Entities, we have access to information on the nationality of the owners, in addition to their correspondence addresses. When the correspondence address is in a tax haven, we then reallocate the owner to their country of citizenship.

Panel B of Appendix Figure A.6 displays the effects of this reallocation on estimates of real estate wealth by country in the identified sample. As expected, the value of properties attributed to owners from tax havens decreases significantly. Individuals from Hong Kong and from Switzerland in particular, are estimation to own a significantly smaller share of real estate than before. The value of properties allocated to individuals from the United Kingdom, on the other hand, increases significantly. This might be an overestimation, as some United Kingdom nationals residing in tax havens such as Switzerland, or Hong Kong, will wrongly be reassigned.

We then turn to owners identified thanks to the Leaks. One caveat of this data is that we do not have information on the country of citizenship of the owners of shell companies. However, we know the names of the owners. We leverage the fact that names credibly predicts the country of origin of a person. We use *Namsor*, a software able to predict the most probable country of origin given the name of a person.⁶⁵ We are able to analyse the names of the 2,915 company

⁶⁵The tool is available here: <https://namsor.app/>.

owners from the beneficial ownership data that appear to own a UK property in 2018.⁶⁶ Panel A of Appendix Figure A.6 shows the results of this reallocation of owners from tax havens. As per the register leaks, the amount of real estate wealth attributed to tax havens decreases substantially. On the other hand, property wealth of UK owners and of other European countries increases significantly.

C.4 Allocating offshore ownership of real estate

We identify 21% of the value of properties held by foreign companies in the United Kingdom. In addition, some individuals with a foreign correspondence address holding properties in the UK provide an address in a tax haven. This section describes our method to allocate the value unidentified properties to the most probable country of residence of their owners.

Corporate ownership First, we focus on properties held through foreign companies. To formalize the problem, consider first that the total value of UK real estate wealth ultimately owned from country c through foreign companies is equal to I^c . Define Ω^O as the set of firms appearing in the OCOD dataset and Ω_h^O a subset of Ω^O composed of all firms incorporated in jurisdiction h . Then, let us define as P_i the total value of UK properties owned by company i and as ϕ_i^c the share of owners of company i who are residents from country c . I^c is equal to $\sum_{i \in \Omega^O} \phi_i^c \cdot P_i$. Decomposing I^c , we get:

$$I^c = \sum_{i \in \Omega^O} \phi_i^c \cdot P_i \quad (6)$$

$$= \frac{\sum_{i \in \Omega^O} \phi_i^c \cdot P_i}{\sum_{i \in \Omega^O} P_i} \cdot \sum_{i \in \Omega^O} P_i \quad (7)$$

$$= \rho^c \cdot \sum_{i \in \Omega^O} P_i \quad (8)$$

With ρ^c , equal to $\frac{\sum_{i \in \Omega^O} \phi_i^c \cdot P_i}{\sum_{i \in \Omega^O} P_i}$, being the share of corporately owned UK real estate in the OCOD dataset that ultimately belongs to owners from country c .

After our matching procedure, we observe the value of ρ^c for a subsample of companies Ω^B matched through the beneficial ownership data. These companies are either from the Register of Overseas Entities (Ω^R) or the Leaks data (Ω^L). In order to estimate I^c , we assume that the unmatched properties follow the country-by-country distribution of ultimate ownership observed in the sample identified through the leaks, i.e. through companies Ω^L . In other words, we assume that for the unidentified sample, ρ^c is equal to $\hat{\rho}^c$, with $\hat{\rho}^c = \frac{\sum_{i \in \Omega^L} \phi_i^c \cdot P_i}{\sum_{i \in \Omega^L} P_i}$. This means that, as in the sample of OCOD companies matched to the leaks, 26.7% of the real estate wealth is owned by the United Kingdom, we allocate

⁶⁶This sample of 2,915 refers to the UBO of all matched companies, except the one from the CNBIOM where we don't observe UBO names.

26.7% of the unidentified wealth to the United Kingdom. We use the country distribution identified through the leaks data but not from the Register of Overseas Entities because the Register of Overseas Entities is likely to provide a biased picture of the offshore real estate market. As it is a UK policy, we are probably over-estimating the share of UK owners purchasing UK properties through shell companies, because UK residents would be more likely to comply with their reporting requirements. It probably also under-estimates the amount of properties held by e.g. Russians, as the Register was enforced following the sanctions taken against Russian individuals in February 2022. On the other hand, the Offshore Leaks are an event that can be considered as random from the point of view of the users of shell companies, who could not predict the disclosure of their ownership of these companies. As a result, they probably provide a more accurate picture of the origin of investments through shell companies in the UK. We show in Appendix figure A.9 that allocating ultimate ownership for each incorporation country barely affects our results.⁶⁷

Direct ownership Next, we “correct” the ownership country of UK properties held by individuals when the correspondence address is in a tax haven by predicting the most probable true residence country of the owner. As shown in figure A.4, an important share of the owners appears to be tax havens residents. However, it is likely that the true investors behind directly owned UK properties are mostly non tax havens residents and that either the “correspondence address” is not the true address of the owner or individuals appearing as tax havens residents are nominees and not the true beneficial owners of the asset. To allocate direct ownership from tax havens to the most probable “true” residence countries of the owners, we exploit the Register of Overseas Entities data. Indeed, the Register provides information on both the nationality of the owners of overseas companies owning properties in the UK, and their correspondence address. Let us define $\hat{D}^c$ as the value of UK properties directly held by residents from country c . Then, let us define ρ_h^{*c} as the share of all individuals with a correspondence address in jurisdiction h that are citizens from country c as observed in the register data. For example, we observe in the Register data that 24.8% of the individuals with a correspondence address in Hong Kong are Chinese nationals. ρ_h^{*c} is thus equal to 0.248 when country c is China and country h is Hong Kong.⁶⁸ Then, defining as Q^c the value of all directly owned properties by residents from country c and TH as the set of tax haven jurisdictions, we estimate D^c for each country c as:

$$\hat{D}^c = Q^c \cdot \mathbb{1}_{\{c \notin TH\}} + \sum_{h \in TH} \rho_h^{*c} \cdot Q^h \quad (9)$$

This means that we reallocate ownership from tax havens to the most likely country of residence of the individuals, but we consider that individuals with a correspondence address in a non-haven are residents from this country.

Combining $\hat{I}^c$ and $\hat{D}^c$, we present in figure VI our estimates of the geographic distribution of UK real estate

⁶⁷Instead of only considering the country distribution of owners identified through the leaks as in the main specification, we consider the country distribution identified through the leaks for each country of incorporation of a company. This means that as according to matches between the OCOD and the leaks e.g. 16% of owners of companies incorporated in the Bahamas are from the United Kingdom, we allocate 16% of the unidentified sample of Bahamas companies to the United Kingdom. We repeat this exercise for each country of incorporation.

⁶⁸Here, we exploit information contained in the full Register data rather than in the matched sample only.

ownership, by ultimate owners' country of residence.

Alternative method for corporate ownership We also estimate the geographical distribution of ultimate ownership using the full information available in the leaks data, including both the country of incorporation of the company and the country of residence of the owner. Instead of only considering the country distribution of owners identified through the leaks as in the main specification, we consider the country distribution identified through the leaks for each country of incorporation of a company. This means that as according to matches between the OCOD and the leaks e.g. 16% of owners of companies incorporated in the Bahamas are from the United Kingdom, we allocate 16% of the unidentified sample of Bahamas companies to the United Kingdom. We repeat this exercise for each country of incorporation. Results—very similar to those obtained with our baseline method—are provided in Appendix Figure [A.8](#).

D Tax reforms

A number of tax reforms affecting the purchase and the ownership of UK *residential* properties by foreign companies have been introduced over the 2012-2017 period.⁶⁹ We provide here some details about the most significant changes. They relate to four taxes, namely the Stamp Duty and Land Tax (SDLT), the Annual Tax on Enveloped Dwellings (ATED), the Capital Gains Tax and the Inheritance Tax.

Stamp Duty and Land Tax If an individual buys a residential property in the UK, Stamp Duty and Land Tax (SDLT) is charged. The rate is progressive and has changed a number of times since 2012, with a top marginal rate of 12% in 2021. In 2012, a 15% rate is applied to purchases made by corporate bodies when the price exceeds £2M (£500K from 2014). However, in the case of a company carrying a property rental business, the 15% higher rate charge does not apply and the rate is the same as for a natural person. Note that this difference in the SDLT rate depending on whether the property is used for letting or not disappears in April 2016 when an additional 3% second home surcharge starts to apply to companies carrying a property rental business.

Annual Tax on Enveloped Dwellings A way to avoid the SDLT is to buy the property through a corporate structure and to buy the shares of the company instead of the property itself. In order to counterbalance this tax privilege, the Annual Tax on Enveloped Dwellings (ATED) is introduced in the Finance Act of 2013. It is an annual tax payable by companies owning UK residential property valued at more than £2M in 2013 (the threshold is now set at £500,000) and occupied rather than let out to an unconnected person.⁷⁰ The amount charged is progressive, lying from £3,700 (property value below £1M) to £237,400 (values above £20M) in 2021-2022.

Capital Gains Tax The rules related to the taxation of capital gains arising from the indirect ownership of UK properties have evolved during the period we consider. In 2013, the ATED-related Capital Gains Tax (CGT) is introduced. It applies to properties also covered by ATED at a rate of 28%. In 2015, a new tax, called the Non-Resident CGT, starts to apply, under which all non-UK resident persons and companies will pay a 20% capital gains tax on any profit realised on their residential property after 6 April 2015.

Inheritance Tax For non-UK residents and non-dom individuals, a common way to avoid inheritance tax on a UK property has been to hold it through an offshore company. Indeed, while the personal representatives or the beneficiaries of a non-dom individual owning UK property *directly* are liable to the inheritance tax in case of death (40% on the value of the property), no inheritance tax is applied to the shares of a foreign company - even though its sole asset is a

⁶⁹Note that our analysis of the responses to the CRS stops in 2017 and that we describe the tax changes happening before 2017 only. An important tax change for commercial properties happens in 2019, with the extension of the Non-Resident Capital Gains Tax to commercial properties. Other changes, both for residential and commercial properties also happen after 2017.

⁷⁰Properties let to unconnected parties qualify for relief and are therefore exempt from the ATED charge.

UK property - and the inheritance tax can therefore be avoided.⁷¹ This tax privilege was however drastically reduced in April 2017, when companies which value are attributable to a UK residential property interest (UK RPI) started to fall within the scope of Inheritance Tax.⁷²

⁷¹This scheme does not work for UK residents.

⁷²If the company has other assets (e.g. located in France), Inheritance tax will only apply on the fraction of assets subject to the English Inheritance tax. Moreover, if the deceased person's stake in the company is too small (that is, less than 5% when combined with the stakes of persons connected to her) then the new rule doesn't apply.